OPTIMAL FEEDBACK LAW RECOVERY BY GRADIENT-AUGMENTED NEURAL NETWORK WITH NONCONVEX REGULARIZATION ON THE UNBOUNDED PARAMETER SPACE

ABSTRACT. We recover optimal feedback laws for nonlinear control problems by approximating Hamilton–Jacobi–Bellman value functions with sparse shallow neural networks. Value and gradient are generated from the open-loop problem by the Pontryagin maximum principle, and the network is trained with a gradient-augmented H^1 fidelity. For nonhomogeneous activation functions on the unbounded parameter space $\Omega = \mathbb{R}^{d+1}$, fix $p > 0$, set $w_p(\omega) = 1 + |\omega|^p$ and $\mu_p = w_p\mu$, and normalize the ridge atom K as $K_p = K/w_p$. We study the normalized objective

$$J(\mu) = L(\mu) + \alpha\Phi_\phi(\mu_p),$$

where the concave scalar penalty has finite positive right slope at zero. Normalization makes the dictionary vanish at infinity under a polynomial growth gap. This yields existence by normalized weak-* compactness. Under strict bounded-interval curvature, every finite-objective local minimizer has finite support and satisfies a covering-number bound, while the global minimum is attained by a finite network. Exact maximization of the weighted Gâteaux derivative at each insertion, followed by a nonincreasing correction, drives its excess above the insertion threshold to zero, with a best-iterate rate of order $n^{-1/2}$. For positively k-homogeneous activations, we retain the equivalent formulation on the unit sphere with its induced fractional-power penalty on the outer coefficients. The resulting algorithms are studied on the Van der Pol oscillator and the pendulum swing-up problem, whose value function has a gradient discontinuity across a switching set.

CONTENTS

1. Introduction

The problem under consideration is the dynamic optimal control problem in the space-time domain

$$
(P) \qquad
\begin{aligned}
&\inf_{u\in L^2(t_0,T;\mathbb{R}^{m_u})} \mathcal{C}(u;t_0,x) := \int_{t_0}^{T} \big(h(y(t)) + \eta|u(t)|^2\big)\,dt, \quad \eta>0,\\
&\text{s.t.}\\
&\dot y = f(y) + g(y)u,\\
&y(t_0) = x,
\end{aligned}
$$

We assume $y(t)\in\mathbb{R}^d$, $u(t)\in\mathbb{R}^{m_u}$, $h\in C^1(\mathbb{R}^d)$, $f\in C^1(\mathbb{R}^d;\mathbb{R}^d)$, and $g\in C^1(\mathbb{R}^d;\mathbb{R}^{d\times m_u})$. The control-theoretic results below are used under the following standing well-posedness assumptions: for every admissible initial pair and every $u\in L^2(t_0,T;\mathbb{R}^{m_u})$, the state equation has a unique absolutely continuous solution; the value is finite and the infimum is attained; every minimizer satisfies the normal Pontryagin maximum principle; and the dynamic programming principle and comparison hypotheses hold, so that the continuous value function is characterized as the viscosity solution of the HJB equation. These assumptions are imposed explicitly rather than inferred from the displayed C^1 regularity alone.

One approach computes an open-loop state-costate-control path for a fixed initial condition x. The Pontryagin maximum principle (PMP) yields a coupled forward–backward two-point boundary value problem:

$$
\text{(TPBVP)} \qquad
\begin{cases}
\dfrac{d}{dt}y^* = f(y^*) + g(y^*)u^*,\\
y^*(t_0) = x,\\
-\dfrac{d}{dt}p^* = \partial_y\Big(f(y^*) + g(y^*)u^*\Big)^\top p^* + \partial_y h(y^*),\\
p^*(T) = 0.
\end{cases}
$$

The stationarity condition with respect to the control gives

$$
u^*(t) = -\frac{1}{2\eta} g(y^*(t))^\top p^*(t) \quad \text{for almost every } t\in(t_0,T).
$$

The open-loop control is not robust with respect to disturbances of the state [1]: it is computed for a single initial condition, and deviations from the optimal path are not corrected. In addition, numerical methods for the two-point boundary value problem are sensitive to the initial guess and may fail to converge. The dynamic programming principle instead introduces the value function

$$
V(t,x) := \inf_u \mathcal{C}(u;t,x)
$$

which satisfies the Hamilton–Jacobi–Bellman (HJB) equation in the viscosity sense:

$$
\text{(HJB)} \qquad
\begin{cases}
\partial_t V - \dfrac{1}{4\eta}\nabla V(t,x)^\top g(x)g(x)^\top \nabla V(t,x) + \nabla V(t,x)^\top f(x) + h(x) = 0,\\
V(T,x) = 0.
\end{cases}
$$

At points where V is differentiable, the equation holds classically and Hamiltonian minimization gives the feedback $u^*(t,x) = -(2\eta)^{-1} g(x)^\top \nabla V(t,x)$. The value function encodes the optimal cost from every state. The feedback law synthesized from it corrects deviations from a nominal path.

For nonlinear control problems one usually seeks a feedback law, which requires solving the HJB equation. Computing this function is notoriously difficult because of the curse of dimensionality. Alongside classical methods, data-driven methods have therefore become increasingly common. One line of work, introduced in [18], generates the training data with (TPBVP) and approximates the optimal value function from it. The optimal feedback law is then synthesized from the fitted value function after validation. Its samples are causality-free: generating one does not require approximate values at neighbouring grid points. Moreover, Azmi, Kalise and Kunisch [1] enrich the dataset with adjoint information: a state-adjoint-control triple is generated by solving the open-loop problem, and at differentiability points the adjoint supplies the value gradient. The value function is then approximated by hyperbolic-cross basis functions with an ℓ^1 penalty, and including the adjoint data targets an H^1 approximation. Neural-network approximations form another growing line of work [16, 23]. Other prior information, such as semiconcavity of the value function, can also be incorporated. Alternatively, one can use an HJB- or physics-informed loss [9, 17, 22, 24, 15].

For $\boldsymbol{\omega} = (\omega_n)_{n=1}^N$, $\omega_n = (a_n, b_n)$, and $c = (c_n)_{n=1}^N$, we use the two-layer network

$$\mathcal{N}_{\boldsymbol{\omega},c}(x) := \sum_{n=1}^{N} c_n \rho(a_n \cdot x + b_n) \in \mathbb{R}, \tag{1}$$

where ρ is a sufficiently smooth activation function and N the width of the network. We associate it with the finite signed measure $\mu_{\boldsymbol{\omega},c} := \sum_{n=1}^N c_n \delta_{\omega_n}$. Its gradient is

$$\nabla_x \mathcal{N}_{\boldsymbol{\omega},c}(x) = \sum_{n=1}^{N} c_n \nabla_x \rho(a_n \cdot x + b_n) \in \mathbb{R}^d.$$

For the finite-horizon problem, we write $V(x) := V(0, x)$. The infinite-horizon pendulum value considered below is stationary, so it has no suppressed time argument. The training data are generated by solving (TPBVP). For the trajectory issued from x^m, let

$$g^m := p^{*,m}(0)$$

denote the costate label in the gradient channel. At every sample where V is differentiable, sensitivity relations give $g^m = \nabla V(x^m)$ [8, 6, 7]. At a switching point, competing optimal paths may supply different costate labels; no classical gradient is asserted there. Thus the data consist of triples $\{x^m, V(x^m), g^m\}_{m=1}^M$. We have the minimization problem

$$\min_{N,\boldsymbol{\omega},c} l^M(\mu_{\boldsymbol{\omega},c}), \qquad l^M(\mu_{\boldsymbol{\omega},c}) := \frac{1}{2M} \sum_{m=1}^{M} \left(|V(x^m) - \mathcal{N}_{\boldsymbol{\omega},c}(x^m)|^2 + |g^m - \nabla \mathcal{N}_{\boldsymbol{\omega},c}(x^m)|^2 \right).$$

Fix a bounded Lipschitz training domain $D \subset \mathbb{R}^d$. Section 3 introduces the corresponding population fidelity L with respect to Lebesgue measure on D.

A traditional sparse shallow network uses the ReLU activation and an ℓ^1 penalty on the outer coefficients [4, 2]. Under gradient-augmented fitting, this formulation has two limitations. 1. The convex penalty may retain redundant neurons whose inner parameters cluster. 2. The second-order weak derivative of ReLU does not exist as a function while it is a distribution, and we cannot use it to approximate high-order derivatives.

Sparsity beyond the convex penalty. The number of neurons N is itself an optimization variable. Although width alone does not generally explain a model's inductive bias [19], sparsity is crucial for computational efficiency in many problems. Furthermore, sparsity is necessary for certain applications, such as knowledge distillation and deconvolution of point-like objects. A standard sparse-learning procedure uses a convex reformulation with a sparsity-promoting penalty on the outer weights, most commonly the ℓ^1 penalty (LASSO). Pieper and Petrosyan [20] showed that convex penalties cannot effectively eliminate this redundancy when neurons cluster in a small neighborhood and introduced a concave nonconvex functional for this purpose. We extend the relaxed concave measure formulation to gradient-augmented fitting with an unbounded, nonhomogeneous dictionary. Placing the parameter weight inside the scalar penalty preserves its strict curvature after normalization and yields both finite support and a quantitative neuron bound. For positively k-homogeneous activations, radial rescaling instead induces a fractional exponent and its corresponding width bound.

Activation functions beyond ReLU. We therefore consider smoother activation functions. This extension requires a treatment of the unbounded inner-parameter space for gradient training. For positively k-homogeneous activations, radial scaling can be absorbed into the outer coefficient, and the network has an equivalent representation with inner parameters on the unit sphere. For nonhomogeneous activations such as tanh, softplus, and the Gaussian, radial scaling changes the atom and cannot be removed by normalization. Without normalization, sequences with $|\omega| \to \infty$ may retain a nonzero contribution to the represented function, as shown in Section 3.3. We therefore introduce

$$w_p(\omega) = 1 + |\omega|^p, \qquad \mu_p = w_p\mu, \qquad K_p = \frac{K}{w_p}.$$

Then $\mathcal{N}\mu = \int_\Omega K_p \, d\mu_p$, and K_p vanishes at infinity when p exceeds the polynomial growth of K. We analyze

$$J(\mu) = L(\mu) + \alpha\Phi_\phi(\mu_p),$$

where $p > 0$ is the moment order, $\alpha > 0$ is the regularization weight, and ϕ and Φ_ϕ are the scalar and measure penalties defined in Section 3. The symbol γ used below is the shape parameter of the log penalty in (32). The finite-network penalty is $\alpha \sum_n \phi(w_p(\omega_n)|c_n|)$ after repeated locations are merged.

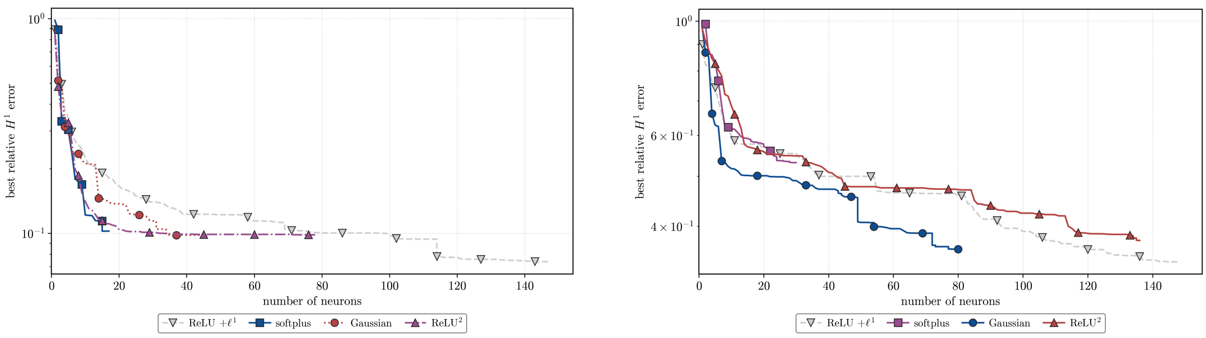
Contributions. The contributions are the following.

(1) *Nonhomogeneous activations on the unbounded parameter domain.* For concave scalar penalties with finite positive right slope, we prove a disjoint-test representation and derive weak-* lower semicontinuity and, under coercivity, bounded total-variation sublevels (Theorem 3.9 and Corollary 3.10). The weighted-measure isometry in Theorem 3.13 moves the parameter growth into the normalized atom K_p. For continuous activations of polynomial value growth $s_0 < p$, this gives a global minimizer on $\Omega = \mathbb{R}^{d+1}$ (Theorem 3.16). The unnormalized fidelity can fail to be weak-* lower semicontinuous, and the unnormalized problem can fail to attain its infimum (Examples 3.11 and 3.12). Under full H^1 growth $s_1 < p$ and strict bounded-interval curvature, every finite-objective local minimizer is finite atomic, its normalized atoms are quantitatively separated, and its width is bounded by a covering number (Theorem 3.20). Corollary 3.21 gives an attained finite-network problem and an a priori neuron bound. The normalized optimality system is Theorem 3.17. Theorem 5.1 gives squared one-step descent, and Corollary 5.2 gives an $n^{-1/2}$ best-iterate rate for the excess above the insertion threshold.

(2) *The k-homogeneous formulation.* Extending the degree-one ReLU rescaling equivalence of Pieper and Petrosyan [20, Appendix F, Proposition 3], we show that for positively k-homogeneous activations, minimizing a separable penalty on the inner and outer parameters over radial representations induces the fractional-power penalty $|c|^{2/(k+1)}$ on the unit sphere (Lemma 4.3), and the resulting problem is posed over all of $\mathcal{M}(\mathbb{S}^d)$. Finite atomic truncations show that this problem and the finite-network problem have the same infimum (Proposition 4.6). We derive the atomwise optimality conditions and a positive lower bound on the weights of every total-variation local minimizer (Theorem 4.7), and prove that a global minimizer exists, that every global minimizer is a finite network obeying an explicit a priori width bound (Theorem 4.9), and that every total-variation local minimizer with finite objective has finite support (Corollary 4.11).

(3) *Gradient-augmented algorithms and feedback recovery.* The first insertion algorithm places the polynomial parameter weight inside the log penalty and searches $|P^M_{p,\mu}|$; the second uses the fractional-exponent penalty on the unit sphere. The numerical examples compare activation regularity, support size, and feedback recovery for a smooth value function and for a value function whose gradient is discontinuous across a switching set.

Figure 1 compares the traditional ReLU network with an ℓ^1 penalty against representative models from the two formulations. The comparison is one of accuracy at a given support size rather than of lowest attainable error. On the Van der Pol problem the traditional ℓ^1 network is the less accurate of the two at every support below 121 atoms — 1.62×10^{-1} against 1.00×10^{-1} at twenty atoms — and overtakes the sparse penalties only past that width, where it eventually reaches 7.4×10^{-2}. The nonconvex penalties therefore buy accuracy per neuron, not a lower floor. On the pendulum, where the value gradient jumps across the switching set, the Gaussian gives the smallest regional errors, while ReLU^2 gives the feedback law closest to the reference. Softplus also reaches an upright from both tested starts, but at a higher cost from the harder start.



(A) Van der Pol oscillator. (B) Pendulum swing-up.

FIGURE 1. Running best relative H^1 validation error as neurons are inserted. The nonhomogeneous activations are fitted by Algorithm 1 with $(\alpha, \gamma, p) = (10^{-4}, 10, 2.01)$; ReLU^2 and the traditional ℓ^1 network are fitted by Algorithm 2, the latter at $k = 1$, where the induced exponent is $q = 1$. All curves use one atom per outer iteration and seed 42.

Section 2 records the measure-theoretic preliminaries, Sections 3–4 develop the nonhomogeneous and homogeneous formulations, Section 5 gives the data and training algorithms, and Section 6 reports the numerical studies.

2. Preliminaries

This section reviews the measure-theoretic facts used in the paper. The learning objectives and problem-specific assumptions are introduced in Section 3.

2.1. **Finite signed Radon measures.** Let Ω be a locally compact Polish space. We write $\mathcal{M}(\Omega)$ for the Banach space of finite signed Radon measures on Ω, equipped with the total-variation norm

$$\|\mu\|_{\mathcal{M}(\Omega)} := |\mu|(\Omega).$$

By the Riesz representation theorem, $\mathcal{M}(\Omega) = C_0(\Omega)^*$ under the pairing

$$\langle f, \mu \rangle := \int_\Omega f \, d\mu.$$

Here $C_0(\Omega)$ denotes the continuous functions vanishing at infinity.

For $\mu \in \mathcal{M}(\Omega)$ we define the *atom set* $\operatorname{atom} \mu := \{\omega \in \Omega : |\mu|(\{\omega\}) > 0\}$, which is at most countable, the purely atomic part $\mu_{\text{atom}} := \sum_{\omega \in \operatorname{atom} \mu} \mu(\{\omega\}) \, \delta_\omega$, and the nonatomic part $\mu_{\text{cont}} := \mu - \mu_{\text{atom}}$ [5]. This gives the decomposition $\mu = \mu_{\text{cont}} + \mu_{\text{atom}}$. The atom set need not be closed because atoms may accumulate at a point that carries no mass. The polar decomposition of a signed measure is written as

$$\mu = \operatorname{sign}(\mu) \, |\mu|, \qquad \operatorname{sign}(\mu) := \frac{d\mu}{d|\mu|} \in \{-1, 1\} \quad |\mu|\text{-almost everywhere.}$$

2.2. **Measures with a finite moment.** In this subsection let $\Omega = \mathbb{R}^{d_\Omega}$, where $d_\Omega \geq 1$. For $p > 0$, set

$$w_p(\omega) := 1 + |\omega|^p$$

and define

$$\mathcal{M}_p(\Omega) := \left\{ \mu \in \mathcal{M}(\Omega) : \int_\Omega w_p \, d|\mu| < \infty \right\}, \qquad \|\mu\|_{\mathcal{M}_p(\Omega)} := \int_\Omega w_p \, d|\mu|. \tag{2}$$

The associated space of continuous functions is

$$C_{w_p^{-1}}(\Omega) := \left\{ f \in C(\Omega) : \frac{f}{w_p} \in C_0(\Omega) \right\}, \qquad \|f\|_{C_{w_p^{-1}}} := \sup_{\omega \in \Omega} \frac{|f(\omega)|}{w_p(\omega)}.$$

Under the dual pairing, $\mathcal{M}_p(\Omega) = C_{w_p^{-1}}(\Omega)^*$ isometrically [3, Section 5]. For an atomic measure,

$$\left\| \sum_{j=1}^N c_j \delta_{\omega_j} \right\|_{\mathcal{M}_p(\Omega)} = \sum_{\eta \in \{\omega_1, \dots, \omega_N\}} \left| \sum_{j : \omega_j = \eta} c_j \right| w_p(\eta).$$

Notation. Throughout the remainder of the paper, $D \subset \mathbb{R}^d$ is a bounded Lipschitz domain, $\nu := \mathcal{L}^d|_D$ is the population measure, and $L^2(D)$ means $L^2(D, \nu)$. We assume that the population target V belongs to $H^1(D)$. The symbol $M \geq 1$ denotes the number of samples in an empirical loss. The inner-parameter domain is $\Omega = \mathbb{R}^{d+1}$, and $\rho \in C(\mathbb{R})$ denotes the fixed activation function of (1), with

$$\omega = (a, b), \qquad [K(\omega)](x) := \rho(a \cdot x + b).$$

With this notation, the network in (1) is $\mathcal{N}_{\boldsymbol{\omega}, c} = \sum_{n=1}^N c_n K(\omega_n)$. The symbol δ_ω denotes the Dirac measure at ω, while μ' denotes a general signed-measure direction. We reserve p for the moment order.

3. The learning problem

3.1. **The value operator and the gradient-augmented loss.** The finite network $\mathcal{N}_{\boldsymbol{\omega}, c}$ corresponds to the finite atomic measure $\mu_{\boldsymbol{\omega}, c}$. This motivates the integral formulation below, which agrees exactly with the finite sum on atomic measures.

Assumption 3.1 (Regularity and polynomial growth of ρ)**.** The activation satisfies $\rho \in C^1(\mathbb{R})$. There are $C_\rho > 0$, $s_0 \geq 0$, and $s_1 \geq \max\{s_0, 1\}$ such that

(1) $|\rho(z)| \leq C_\rho (1 + |z|)^{s_0}$ for every $z \in \mathbb{R}$;
(2) $|\rho'(z)| \leq C_\rho (1 + |z|)^{s_1 - 1}$ for every $z \in \mathbb{R}$.

Since D is bounded, Assumption 3.1 gives constants $C_D, C_K > 0$ such that

$$\|K(\omega)\|_{L^2(D)} \leq C_D (1 + |\omega|)^{s_0}. \tag{3}$$

Moreover, $\nabla K(a, b)(x) = a \rho'(a \cdot x + b)$, and hence

$$\|K(\omega)\|_{H^1(D)} \leq C_K (1 + |\omega|)^{s_1}. \tag{4}$$

If $\omega_n \to \omega$, their pre-activations lie in a common compact interval on D. Uniform continuity of ρ and ρ' on that interval, together with $a_n \to a$, therefore gives $K(\omega_n) \to K(\omega)$ in $H^1(D)$. Thus $K : \Omega \to H^1(D)$ is continuous and, in particular, strongly measurable.

For $\mu \in \mathcal{M}(\Omega)$ satisfying

$$\int_\Omega \|K(\omega)\|_{H^1(D)} \, d|\mu|(\omega) < \infty, \tag{5}$$

define the $H^1(D)$-valued Bochner integral

$$\mathcal{N}\mu := \int_\Omega K(\omega) \, d\mu(\omega).$$

By (4), condition (5) holds for $\mu \in \mathcal{M}_p(\Omega)$ when $p > s_1$. For $\psi \in C_c^\infty(D)$ and $i \in \{1, \dots, d\}$, the growth bound permits Fubini's theorem and gives

$$\int_D (\mathcal{N}\mu)\, \partial_i \psi \, dx = \int_\Omega \int_D K(\omega)\, \partial_i \psi \, dx \, d\mu(\omega)$$
$$= -\int_D \left(\int_\Omega \partial_i K(\omega)\, d\mu(\omega) \right) \psi \, dx.$$

Consequently,

$$\nabla(\mathcal{N}\mu) = \int_\Omega \nabla K(\omega)\, d\mu(\omega) \quad \text{in } L^2(D; \mathbb{R}^d).$$

For measures satisfying only $\int_\Omega \|K(\omega)\|_{L^2(D)}\, d|\mu|(\omega) < \infty$, the same definition is valid as an $L^2(D)$-valued Bochner integral. By (3), this includes $\mathcal{M}_p(\Omega)$ when $p > s_0$. In particular, $\mathcal{N}\mu_{\boldsymbol{\omega},c} = \mathcal{N}_{\boldsymbol{\omega},c}$. Whenever $\mathcal{N}\mu \in H^1(D)$, set $r_\mu := \mathcal{N}\mu - V$.

The fidelity term is the gradient-augmented loss with the distributional gradient convention. If the $L^2(D)$-valued integral defining $\mathcal{N}\mu$ does not exist, we set $L(\mu) = +\infty$; otherwise,

$$L(\mu) := \begin{cases} \frac{1}{2}\|\mathcal{N}\mu - V\|_{L^2(D)}^2 + \frac{1}{2}\|\nabla(\mathcal{N}\mu) - \nabla V\|_{L^2(D)}^2, & \mathcal{N}\mu \in H^1(D), \\ +\infty, & \text{otherwise.} \end{cases}$$

For sampled labels $(V(x^m), g^m)_{m=1}^M$ and a finite network for which the displayed point values exist, the corresponding empirical fidelity is

$$l^M(\mu) := \frac{1}{2M} \sum_{m=1}^M \left(|\mathcal{N}\mu(x^m) - V(x^m)|^2 + |\nabla \mathcal{N}\mu(x^m) - g^m|^2 \right). \tag{6}$$

3.2. **The nonconvex functional.** Let $\phi : \mathbb{R}_+ \to \mathbb{R}_+$ satisfy the following assumptions.

Assumption 3.2. The finite right derivative

$$L_\phi := \phi'(0+)$$

exists and satisfies $0 < L_\phi < \infty$. We use the convention $\phi'(0) := L_\phi$.

Assumption 3.3. The function ϕ is concave, nondecreasing, and coercive, with $\phi(0) = 0$, and belongs to $C^1((0, \infty))$.

Assumptions 3.2 and 3.3 imply that $\phi'(t) > 0$ for every $t > 0$. Indeed, if $\phi'(t_0) = 0$ at some $t_0 > 0$, concavity and monotonicity would give $\phi'(t) = 0$ for every $t \geq t_0$, making ϕ constant there and contradicting coercivity. Hence ϕ is continuous and strictly increasing from $[0, \infty)$ onto $[0, \infty)$. We denote its inverse by ϕ^{-1}.

Assumption 3.4 (Local curvature)**.** The function ϕ belongs to $C^2((0, \infty))$ and satisfies both of the following conditions:

(1) there are $\gamma_1 \geq 0$ and $\hat{z}_1 > 0$ such that $-\gamma_1(z_2 - z_1) \leq \phi'(z_2) - \phi'(z_1) \leq 0$ for $0 \leq z_1 \leq z_2 \leq \hat{z}_1$;

(2) for every $A > 0$,

$$\gamma_A := -\sup_{0 < t \leq A} \phi''(t) > 0.$$

Remark 3.5. The lower curvature control in Assumption 3.4(1) is local and does not follow from global concavity. For example, $\phi(z) = z - \frac{2}{3}z^{3/2}$ near the origin has $\phi''(z) = -\frac{1}{2}z^{-1/2}$; it can be extended linearly after a small positive point so as to remain nondecreasing, concave, and coercive.

Lemma 3.6. *Let $\phi : [0, \infty) \to [0, \infty)$ be concave with $\phi(0) = 0$. Then*

(1) *$\phi(z_1 + z_2) \leq \phi(z_1) + \phi(z_2)$ for all $z_1, z_2 \geq 0$.*

If in addition $L_\phi := \phi'(0+)$ is finite, then

(2) *$\phi(z) \le L_\phi z$ for all $z \ge 0$;*
(3) *$\phi(z)/z \longrightarrow L_\phi$ as $z \downarrow 0$;*
(4) *for every nonnegative sequence $(t_j)_{j\ge1}$ with $\sum_{j\ge1} t_j < \infty$,*

$$\phi\left(\sum_{j\ge1} t_j\right) \le \sum_{j\ge1} \phi(t_j).$$

Proof. Concavity and $\phi(0) = 0$ make $z \mapsto \phi(z)/z$ nonincreasing on $(0,\infty)$: for $0 < z_1 \le z_2$, writing $z_1 = \frac{z_1}{z_2} z_2 + \big(1 - \frac{z_1}{z_2}\big) \cdot 0$ and using concavity together with $\phi(0) = 0$ gives $\phi(z_1) \ge \frac{z_1}{z_2}\phi(z_2)$.

(1) Suppose without loss of generality that $0 < z_1 \le z_2$. Then

$$\begin{aligned}
\phi(z_1+z_2) &\le \frac{z_1+z_2}{z_2}\phi(z_2) && \text{because } \phi(z)/z \text{ is nonincreasing}\\
&= \phi(z_2) + z_1\frac{\phi(z_2)}{z_2}\\
&\le \phi(z_2) + \phi(z_1) && \text{by the same monotonicity.}
\end{aligned}$$

The cases with $z_1 = 0$ or $z_2 = 0$ follow from $\phi(0) = 0$.

(2) and (3) By the same monotonicity and $\phi(0) = 0$,

$$\frac{\phi(z)}{z} \le \lim_{s\downarrow0} \frac{\phi(s)}{s} = \lim_{s\downarrow0}\frac{\phi(s)-\phi(0)}{s} = L_\phi \qquad (z>0),$$

and the displayed limit is precisely (3).

(4) By (2) the function ϕ is continuous at 0, and a finite concave function is continuous on $(0,\infty)$. Iterating (1) over partial sums gives

$$\phi\Big(\sum_{j\le N} t_j\Big) \le \sum_{j\le N}\phi(t_j) \le \sum_{j\ge1}\phi(t_j),$$

the last step because the terms are nonnegative. Letting $N \to \infty$ and using continuity of ϕ at $\sum_{j\ge1} t_j$ proves the claim. ☐

Assumption 3.3 implies the standing hypotheses of Lemma 3.6, and Assumption 3.2 supplies the finite right slope needed for parts (2)–(4). Neither coercivity nor differentiability on $(0,\infty)$ is used, so the lemma also applies under the weaker hypotheses of Theorem 3.9.

Proposition 3.7 (Second-order bounds)**.** *Suppose that $\phi(0) = 0$ and Assumption 3.2 holds.*

(1) *If the derivative bound in Assumption 3.4(1) holds, then $\phi(z) \ge L_\phi z - \frac{\gamma_1}{2}z^2$ for $0 \le z \le \hat z_1$;*
(2) *if $\phi \in C^2((0,\infty))$ and Assumption 3.4(2) holds, then, for every $A > 0$, $\phi(z) \le L_\phi z - \frac{\gamma_A}{2} z^2$ for $0 \le z \le A$.*

Proof. For $\varepsilon > 0$, apply the fundamental theorem of calculus on $[\varepsilon, z]$ and then let $\varepsilon \downarrow 0$. The definition of the finite right derivative and $\phi(0) = 0$ give

$$\phi(z) - L_\phi z = \int_0^z [\phi'(s) - L_\phi]\, ds.$$

The first conclusion follows from Assumption 3.4(1). For the second, Assumption 3.4(2) gives $\phi''(t) \le -\gamma_A$ on $(0, A]$. Integrating from ε to s and then letting $\varepsilon \downarrow 0$ gives $\phi'(s) - L_\phi \le -\gamma_A s$ for $0 < s \le A$. Integration proves the claim. ☐

For an open set $U \subset \Omega$, let $C_c(U)$ denote the continuous functions with compact support in U. Define

$$\Phi_\phi(\mu) := L_\phi \|\mu_{\mathrm{cont}}\|_{\mathcal{M}(\Omega)} + \sum_{\omega\in\operatorname{atom}\mu} \phi\big(|\mu(\{\omega\})|\big), \qquad \mu \in \mathcal{M}(\Omega). \tag{7}$$

Lemma 3.8. *Let $U \subset \Omega = \mathbb{R}^{d+1}$ be open, let $\mu \in \mathcal{M}(U)$, and let $\delta > 0$. Suppose that*

$$|\mu(\{\omega\})| \leq \frac{\delta}{4} \qquad \text{for every } \omega \in U.$$

Then, for every $\varepsilon > 0$, there are $m \geq 1$ and functions $h_1, \ldots, h_m \in C_c(U)$ with pairwise disjoint supports and $|h_i| \leq 1$ such that

$$\left| \int_U h_i \, d\mu \right| \leq \delta \quad (i = 1, \ldots, m), \qquad \sum_{i=1}^{m} \left| \int_U h_i \, d\mu \right| \geq \|\mu\|_{\mathcal{M}(U)} - \varepsilon.$$

Proof. If $\varepsilon \geq \|\mu\|_{\mathcal{M}(U)}$, take $m = 1$ and $h_1 = 0$. Henceforth assume $0 < \varepsilon < \|\mu\|_{\mathcal{M}(U)}$. Let $U = P \cup N$ be a Hahn decomposition of μ. By inner regularity, there are disjoint compact sets $K_+ \subset P$ and $K_- \subset N$ such that

$$|\mu|\big(U \setminus (K_+ \cup K_-)\big) < \frac{\varepsilon}{4}. \tag{8}$$

We now construct the required finite small-mass partition, including both the atomic and nonatomic parts. For every $\omega \in K_+ \cup K_-$, continuity from above gives

$$|\mu|\big(B_r(\omega)\big) \longrightarrow |\mu|(\{\omega\}) \leq \frac{\delta}{4} \qquad (r \downarrow 0).$$

Choose $r_\omega > 0$ such that $|\mu|(B_{r_\omega}(\omega)) < \delta/2$. Compactness gives a finite subcover $B_1, \ldots, B_\ell$ of $K_+ \cup K_-$. Intersect these balls separately with K_+ and K_- and disjointize the resulting finite covers by replacing the i-th set with the part not covered by the preceding sets. This gives, for some $m_0 \geq 1$, a finite Borel partition $E_1, \ldots, E_{m_0}$ of $K_+ \cup K_-$ such that each E_i lies in either P or N and $|\mu|(E_i) < \delta/2$.

By inner regularity, choose compact sets $C_i \subset E_i$ whose total omitted mass is less than $\varepsilon/4$, and discard the empty sets. At least one set remains because $\varepsilon < \|\mu\|_{\mathcal{M}(U)}$ and (8). Relabel the remaining sets $C_1, \ldots, C_m$. They are pairwise disjoint and satisfy

$$\sum_{i=1}^{m} |\mu|(C_i) \geq |\mu|(K_+ \cup K_-) - \frac{\varepsilon}{4}, \qquad |\mu|(C_i) \leq \frac{\delta}{2}. \tag{9}$$

Each C_i lies entirely in either P or N.

Because the compact sets C_i are finitely many, pairwise disjoint, and contained in the open set U, they have pairwise disjoint relatively compact open neighborhoods. Outer regularity lets us shrink these neighborhoods, while keeping each C_i inside, to obtain open sets U_i satisfying

$$\sum_{i=1}^{m} |\mu|(U_i \setminus C_i) < \min\left\{\frac{\varepsilon}{4}, \frac{\delta}{2}\right\}. \tag{10}$$

For each i, the continuous function

$$\chi_i(\omega) := \frac{\operatorname{dist}(\omega, U_i^c)}{\operatorname{dist}(\omega, C_i) + \operatorname{dist}(\omega, U_i^c)}$$

equals one on C_i, vanishes outside U_i, and has compact support in U. Set $h_i = \chi_i$ when $C_i \subset P$ and $h_i = -\chi_i$ when $C_i \subset N$. Their supports are pairwise disjoint and $|h_i| \leq 1$. Moreover,

$$\left| \int_U h_i \, d\mu \right| \leq |\mu|(U_i) \leq |\mu|(C_i) + |\mu|(U_i \setminus C_i) \leq \delta, \qquad \text{by (9)–(10)},$$

and

$$\begin{aligned} \sum_{i=1}^{m} \left| \int_U h_i \, d\mu \right| &\geq \sum_{i=1}^{m} \big(|\mu|(C_i) - |\mu|(U_i \setminus C_i)\big) \\ &\geq \|\mu\|_{\mathcal{M}(U)} - \frac{3\varepsilon}{4} \qquad \text{by (8)–(10)}. \end{aligned}$$

This is stronger than the required lower bound. □

Theorem 3.9 (Representation formula for the nonconvex functional)**.** *Suppose that $\phi : [0,\infty) \to [0,\infty)$ is nondecreasing and concave, $\phi(0) = 0$, and*

$$L_\phi := \phi'(0+) \in (0,\infty).$$

Then, for every $\mu \in \mathcal{M}(\Omega)$,

$$\Phi_\phi(\mu) = \sup\left\{ \sum_{i=1}^{m} \phi\left(\left| \int_\Omega f_i \, d\mu \right| \right) : \begin{array}{l} m \geq 1, \quad f_i \in C_c(\Omega), \quad |f_i| \leq 1, \\ \mathrm{supp}(f_i) \text{ are pairwise disjoint} \end{array} \right\}. \tag{11}$$

Proof. We prove the representation by first bounding every admissible finite sum from above and then constructing sums that approach $\Phi_\phi(\mu)$ from below. The hypotheses are exactly the standing hypotheses of Lemma 3.6 together with $L_\phi \in (0,\infty)$, so parts (1)–(3) of that lemma give

$$\phi(r+s) \leq \phi(r) + \phi(s), \qquad \phi(r) \leq L_\phi r, \qquad \frac{\phi(r)}{r} \longrightarrow L_\phi \quad (r \downarrow 0), \tag{12}$$

and part (4) gives the countable form $\phi\big(\sum_{j\geq 1} t_j\big) \leq \sum_{j\geq 1} \phi(t_j)$.

Write the atomic part of μ as

$$\mu_{\text{atom}} = \sum_{j\geq 1} c_j \delta_{\omega_j},$$

where the ω_j are distinct and $c_j \neq 0$. For an admissible family $(f_i)_{i=1}^m$, set

$$A_i := \sum_{j\geq 1} |c_j| \, |f_i(\omega_j)|, \qquad B_i := \int_\Omega |f_i| \, d|\mu_{\text{cont}}|.$$

$$\begin{aligned} \phi\left(\left| \int_\Omega f_i \, d\mu \right| \right) &\leq \phi(A_i + B_i) && \text{by the triangle inequality} \\ &\leq \sum_{\omega_j \in \mathrm{supp}(f_i)} \phi(|c_j|) + L_\phi B_i && \text{by Lemma 3.6(2), (4) and } |f_i| \leq 1. \end{aligned}$$

The disjoint supports count each atom at most once and satisfy $\sum_i |f_i| \leq 1$. Therefore

$$\begin{aligned} \sum_{i=1}^{m} \phi\left(\left| \int_\Omega f_i \, d\mu \right| \right) &\leq \sum_{j\geq 1} \phi(|c_j|) + L_\phi \int_\Omega \sum_{i=1}^{m} |f_i| \, d|\mu_{\text{cont}}| \\ &\leq \sum_{j\geq 1} \phi(|c_j|) + L_\phi \|\mu_{\text{cont}}\|_{\mathcal{M}(\Omega)} = \Phi_\phi(\mu). \end{aligned}$$

This proves the upper bound in (11).

For the reverse inequality, the claim is immediate when $\mu = 0$. Otherwise, fix $\varepsilon > 0$. Choose $\theta \in (0, L_\phi)$ such that $\theta \|\mu\|_{\mathcal{M}(\Omega)} < \varepsilon/4$. By the last limit in (12), choose $\delta > 0$ such that

$$\phi(r) \geq (L_\phi - \theta) r \qquad (0 \leq r \leq \delta). \tag{13}$$

Let

$$F := \{ j \geq 1 : |c_j| > \delta/8 \}, \qquad \mu_{\text{rem}} := \mu - \sum_{j \in F} c_j \delta_{\omega_j}.$$

The set F is finite and every point mass of μ_{rem} has magnitude at most $\delta/8$.

Because F is finite, its points have pairwise disjoint relatively compact neighborhoods. Moreover, $\mu_{\text{rem}}(\{\omega_j\}) = 0$ for $j \in F$, so continuity of the measure from above allows these neighborhoods to be chosen with arbitrarily small total $|\mu_{\text{rem}}|$-mass. Using the distance-function construction from Lemma 3.8, choose $g_j \in C_c(\Omega)$, $j \in F$, with pairwise disjoint supports, $|g_j| \leq 1$, and

$$g_j(\omega_j) = \mathrm{sign}(c_j), \qquad \sum_{j \in F} |\mu_{\text{rem}}|(\mathrm{supp}(g_j)) < \tau,$$

where $\tau > 0$ can be chosen arbitrarily small. For every $j \in F$,

$$\left| \left| \int_\Omega g_j \, d\mu \right| - |c_j| \right| \leq |\mu_{\mathrm{rem}}|(\mathrm{supp}(g_j)).$$

By continuity of ϕ and finiteness of F, first choose τ sufficiently small and then choose the supports so that

$$\sum_{j\in F} \phi\left(\left| \int_\Omega g_j \, d\mu \right| \right) \geq \sum_{j \in F} \phi(|c_j|) - \frac{\varepsilon}{4}, \qquad 2L_\phi \tau < \frac{\varepsilon}{4}. \tag{14}$$

Set

$$U := \Omega \setminus \bigcup_{j\in F} \mathrm{supp}(g_j) \qquad \text{and} \qquad \lambda := \mu_{\mathrm{rem}}|_U.$$

Lemma 3.8 applied on the open set U gives $h_1, \ldots, h_m \in C_c(U)$ whose zero extensions belong to $C_c(\Omega)$, have pairwise disjoint supports, and satisfy

$$\left| \int_U h_i \, d\lambda \right| \leq \delta, \qquad \sum_{i=1}^m \left| \int_U h_i \, d\lambda \right| \geq \|\lambda\|_{\mathcal{M}(U)} - \tau \geq \|\mu_{\mathrm{rem}}\|_{\mathcal{M}(\Omega)} - 2\tau.$$

Together with the g_j, the zero extensions form an admissible family in (11). Since every h_i is supported in U and $\mu|_U = \lambda$, its integrals against μ and λ agree. Therefore (13) and (14) yield

$$\begin{aligned}
&\sum_{j\in F} \phi\left(\left| \int_\Omega g_j \, d\mu \right| \right) + \sum_{i=1}^m \phi\left(\left| \int_\Omega h_i \, d\mu \right| \right) \\
&\quad \geq \sum_{j\in F} \phi(|c_j|) - \frac{\varepsilon}{4} + (L_\phi - \theta)\big(\|\mu_{\mathrm{rem}}\|_{\mathcal{M}(\Omega)} - 2\tau\big) \\
&\quad \geq \Phi_\phi(\mu) - \frac{\varepsilon}{4} - \theta\|\mu_{\mathrm{rem}}\|_{\mathcal{M}(\Omega)} - 2L_\phi\tau \qquad\qquad \text{by } \Phi_\phi(\mu_{\mathrm{rem}}) \leq L_\phi \|\mu_{\mathrm{rem}}\|_{\mathcal{M}(\Omega)} \\
&\quad > \Phi_\phi(\mu) - \varepsilon.
\end{aligned}$$

Taking the supremum and then letting $\varepsilon \downarrow 0$ proves the reverse inequality. $\square$

Corollary 3.10. *Under the assumptions of Theorem 3.9, the functional Φ_ϕ is weak-* lower semicontinuous on $\mathcal{M}(\Omega) = C_0(\Omega)^*$. If ϕ is also coercive, then every sublevel of Φ_ϕ is bounded in total variation.*

Proof. We use the representation in Theorem 3.9 for lower semicontinuity and the atomic–continuous decomposition for the sublevel bound. For each fixed admissible family in (11), weak-* convergence gives convergence of all its integrals, and continuity of ϕ gives convergence of the corresponding finite sum. The representation is a supremum of weak-* continuous functions and is therefore weak-* lower semicontinuous.

For the sublevel bound, apply repeated subadditivity to the first N atoms and let $N \to \infty$. The atom magnitudes are summable, so continuity of ϕ gives

$$\phi\big(\|\mu_{\mathrm{atom}}\|_{\mathcal{M}(\Omega)}\big) \leq \sum_{\omega \in \mathrm{atom}\,\mu} \phi\big(|\mu(\{\omega\})|\big) \leq \Phi_\phi(\mu), \qquad L_\phi \|\mu_{\mathrm{cont}}\|_{\mathcal{M}(\Omega)} \leq \Phi_\phi(\mu).$$

If $\Phi_\phi(\mu) \leq C$, coercivity bounds $\|\mu_{\mathrm{atom}}\|_{\mathcal{M}(\Omega)}$ in terms of C, while $\|\mu_{\mathrm{cont}}\|_{\mathcal{M}(\Omega)} \leq C/L_\phi$. The atomic part is supported on the countable atom set, while the continuous part assigns that set zero variation. The two parts are therefore mutually singular, so their total-variation norms add to $\|\mu\|_{\mathcal{M}(\Omega)}$. Hence the sublevel is bounded in total variation. $\square$

For $\alpha > 0$, define

$$J_0(\mu) := L(\mu) + \alpha \Phi_\phi(\mu)$$

for the unnormalized problem on the parameter domain.

3.3. **The normalized problem.**

Escape on the unbounded parameter domain. For positively homogeneous profiles the radial scale of ω is absorbed by the outer coefficient, and the parameter can be restricted to the unit sphere $\mathbb{S}^d$ (Section 4). But in general the domain is $\Omega = \mathbb{R}^{d+1}$, which is not compact. The functional J_0 need not be well posed. The regularizer Φ_ϕ depends on the sizes of the coefficients but not on the locations of the atoms. Along a sequence with $|\omega| \to \infty$, the fidelity need not be lower semicontinuous.

Example 3.11 (Failure of weak-* lower semicontinuity for tanh). Let $\rho = \tanh$, $V \equiv 1$, fix $a \in \mathbb{R}^d$, and set $\mu_n = \delta_{(a,b_n)}$ with $b_n \to \infty$. Then $\mu_n \overset{*}{\rightharpoonup} 0$, yet

$$\mathcal{N}\mu_n = \tanh(a \cdot x + b_n) \to 1$$

strongly in $H^1(D)$. Hence $\liminf_{n\to\infty} L(\mu_n) = 0 < L(0)$, so the fidelity term is not weak-* lower semicontinuous. Mass escaping to infinity carries a non-vanishing value contribution unless $K(\omega)$ tends to zero as $|\omega| \to \infty$. Note that $\Phi_\phi(\mu_n) = \phi(1)$ is constant along this sequence, so the unnormalized coefficient penalty does not prevent this escape. The defect is that the unnormalized kernel does not vanish at parameter infinity. The normalized atom introduced in Theorem 3.13 does vanish under the corresponding polynomial growth gap.

Example 3.12 (Nonattainment for softplus). For the profile $\rho(z) = \log(1 + e^z)$ (softplus), let $b_n \to \infty$. The measures $\mu_n = \rho(b_n)^{-1}\delta_{(0,b_n)}$ fit $V \equiv 1$ exactly for every n at penalty $\alpha\,\phi(1/\rho(b_n)) \to 0$, so $\inf J_0 = 0$ is not attained. Indeed, if $J_0(\mu) = 0$, then $\Phi_\phi(\mu) = 0$. Strict positivity of ϕ on $(0,\infty)$ and $L_\phi > 0$ force both the atomic and continuous parts of μ to vanish. Thus $\mu = 0$, but $L(0) = \frac{1}{2}\|1\|^2_{H^1(D)} > 0$, a contradiction.

The boundary behavior can also differ from the interior behavior. Let $F = \partial D \cap \{x : n_F \cdot x = c_F\}$ be a flat face of positive surface measure, assume that D lies on one side of its supporting hyperplane, and let $\rho \in C_0(\mathbb{R})$. Choose z_* with $\rho(z_*) \neq 0$, let $t_j \to \infty$, and set

$$a_j = t_j n_F, \qquad b_j = -t_j c_F + z_*.$$

Then

$$[K(a_j, b_j)](x) = \rho\big(t_j(n_F \cdot x - c_F) + z_*\big).$$

Since D lies on one side of the supporting hyperplane, $n_F \cdot x - c_F$ has a constant sign on D and vanishes only on $D \cap \{x : n_F \cdot x = c_F\} \subset \partial D$, a Lebesgue-null set. Hence for almost every $x \in D$ the argument $t_j(n_F \cdot x - c_F) + z_*$ tends to a single infinity, with the sign fixed by the orientation of n_F. Because ρ is bounded and D is bounded, the constant $\|\rho\|_\infty$ is an integrable dominating function, and since ρ vanishes at infinity, dominated convergence gives $K(a_j, b_j) \to 0$ in $L^2(D)$. The trace on the face F, however, satisfies $[K(a_j, b_j)]|_F = \rho(z_*)$ for every j.

Therefore, for $p > 0$, let w_p and $\mathcal{M}_p(\Omega)$ be as in (2). For $\mu \in \mathcal{M}_p(\Omega)$, we define the normalized measure and normalized atom by

$$d\mu_p := w_p\, d\mu, \qquad K_p(\omega) := \frac{K(\omega)}{w_p(\omega)}.$$

Theorem 3.13. *Multiplication by w_p is a linear isometric bijection*

$$\mathcal{M}_p(\Omega) \longrightarrow \mathcal{M}(\Omega), \qquad \mu \longmapsto \mu_p,$$

with inverse $d\mu = w_p^{-1} d\mu_p$, and

$$\|\mu\|_{\mathcal{M}_p(\Omega)} = \|\mu_p\|_{\mathcal{M}(\Omega)}. \tag{15}$$

Moreover, μ_n converges weak- to μ in $C_{w_p^{-1}}(\Omega)^*$ if and only if $(\mu_n)_p \overset{*}{\rightharpoonup} \mu_p$ in $C_0(\Omega)^*$. The multiplication preserves the atom set and the polar sign, and*

$$(\mu_p)_{\text{atom}} = w_p \mu_{\text{atom}}, \qquad (\mu_p)_{\text{cont}} = w_p \mu_{\text{cont}}. \tag{16}$$

If part (1) of Assumption 3.1 holds and $p > s_0$, then $K_p \in C_0(\Omega; L^2(D))$ and

$$\mathcal{N}\mu = \int_\Omega K\, d\mu = \int_\Omega K_p\, d\mu_p. \tag{17}$$

Finally,

$$\Phi_\phi(\mu_p) = L_\phi \int_\Omega w_p\, d|\mu_{\mathrm{cont}}| + \sum_{\omega \in \mathrm{atom}\,\mu} \phi\big(w_p(\omega)|\mu(\{\omega\})|\big). \tag{18}$$

Proof. We use multiplication by a positive continuous weight to prove every claim. If $\mu = \mathrm{sign}(\mu)|\mu|$, then

$$\mu_p = \mathrm{sign}(\mu) w_p |\mu|, \qquad |\mu_p| = w_p|\mu|.$$

This proves (15), preserves the polar sign, and shows that the inverse is $w_p^{-1}\mu_p$. Conversely, if $\xi \in \mathcal{M}(\Omega)$ and $d\mu = w_p^{-1} d\xi$, then

$$\int_\Omega w_p\, d|\mu| = |\xi|(\Omega),$$

so $\mu \in \mathcal{M}_p(\Omega)$ and the two multiplication maps are inverse.

For $f \in C_{w_p^{-1}}(\Omega)$, set $h = f/w_p \in C_0(\Omega)$. Then

$$\int_\Omega f\, d\mu = \int_\Omega h\, d\mu_p.$$

The bijection $f = w_p h$ between the two test-function spaces proves the weak-* equivalence.

At each $\omega \in \Omega$,

$$\mu_p(\{\omega\}) = w_p(\omega)\mu(\{\omega\}).$$

Since w_p is strictly positive, the atom sets agree. Multiplication by w_p also preserves nonatomicity, which proves (16).

It remains to treat the network map. The continuity of K_p follows from that of K and w_p. For $r = |\omega| \geq 1$, the growth estimate (3) gives

$$\begin{aligned} \|K_p(\omega)\|_{L^2(D)} &\leq C_D \frac{(1+r)^{s_0}}{1+r^p} \\ &\leq 2^{s_0} C_D r^{s_0 - p} \longrightarrow 0, \qquad r \to \infty, \end{aligned}$$

because $p > s_0$. Thus $K_p \in C_0(\Omega; L^2(D))$. The change-of-measure identity for the Bochner integral now gives (17). Finally, (16) and $|w_p\mu_{\mathrm{cont}}| = w_p|\mu_{\mathrm{cont}}|$ give (18) directly from (7). ☐

For $\alpha > 0$, define the adopted objective

$$J(\mu) := L(\mu) + \alpha \Phi_\phi(\mu_p), \qquad \mu \in \mathcal{M}_p(\Omega).$$

For a finite atomic measure $\mu_{\boldsymbol{\omega},c} = \sum_{n=1}^N c_n \delta_{\omega_n}$ with distinct support points, this becomes

$$J(\mu_{\boldsymbol{\omega},c}) = L(\mu_{\boldsymbol{\omega},c}) + \alpha \sum_{n=1}^N \phi\big(w_p(\omega_n)|c_n|\big).$$

Throughout, finite-network representations are understood after repeated locations are merged and zero coefficients are discarded; their width is therefore $\#\,\mathrm{atom}\,\mu$.

Definition 3.14 (Local minimizer)**.** A measure $\bar\mu \in \mathcal{M}_p(\Omega)$ is a local minimizer of J if $J(\bar\mu) < \infty$ and there is $\varepsilon > 0$ such that

$$J(\bar\mu) \leq J(\mu) \qquad \text{for all} \qquad \mu \in \mathcal{M}_p(\Omega), \quad \|\mu - \bar\mu\|_{\mathcal{M}_p(\Omega)} < \varepsilon.$$

By Theorem 3.13, this is exactly total-variation local minimality in the normalized coordinate μ_p.

3.4. **Existence of a global minimizer.** We apply the direct method in the normalized coordinate. Weak-* compactness controls μ_p, Corollary 3.10 supplies the penalty lower semicontinuity and sublevel bound, and Lemma 3.15 gives the strong convergence needed for the value part of the fidelity.

Lemma 3.15. *Assume (3), let $p > s_0$, and suppose that*

$$(\mu_n)_p \overset{*}{\rightharpoonup} \mu_p \quad \textit{in } \mathcal{M}(\Omega).$$

Then $\mathcal{N}\mu_n \to \mathcal{N}\mu$ strongly in $L^2(D)$.

Proof. We use a compactly supported cutoff to pass to the weak-* limit locally and then control the tails uniformly. By Theorem 3.13,

$$\mathcal{N}\mu_n = \int_\Omega K_p \, d(\mu_n)_p, \qquad \mathcal{N}\mu = \int_\Omega K_p \, d\mu_p.$$

Set

$$C := \sup_n \|(\mu_n)_p\|_{\mathcal{M}(\Omega)} < \infty,$$

where finiteness follows from weak-* convergence and the uniform boundedness principle. Weak-* lower semicontinuity of total variation gives $\|\mu_p\|_{\mathcal{M}(\Omega)} \le C$.

For $R \ge 1$, choose $\chi_R \in C_c(\Omega)$ with $0 \le \chi_R \le 1$, $\chi_R = 1$ on B_R, and $\operatorname{supp} \chi_R \subset B_{R+1}$. For fixed R and $x \in D$, the function

$$\omega \longmapsto \chi_R(\omega)[K_p(\omega)](x)$$

belongs to $C_c(\Omega)$. Hence weak-* convergence gives

$$\int_\Omega \chi_R(\omega)[K_p(\omega)](x) \, d(\mu_n)_p(\omega) \longrightarrow \int_\Omega \chi_R(\omega)[K_p(\omega)](x) \, d\mu_p(\omega)$$

for every $x \in D$.

Since D is bounded and $\operatorname{supp} \chi_R$ is compact,

$$M_R := \sup_{\omega \in \operatorname{supp} \chi_R, \ x \in D} |[K_p(\omega)](x)| < \infty.$$

The absolute values of both truncated integrals are bounded by CM_R. The integrand $(\omega, x) \mapsto \chi_R(\omega)[K_p(\omega)](x)$ is jointly Borel measurable and bounded by M_R; Fubini's theorem therefore identifies the pointwise integrals above, almost everywhere in x, with representatives of the $L^2(D)$-valued Bochner integrals below. The constant function $2CM_R$ belongs to $L^2(D)$ because D is bounded. Dominated convergence therefore yields

$$\int_\Omega \chi_R K_p \, d(\mu_n)_p \longrightarrow \int_\Omega \chi_R K_p \, d\mu_p \qquad \text{strongly in } L^2(D).$$

It remains to remove the cutoff. If $r := |\omega| > R \ge 1$, then

$$\begin{aligned}
\|K_p(\omega)\|_{L^2(D)} &\le C_D \frac{(1+r)^{s_0}}{1+r^p} && \text{by (3)} \\
&\le 2^{s_0} C_D r^{s_0 - p} \le 2^{s_0} C_D R^{s_0 - p}, && \text{since } p > s_0 \text{ and } r > R.
\end{aligned}$$

Consequently,

$$\begin{aligned}
\left\| \int_\Omega (1 - \chi_R) K_p \, d(\mu_n)_p \right\|_{L^2(D)} &\le 2^{s_0} C_D C R^{s_0 - p}, \\
\left\| \int_\Omega (1 - \chi_R) K_p \, d\mu_p \right\|_{L^2(D)} &\le 2^{s_0} C_D C R^{s_0 - p}.
\end{aligned}$$

For fixed R, let $n \to \infty$ in the truncated integrals. Then let $R \to \infty$; the two tail bounds vanish because $s_0 - p < 0$. This proves the claim. $\square$

Theorem 3.16 (Existence on the unbounded parameter domain)**.** *Let $V \in H^1(D)$, let ϕ satisfy Assumptions 3.2 and 3.3, and let $\rho \in C(\mathbb{R})$ satisfy Assumption 3.1(1). If $p > s_0$ and $\alpha > 0$, then*

$$\inf_{\mu \in \mathcal{M}_p(\Omega)} J(\mu)$$

admits at least one global minimizer.

Proof. We use the direct method in the normalized measure coordinate. Only the stated continuity and value-growth bound for ρ are needed. Since $V \in H^1(D)$,

$$J(0) = \frac{1}{2}\|V\|^2_{H^1(D)} =: C < \infty.$$

Choose a minimizing sequence $(\mu_n) \subset \mathcal{M}_p(\Omega)$ with $J(\mu_n) \le C + 1$. The fidelity and penalty are nonnegative, so

$$\Phi_\phi((\mu_n)_p) \le \frac{C+1}{\alpha}.$$

The total-variation sublevel bound in Corollary 3.10 shows that $((\mu_n)_p)$ is bounded in $\mathcal{M}(\Omega)$. Weak-* compactness and separability of $C_0(\Omega)$ give, after passing to a subsequence,

$$(\mu_n)_p \xrightharpoonup{*} \bar{\mu}_p.$$

Theorem 3.13 defines a unique $\bar{\mu} \in \mathcal{M}_p(\Omega)$ with normalized coordinate $\bar{\mu}_p$. Lemma 3.15 gives

$$\mathcal{N}\mu_n \longrightarrow \mathcal{N}\bar{\mu} \qquad \text{strongly in } L^2(D). \tag{19}$$

The fidelity bound gives

$$\sup_n \|\nabla(\mathcal{N}\mu_n)\|_{L^2(D)} \le \|\nabla V\|_{L^2(D)} + \sqrt{2(C+1)}.$$

After another subsequence, $\nabla(\mathcal{N}\mu_n) \rightharpoonup g$ weakly in $L^2(D)^d$. For every $\psi \in C_c^\infty(D)^d$,

$$\int_D g \cdot \psi \, dx = \lim_{n\to\infty} \int_D \nabla(\mathcal{N}\mu_n) \cdot \psi \, dx = -\lim_{n\to\infty} \int_D \mathcal{N}\mu_n \operatorname{div} \psi \, dx = -\int_D \mathcal{N}\bar{\mu} \operatorname{div} \psi \, dx,$$

the last step by (19). Hence $g = \nabla(\mathcal{N}\bar{\mu})$ distributionally and $\mathcal{N}\bar{\mu} \in H^1(D)$.

The value part of the fidelity is continuous by (19), and the gradient part is weakly lower semicontinuous. Hence

$$L(\bar{\mu}) \le \liminf_{n\to\infty} L(\mu_n).$$

Corollary 3.10 gives

$$\Phi_\phi(\bar{\mu}_p) \le \liminf_{n\to\infty} \Phi_\phi((\mu_n)_p).$$

Combining the two inequalities proves $J(\bar{\mu}) \le \liminf_n J(\mu_n) = \inf J$. □

3.5. **Optimality conditions and finite support.** Let $\bar{\mu}$ be a finite-objective local minimizer. For a measure μ and an admissible direction μ', the Gâteaux derivative is

$$DL(\mu)[\mu'] := \lim_{t\to 0} \frac{L(\mu + t\mu') - L(\mu)}{t} = \langle r_\mu, \mathcal{N}\mu' \rangle_{H^1(D)}.$$

The corresponding derivative profile is

$$P_\mu(\omega) := DL(\mu)[\delta_\omega] = \langle r_\mu, K(\omega) \rangle_{H^1(D)}.$$

Define its normalized form and the uniform normalized feature bound by

$$P_{p,\mu}(\omega) := \frac{P_\mu(\omega)}{w_p(\omega)} = \langle r_\mu, K_p(\omega) \rangle_{H^1(D)}, \qquad B_p := \sup_{\omega\in\Omega} \|K_p(\omega)\|_{H^1(D)}.$$

At the local minimizer write $\bar{P} := P_{\bar{\mu}}$ and $\bar{P}_p := P_{p,\bar{\mu}}$. Thus $\bar{P}$ is the function representing the Gâteaux derivative. For differentiable ρ,

$$\bar{P}(a, b) = \int_D r_{\bar{\mu}}(x) \rho(a \cdot x + b) \, dx + \int_D \nabla r_{\bar{\mu}}(x) \cdot a\rho'(a \cdot x + b) \, dx.$$

The second term carries the factor a, so on the unbounded domain $\bar{P}$ is defined pointwise but need not be bounded.

By Cauchy–Schwarz, (4) gives

$$|\bar{P}(\omega)| \le \|r_{\bar{\mu}}\|_{H^1(D)} \|K(\omega)\|_{H^1(D)} \le C_P(1 + |\omega|)^{s_1}, \qquad C_P := C_K \|r_{\bar{\mu}}\|_{H^1(D)}.$$

The constant C_P depends on the fixed minimizer through its residual, but not on ω. If $p > s_1$, then

$$K_p \in C_0(\Omega; H^1(D)), \qquad \bar{P}_p \in C_0(\Omega), \qquad \bar{P} \in C_{w_p^{-1}}(\Omega).$$

Consequently, for every $\mu' \in \mathcal{M}_p(\Omega)$,

$$DL(\bar{\mu})[\mu'] = \int_\Omega \bar{P}\, d\mu' = \int_\Omega \bar{P}_p\, d\mu'_p.$$

The pointwise definition of $\bar{P}$ as a Gâteaux derivative does not require $\bar{P} \in C_0(\Omega)$.

Theorem 3.17 (The optimality condition)**.** *Let $\phi : [0,\infty) \to [0,\infty)$ be concave and nondecreasing, with $\phi(0) = 0$, $\phi \in C^1((0,\infty))$, and $0 < L_\phi := \phi'(0+) < \infty$. Let $\bar{\mu} \in \mathcal{M}_p(\Omega)$ be a local minimizer of J with $J(\bar{\mu}) < \infty$, and suppose that Assumption 3.1 holds and $p > s_1$. Then*

$$|\bar{P}_p(\omega)| \le \alpha L_\phi \qquad \textit{for every } \omega \in \Omega, \tag{20}$$

$$\bar{P}_p(\omega) = -\alpha\phi'\big(w_p(\omega)|\bar{\mu}(\{\omega\})|\big)\operatorname{sign}\big(\bar{\mu}(\{\omega\})\big) \qquad \textit{for } \omega \in \operatorname{atom}\bar{\mu}. \tag{21}$$

Moreover,

$$\bar{P}_p = -\alpha L_\phi \operatorname{sign}(\bar{\mu}_{\mathrm{cont}}) \qquad |(\bar{\mu}_p)_{\mathrm{cont}}|\textit{-almost everywhere.} \tag{22}$$

Proof. We use Theorem 3.13 to work with $\bar{\mu}_p$ in the total-variation norm. The normalized fidelity derivative in the direction $\xi \in \mathcal{M}(\Omega)$ is

$$\left\langle r_{\bar{\mu}}, \int_\Omega K_p\, d\xi \right\rangle_{H^1(D)} = \int_\Omega \bar{P}_p\, d\xi.$$

Fix $\omega \in \operatorname{atom}\bar{\mu}$ and put $z := \bar{\mu}_p(\{\omega\}) = w_p(\omega)\bar{\mu}(\{\omega\}) \neq 0$. For $s \in \{-1, 1\}$ and sufficiently small $t > 0$, the sign of $z + ts$ equals the sign of z. Dividing the local-minimality inequality by t and letting $t \downarrow 0$ gives

$$0 \le s\left[\bar{P}_p(\omega) + \alpha\phi'(|z|)\operatorname{sign}(z)\right].$$

Using both choices of s makes the bracket vanish, and hence

$$\bar{P}_p(\omega) = -\alpha\phi'(|z|)\operatorname{sign}(z).$$

Because $w_p > 0$, $\operatorname{sign}(z) = \operatorname{sign}(\bar{\mu}(\{\omega\}))$, so this is (21). Concavity and monotonicity give $0 \le \phi'(|z|) \le L_\phi$, and hence $|\bar{P}_p(\omega)| \le \alpha L_\phi$ at every atom.

If $\omega \notin \operatorname{atom}\bar{\mu}$, insertion of $ts\delta_\omega$ into the normalized measure changes the penalty by $\phi(t)$. Dividing the local minimality inequality by $t > 0$ and letting $t \downarrow 0$ yields

$$0 \le s\bar{P}_p(\omega) + \alpha L_\phi.$$

The two choices of s prove (20) away from the atoms, and the preceding atom estimate completes its proof.

It remains to identify the continuous part. Set $\xi := (\bar{\mu}_p)_{\mathrm{cont}}$. Since $K_p \in C_0(\Omega; H^1(D))$, the directions $\pm\xi$ are admissible. For $s \in \{-1, 1\}$ and $0 < t < 1$, the perturbation $\bar{\mu}_p + ts\xi$ multiplies the continuous part by $1 + ts > 0$. Its one-sided directional derivative therefore gives

$$0 \le s\left[\int_\Omega \bar{P}_p\, d\xi + \alpha L_\phi \|\xi\|_{\mathcal{M}(\Omega)}\right].$$

Using both signs shows that the bracket vanishes. On the other hand, (20) makes

$$\bar{P}_p \operatorname{sign}(\xi) + \alpha L_\phi \ge 0$$

$|\xi|$-almost everywhere. Its integral is zero, so the integrand vanishes $|\xi|$-almost everywhere. The sign preservation in Theorem 3.13 gives (22). □

Remark 3.18. Because $\bar{P}_p \in C_0(\Omega)$ and $\alpha L_\phi > 0$, every closed positive superlevel set of $|\bar{P}_p|$ is compact. Thus any violation of (20) is confined to a compact set, even when the unnormalized function $\bar{P}$ does not decay at parameter infinity. Lemma 3.19 uses a lower superlevel to localize the entire support.

Lemma 3.19. *Let ϕ satisfy Assumptions 3.2 and 3.3, let $p > s_1$, and suppose that Assumption 3.1 holds. Suppose that $\bar{\mu} \in \mathcal{M}_p(\Omega)$ is an $\mathcal{M}_p(\Omega)$-local minimizer of J such that $0 < J(\bar{\mu}) < \infty$. Define*

$$T(\bar{\mu}) := \phi^{-1}\left(\frac{J(\bar{\mu})}{\alpha}\right)$$

and

$$\mathcal{K}(\bar{\mu}) := \left\{\omega \in \Omega : |\bar{P}_p(\omega)| \geq \alpha\phi'\big(T(\bar{\mu})\big)\right\}.$$

Then $\mathcal{K}(\bar{\mu})$ is compact and

$$\operatorname{supp}\bar{\mu}_{\mathrm{atom}} \cup \operatorname{supp}\bar{\mu}_{\mathrm{cont}} \subset \mathcal{K}(\bar{\mu}). \tag{23}$$

Moreover, every atom $\omega \in \operatorname{atom}\bar{\mu}$ satisfies

$$w_p(\omega)|\bar{\mu}(\{\omega\})| \leq T(\bar{\mu}).$$

Proof. We use Theorem 3.17 and the fact that $\bar{P}_p$ vanishes at infinity to prove the localization. Concavity and monotonicity make ϕ' positive and nonincreasing, as observed after Assumption 3.3. Since $\bar{P}_p \in C_0(\Omega)$, its closed superlevel set at the strictly positive level $\alpha\phi'(T(\bar{\mu}))$ is compact. This proves the first assertion.

Let $\omega \in \operatorname{atom}\bar{\mu}$ and put

$$t_\omega := |\bar{\mu}_p(\{\omega\})| = w_p(\omega)|\bar{\mu}(\{\omega\})|.$$

The contribution of this atom to the objective gives

$$\begin{aligned}\alpha\phi(t_\omega) &\leq \alpha\Phi_\phi(\bar{\mu}_p) && \text{by nonnegativity of the other terms}\\ &\leq J(\bar{\mu}) = \alpha\phi\big(T(\bar{\mu})\big).\end{aligned}$$

Strict monotonicity of ϕ gives $t_\omega \leq T(\bar{\mu})$. The atom condition in Theorem 3.17 and monotonicity of ϕ' now give

$$|\bar{P}_p(\omega)| = \alpha\phi'(t_\omega) \geq \alpha\phi'\big(T(\bar{\mu})\big).$$

Thus every atom lies in $\mathcal{K}(\bar{\mu})$.

On the continuous part, Theorem 3.17 gives

$$|\bar{P}_p(\omega)| = \alpha L_\phi \geq \alpha\phi'\big(T(\bar{\mu})\big) \qquad |(\bar{\mu}_p)_{\mathrm{cont}}|\text{-almost everywhere.}$$

Hence $(\bar{\mu}_p)_{\mathrm{cont}}$ is concentrated on the closed set $\mathcal{K}(\bar{\mu})$. Theorem 3.13 preserves the atomic and continuous decompositions. Since w_p is strictly positive, each component of $\bar{\mu}_p = w_p\bar{\mu}$ has the same null sets, and hence the same support, as the corresponding component of $\bar{\mu}$. Taking closures for the atomic part and supports for the continuous part proves (23). □

Theorem 3.20 (Finite support and separation of local minimizers)**.** *Let ϕ satisfy Assumptions 3.2 and 3.3, suppose that $\phi \in C^2((0,\infty))$ satisfies Assumption 3.4(2), let $p > s_1$, and suppose that Assumption 3.1 holds. Then every $\mathcal{M}_p(\Omega)$-local minimizer $\bar{\mu} \in \mathcal{M}_p(\Omega)$ of J with $J(\bar{\mu}) < \infty$ is finite atomic. If $J(\bar{\mu}) > 0$, set*

$$T := T(\bar{\mu}), \qquad \mathcal{K} := \mathcal{K}(\bar{\mu}).$$

Then $\bar{\mu}_{\mathrm{cont}} = 0$, every atom belongs to the compact set $\mathcal{K}$, and any two distinct atoms ω_i and ω_j satisfy

$$\|K_p(\omega_i) - K_p(\omega_j)\|^2_{H^1(D)} \geq 2\alpha\gamma_T. \tag{24}$$

Consequently,

$$\#\,\mathrm{atom}(\bar{\mu}) \le N_{\mathrm{cov}}\Big(K_p(\mathcal{K}), \frac{1}{2}\sqrt{\alpha\gamma_T}\Big) < \infty, \tag{25}$$

where the covering number $N_{\mathrm{cov}}(A, r)$ is the least number of balls of radius r needed to cover A.

Proof. We use the isometry of Theorem 3.13, the first-order identities of Theorem 3.17, and local coefficient variations to prove the assertions. Concavity, coercivity, and $L_\phi > 0$ imply that $\phi(t) > 0$ for $t > 0$.

The zero-objective branch is immediate:

$$J(\bar{\mu}) = 0 \quad\Longrightarrow\quad \Phi_\phi(\bar{\mu}_p) = 0 \quad\Longrightarrow\quad \bar{\mu}_p = 0 \quad\Longrightarrow\quad \bar{\mu} = 0,$$

where the middle implication follows from $L_\phi > 0$ and $\phi(t) > 0$ for $t > 0$. We may therefore suppose $J(\bar{\mu}) > 0$. Lemma 3.19 gives the compact set $\mathcal{K}$ and already localizes the atomic and continuous supports. Every normalized atom magnitude

$$t_i := |\bar{\mu}_p(\{\omega_i\})|$$

satisfies $t_i \le T$ by Lemma 3.19. The curvature assumption gives $\phi''(t) \le -\gamma_T$ for $0 < t \le T$. Integrating this inequality from the origin, using $\phi'(0+) = L_\phi$, and then integrating once more gives

$$L_\phi t - \phi(t) \ge \frac{\gamma_T}{2} t^2 \qquad (0 \le t \le T). \tag{26}$$

We first separate distinct atoms. For two such atoms, Theorem 3.13 defines a unique measure $\mu_s \in \mathcal{M}_p(\Omega)$ by

$$(\mu_s)_p := \bar{\mu}_p + s\delta_{\omega_i} - s\delta_{\omega_j}.$$

For sufficiently small $|s|$, the signs of both atom coefficients remain fixed, and

$$\begin{aligned} \|\mu_s - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} &= 2|s|, && \text{by Theorem 3.13,} \\ \mathcal{N}\mu_s &= \mathcal{N}\bar{\mu} + s\big[K_p(\omega_i) - K_p(\omega_j)\big], && \text{by the normalized representation of } \mathcal{N}. \end{aligned}$$

Thus $s \mapsto J(\mu_s)$ is twice differentiable near zero and has a local minimum there. Its second derivative satisfies

$$\begin{aligned} 0 &\le \frac{d^2}{ds^2} J(\mu_s)\Big|_{s=0} \\ &= \|K_p(\omega_i) - K_p(\omega_j)\|_{H^1(D)}^2 + \alpha\phi''(t_i) + \alpha\phi''(t_j) && \text{by the quadratic fidelity} \\ &\le \|K_p(\omega_i) - K_p(\omega_j)\|_{H^1(D)}^2 - 2\alpha\gamma_T, && \text{since } t_i, t_j \le T. \end{aligned}$$

This proves (24).

We next rule out the continuous part by concentrating a local piece of one sign. Suppose $(\bar{\mu}_p)_{\mathrm{cont}} \ne 0$ and write

$$d(\bar{\mu}_p)_{\mathrm{cont}} = \sigma\, d\lambda, \qquad \lambda := |(\bar{\mu}_p)_{\mathrm{cont}}|,$$

where σ takes the values -1 and 1 λ-almost everywhere. Choose $\sigma_0 \in \{-1, 1\}$ such that the restriction of λ to $\{\sigma = \sigma_0\}$ is nonzero, and denote that restriction again by λ. Theorem 3.17 gives $\bar{P}_p = -\alpha L_\phi \sigma_0$ on a set of full λ-measure. The support of a Radon measure has full measure, while the countable set $\mathrm{atom}\,\bar{\mu}$ has zero λ-measure because λ is nonatomic. The intersection of these three full-measure sets is nonempty, so choose

$$\bar{\omega} \in \operatorname{supp}\lambda \setminus \mathrm{atom}\,\bar{\mu} \qquad \text{such that} \qquad \bar{P}_p(\bar{\omega}) = -\alpha L_\phi \sigma_0.$$

For $\delta > 0$, put

$$r_\delta := \lambda(B_\delta(\bar{\omega})), \qquad \varepsilon_\delta := \sup_{\omega \in B_\delta(\bar{\omega})} \|K_p(\omega) - K_p(\bar{\omega})\|_{H^1(D)}.$$

Because $\bar\omega \in \operatorname{supp}\lambda$, one has $r_\delta > 0$. Moreover, $r_\delta \to \lambda(\{\bar\omega\}) = 0$ by continuity from above and the nonatomicity of λ, while $\varepsilon_\delta \to 0$ by continuity of K_p at $\bar\omega$. Define μ_δ through its normalized coordinate by

$$(\mu_\delta)_p := \bar\mu_p - \sigma_0 \lambda|_{B_\delta(\bar\omega)} + \sigma_0 r_\delta \delta_{\bar\omega}.$$

The target point is not an atom, and Theorem 3.13 gives

$$\|\mu_\delta - \bar\mu\|_{\mathcal{M}_p(\Omega)} = 2r_\delta \longrightarrow 0.$$

If $e_\delta := \mathcal{N}(\mu_\delta - \bar\mu)$, then

$$\begin{aligned} &\|e_\delta\|_{H^1(D)} \le \varepsilon_\delta r_\delta, && \text{by the Bochner-integral triangle inequality,}\\ &\langle r_{\bar\mu}, e_\delta\rangle_{H^1(D)} = \sigma_0 r_\delta \bar P_p(\bar\omega) - \sigma_0 \int_{B_\delta(\bar\omega)} \bar P_p \, d\lambda = 0, && \text{by Theorem 3.17.} \end{aligned}$$

The exact quadratic expansion of the fidelity and the definition of Φ_ϕ therefore give, for all sufficiently small δ such that $r_\delta \le T$,

$$\begin{aligned} J(\mu_\delta) - J(\bar\mu) &= \frac{1}{2}\|e_\delta\|^2_{H^1(D)} + \alpha\big[\phi(r_\delta) - L_\phi r_\delta\big] \\ &\le \frac{1}{2}\big(\varepsilon_\delta^2 - \alpha\gamma_T\big) r_\delta^2 && \text{by (26)}\\ &< 0, && \text{for sufficiently small } \delta. \end{aligned}$$

This contradicts local minimality. Hence $\bar\mu_{\mathrm{cont}} = 0$.

It remains to count the atoms. The set $K_p(\mathcal{K})$ is compact because $\mathcal{K}$ is compact and K_p is continuous. Cover it by balls of radius $\frac{1}{2}\sqrt{\alpha\gamma_T}$. Each such ball has diameter $\sqrt{\alpha\gamma_T}$, which is strictly smaller than the separation $\sqrt{2\alpha\gamma_T}$ in (24). Hence each ball contains the image of at most one atom. A compact set has a finite cover at every positive radius, and (25) follows. $\square$

Corollary 3.21. *Assume the hypotheses of Theorems 3.16 and 3.20. Then J admits a global minimizer, every global minimizer is finite atomic, and*

$$\min_{\mu\in\mathcal{M}_p(\Omega)} J(\mu) = \min_{\substack{N\ge 1,\ \omega_1,\dots,\omega_N\in\Omega\\ c_1,\dots,c_N\in\mathbb{R}}} \left\{ \frac{1}{2}\left\| \sum_{n=1}^N c_n K(\omega_n) - V\right\|^2_{H^1(D)} + \alpha\sum_{n=1}^N \phi\big(w_p(\omega_n)|c_n|\big)\right\}. \tag{27}$$

The minimum on the right is attained. If μ_* is any global minimizer with $J(\mu_*) > 0$, then it satisfies the data-dependent bound

$$\#\,\mathrm{atom}(\mu_*) \le N_{\mathrm{cov}}\left(K_p\big(\mathcal{K}(\mu_*)\big), \frac{1}{2}\sqrt{\alpha\gamma_{T(\mu_*)}}\right). \tag{28}$$

If $V \ne 0$, define

$$T_0 := \phi^{-1}\left(\frac{\|V\|^2_{H^1(D)}}{2\alpha}\right)$$

and

$$\mathcal{K}_0 := \left\{\omega\in\Omega : \|V\|_{H^1(D)}\|K_p(\omega)\|_{H^1(D)} \ge \alpha\phi'(T_0)\right\}. \tag{29}$$

Then $\mathcal{K}_0$ is compact, every atom of every global minimizer lies in $\mathcal{K}_0$, and

$$\#\,\mathrm{atom}(\mu_*) \le N_{\mathrm{cov}}\left(K_p(\mathcal{K}_0), \frac{1}{2}\sqrt{\alpha\gamma_{T_0}}\right) < \infty. \tag{30}$$

If $V = 0$, then the zero measure is the unique global minimizer.

Proof. We combine the direct method in Theorem 3.16 with the local result in Theorem 3.20. Theorem 3.16 gives a global minimizer. Every global minimizer is $\mathcal{M}_p(\Omega)$-local, so Theorem 3.20 makes it finite atomic; when its objective is positive, Theorem 3.20 also gives (28). Its distinct atom representation has objective equal to the expression on the right of (27), so the right-hand infimum is no larger than the measure minimum.

Conversely, consider an arbitrary finite network, possibly with repeated locations. For each distinct location η_j, merge its coefficients into $d_j := \sum_{n\colon \omega_n=\eta_j} c_n$. The represented function, and hence the fidelity, is unchanged. Monotonicity and repeated subadditivity of ϕ give

$$\phi\big(w_p(\eta_j)|d_j|\big) \le \phi\left(w_p(\eta_j)\sum_{n\colon \omega_n=\eta_j}|c_n|\right) \le \sum_{n\colon \omega_n=\eta_j}\phi\big(w_p(\eta_j)|c_n|\big).$$

After summing over the distinct locations, the measure objective is no larger than the displayed finite-network objective. Thus the measure minimum is no larger than the right-hand infimum in (27). The two infima are equal, and the finite atom representation of a global minimizer attains the right-hand side, so both infima are minima.

Suppose $V \neq 0$. Then a global minimizer μ_* has positive objective, because Theorem 3.20 would otherwise give $\mu_* = 0$, whose fidelity is positive. By comparison with the zero measure,

$$\begin{aligned} J(\mu_*) &\le J(0) = \frac{1}{2}\|V\|^2_{H^1(D)} = \alpha\phi(T_0), \\ \|r_{\mu_*}\|_{H^1(D)} &\le \|V\|_{H^1(D)}, && \text{since } L(\mu_*) \le J(\mu_*). \end{aligned}$$

Hence

$$T(\mu_*) \le T_0, \qquad \gamma_{T(\mu_*)} \ge \gamma_{T_0}. \tag{31}$$

For an atom ω_i of normalized magnitude $t_i = |\mu_{*,p}(\{\omega_i\})|$, Lemma 3.19 gives $t_i \le T(\mu_*)$. Theorem 3.17 then gives

$$\begin{aligned} \alpha\phi'(T_0) \le \alpha\phi'(t_i) &= |\langle r_{\mu_*}, K_p(\omega_i)\rangle_{H^1(D)}| && \text{since } t_i \le T(\mu_*) \le T_0 \\ &\le \|r_{\mu_*}\|_{H^1(D)}\|K_p(\omega_i)\|_{H^1(D)} && \text{by Cauchy–Schwarz} \\ &\le \|V\|_{H^1(D)}\|K_p(\omega_i)\|_{H^1(D)}. \end{aligned}$$

Thus every atom belongs to $\mathcal{K}_0$. The level in (29) is strictly positive, while $K_p \in C_0(\Omega; H^1(D))$, so $\mathcal{K}_0$ is compact. Combining (31) with (24) gives pairwise separation at least $\sqrt{2\alpha\gamma_{T_0}}$. The covering argument from Theorem 3.20 on $K_p(\mathcal{K}_0)$ proves (30).

If $V = 0$, then $J(0) = 0$. Nonnegativity makes zero the minimum, and the zero-objective conclusion of Theorem 3.20 shows that no other global minimizer exists. ☐

The hypotheses used so far are jointly satisfiable, and the following family makes the constants explicit.

Example 3.22 (Admissible penalty family). For $\gamma \ge 0$, define

$$\phi_\gamma(z) := \begin{cases} \dfrac{z}{2} + \dfrac{1}{4\gamma}\log(1+2\gamma z), & \gamma > 0, \\ z, & \gamma = 0. \end{cases} \tag{32}$$

Then $\phi_\gamma(0) = 0$ and, for $\gamma > 0$,

$$\phi_\gamma'(z) = \frac{1}{2} + \frac{1}{2(1+2\gamma z)} \in \left(\tfrac{1}{2}, 1\right], \qquad \phi_\gamma''(t) = -\frac{\gamma}{(1+2\gamma t)^2}.$$

Hence $L_{\phi_\gamma} = \phi_\gamma'(0) = 1 \in (0,\infty)$, which is Assumption 3.2; the derivative is positive and nonincreasing and $\phi_\gamma(z) \ge z/2 \to \infty$, which is Assumption 3.3. For Assumption 3.4, the second derivative is continuous and negative on $(0,\infty)$, part (1) holds with $\gamma_1 = \gamma$ and any $\hat{z}_1 > 0$, and part (2) holds with

$$\gamma_A = \frac{\gamma}{(1+2\gamma A)^2} > 0. \tag{33}$$

Thus ϕ_γ with $\gamma > 0$ satisfies Assumptions 3.2–3.4, and (33) turns the atom separation $\sqrt{2\alpha\gamma_{T_0}}$ and the neuron bound (30) into computable quantities. This is the penalty used by Algorithm 1.

The limit $\gamma \to 0$ is the convex endpoint. Assumption 3.4 fails at $\gamma = 0$, so Theorem 3.20 and Corollary 3.21 do not apply; the weaker conclusions of Theorems 3.16, 3.17, and 5.1 remain available. The degeneration is quantitative rather than abrupt: by (33), $0 < \gamma_{T_0} \le \gamma \to 0$ as $\gamma \to 0$. Thus the guaranteed separation scale tends to zero; the corresponding covering-number bound may deteriorate or become uninformative.

For fixed, distinct locations $\boldsymbol{\omega} = (\omega_n)_{n=1}^N \in \Omega^N$, define the reduced coefficient objective

$$J_{\boldsymbol{\omega}}(c) := L\left(\sum_{n=1}^N c_n \delta_{\omega_n}\right) + \alpha \sum_{n=1}^N \phi\big(w_p(\omega_n)|c_n|\big), \qquad c \in \mathbb{R}^N. \tag{34}$$

Theorem 3.23 (Sufficient condition for local optimality)**.** *Let $\phi : [0,\infty) \to [0,\infty)$ satisfy $\phi(0) = 0$ and $\phi \in C^1((0,\infty))$, with $0 < L_\phi := \phi'(0+) < \infty$. Suppose there are $\gamma_1 \ge 0$ and $\hat{z}_1 > 0$ such that*

$$\phi(z) \ge L_\phi z - \frac{\gamma_1}{2} z^2 \qquad (0 \le z \le \hat{z}_1). \tag{35}$$

Let $p > s_1$, let $V \in H^1(D)$, and suppose that Assumption 3.1 holds. Let

$$\bar{\boldsymbol{\omega}} := (\bar{\omega}_n)_{n=1}^N, \qquad \bar{c} := (\bar{c}_n)_{n=1}^N \in (\mathbb{R} \setminus \{0\})^N, \qquad \bar{\mu} := \sum_{n=1}^N \bar{c}_n \delta_{\bar{\omega}_n},$$

where the points $\bar{\omega}_n$ are distinct and every $\bar{c}_n$ is nonzero. Suppose also that

$$\big|\phi'\big(w_p(\bar{\omega}_n)|\bar{c}_n|\big)\big| < L_\phi \qquad (n = 1, \dots, N). \tag{36}$$

Define

$$\bar{P}_p(\omega) := \langle \mathcal{N}\bar{\mu} - V, K_p(\omega) \rangle_{H^1(D)}.$$

Assume that

1. *$\bar{c} \in (\mathbb{R} \setminus \{0\})^N$ is a local minimizer of $J_{\bar{\boldsymbol{\omega}}}$;*
2. *the strict inequality*
$$|\bar{P}_p(\omega)| < \alpha L_\phi$$
holds for every $\omega \in \Omega \setminus \{\bar{\omega}_1, \dots, \bar{\omega}_N\}$.

Then $\bar{\mu}$ is an $\mathcal{M}_p(\Omega)$-local minimizer of J.

Proof. We use the normalized coordinate to reduce the claim to total-variation estimates. Set

$$\bar{z} := (\bar{z}_n)_{n=1}^N, \qquad \bar{z}_n := w_p(\bar{\omega}_n)\bar{c}_n.$$

For $z = (z_n)_{n=1}^N \in \mathbb{R}^N$, define

$$\widehat{J}(z) := \frac{1}{2}\left\|\sum_{n=1}^N z_n K_p(\bar{\omega}_n) - V\right\|_{H^1(D)}^2 + \alpha \sum_{n=1}^N \phi(|z_n|).$$

The invertible diagonal change $z_n = w_p(\bar{\omega}_n)c_n$ shows that $\bar{z}$ is a local minimizer of $\widehat{J}$.

We first obtain a uniform strict gap. Stationarity of $J_{\bar{\boldsymbol{\omega}}}$ at $\bar{c}$ gives

$$\bar{P}_p(\bar{\omega}_n) = -\alpha\phi'(|\bar{z}_n|)\,\mathrm{sign}(\bar{z}_n).$$

The active-slope gap (36) and condition (2) give

$$|\bar{P}_p(\omega)| < \alpha L_\phi \qquad (\omega \in \Omega).$$

Since $\bar{P}_p \in C_0(\Omega)$, the positive continuous function $\alpha L_\phi - |\bar{P}_p|$ tends to αL_ϕ at infinity. Hence there is $m > 0$ such that

$$|\bar{P}_p(\omega)| \le \alpha L_\phi - m \qquad (\omega \in \Omega). \tag{37}$$

Choose $\varepsilon_1 > 0$ such that $\widehat{J}(z) \ge \widehat{J}(\bar{z})$ whenever $\|z - \bar{z}\|_{\ell^1} \le \varepsilon_1$. Put

$$B_{\bar{\omega}} := \max_{1 \le n \le N} \|K_p(\bar{\omega}_n)\|_{H^1(D)}.$$

Choose $\varepsilon > 0$ so that

$$\varepsilon \le \varepsilon_1, \qquad \varepsilon \le \hat{z}_1, \qquad B_p B_{\bar{\omega}} \varepsilon \le \frac{m}{2}, \qquad \frac{\alpha\gamma_1\varepsilon}{2} \le \frac{m}{4}.$$

A condition with zero left-hand side is void.

Let $\mu \in \mathcal{M}_p(\Omega)$ satisfy $\|\mu - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} \le \varepsilon$. There is nothing to prove if $J(\mu) = +\infty$. Otherwise, write $\xi := \mu_p$ and, for $1 \le n \le N$, set

$$z_n := \xi(\{\bar{\omega}_n\}), \qquad z := (z_n)_{n=1}^N, \qquad \xi_0 := \sum_{n=1}^N z_n \delta_{\bar{\omega}_n}, \qquad \xi = \xi_0 + \widetilde{\xi}.$$

Then $\widetilde{\xi}$ has no mass at the support points, and the normalization isometry gives

$$\|z - \bar{z}\|_{\ell^1} + \|\widetilde{\xi}\|_{\mathcal{M}(\Omega)} = \|\mu - \bar{\mu}\|_{\mathcal{M}_p(\Omega)} \le \varepsilon.$$

Define

$$r' := \sum_{n=1}^N (z_n - \bar{z}_n) K_p(\bar{\omega}_n), \qquad Q_p(\omega) := \langle r', K_p(\omega)\rangle_{H^1(D)}.$$

Then

$$\|r'\|_{H^1(D)} \le B_{\bar{\omega}} \|z - \bar{z}\|_{\ell^1} \le B_{\bar{\omega}}\varepsilon,$$
$$|Q_p(\omega)| \le B_p B_{\bar{\omega}} \varepsilon \le \frac{m}{2}.$$

Combining this with (37) gives

$$|\bar{P}_p(\omega) + Q_p(\omega)| \le \alpha L_\phi - \frac{m}{2} \qquad (\omega \in \Omega). \tag{38}$$

The exact quadratic expansion of the fidelity in the direction $\widetilde{\xi}$ and the nonnegativity of its quadratic remainder give

$$L(\mu) \ge \frac{1}{2}\left\|\sum_{n=1}^N z_n K_p(\bar{\omega}_n) - V\right\|_{H^1(D)}^2 + \int_\Omega (\bar{P}_p + Q_p)\, d\widetilde{\xi}. \tag{39}$$

The measures ξ_0 and $\widetilde{\xi}$ are mutually singular. Every atom of $\widetilde{\xi}$ has magnitude at most $\|\widetilde{\xi}\|_{\mathcal{M}(\Omega)} \le \varepsilon \le \hat{z}_1$. Moreover,

$$\Phi_\phi(\xi) = \sum_{n=1}^N \phi(|z_n|) + \Phi_\phi(\widetilde{\xi}).$$

The local lower bound (35), applied to the atoms and combined with the definition on the continuous part, gives the linear term $L_\phi \|\widetilde{\xi}\|_{\mathcal{M}(\Omega)}$ and an atomic remainder bounded below by

$$-\frac{\gamma_1}{2} \sum_{\omega \in \text{atom}\,\widetilde{\xi}} |\widetilde{\xi}(\{\omega\})|^2 \ge -\frac{\gamma_1\varepsilon}{2} \sum_{\omega \in \text{atom}\,\widetilde{\xi}} |\widetilde{\xi}(\{\omega\})|,$$

because every atom magnitude is at most ε. Therefore

$$\Phi_\phi(\widetilde{\xi}) \ge \left(L_\phi - \frac{\gamma_1\varepsilon}{2}\right) \|\widetilde{\xi}\|_{\mathcal{M}(\Omega)}. \tag{40}$$

Using (39), (38), and (40), we obtain

$$\begin{aligned} J(\mu) &\ge \widehat{J}(z) + \int_\Omega (\bar{P}_p + Q_p)\, d\widetilde{\xi} + \alpha \Phi_\phi(\widetilde{\xi}) \\ &\ge \widehat{J}(z) + \left(\frac{m}{2} - \frac{\alpha\gamma_1\varepsilon}{2}\right) \|\widetilde{\xi}\|_{\mathcal{M}(\Omega)} \\ &\ge \widehat{J}(\bar{z}) + \frac{m}{4}\|\widetilde{\xi}\|_{\mathcal{M}(\Omega)} \ge J(\bar{\mu}). \end{aligned}$$

Thus $\bar{\mu}$ is an $\mathcal{M}_p(\Omega)$-local minimizer. $\square$

Admissible activations. The two exponents in Assumption 3.1 play different roles: s_0 controls the value atom in $L^2(D)$ and is enough for Theorem 3.16; s_1 controls the full atom in $H^1(D)$ and is used by Theorems 3.17, 3.20, and 5.1. Since $\|K(\omega)\|_{L^2(D)} \le \|K(\omega)\|_{H^1(D)}$, it is consistent to impose $s_0 \le s_1$, as Assumption 3.1 does. The exponent s_1 is obtained by estimating $a\rho'(a \cdot x + b)$ separately. The condition $p > s_1$ normalizes the full H^1 atom; together with Assumption 3.4(2), it is sufficient for the finite-support conclusion in Theorem 3.20. The following exponents satisfy the pointwise bounds on a bounded domain.

activation	s_0	s_1	sufficient condition
$\tanh z$	0	1	$p > 1$
e^{-z^2}	0	1	$p > 1$
$\log(1 + e^z)$	1	1	$p > 1$
$\big(z\Phi(z)\big)^2$	2	2	$p > 2$
$(z_+)^k$	k	k	$p > k$

For bounded activations such as tanh and the Gaussian, existence alone requires only $p > 0$; the condition $p > 1$ in the table is the one that also dominates the elementary global H^1 growth estimate. The fourth row is the squared Gaussian error linear unit, written GELU^2 below, with Φ the standard normal distribution function; since Φ takes values in $(0, 1)$ we have $|z\Phi(z)| \le |z|$, and its derivative $2z\Phi(z)\big(\Phi(z) + z\varphi(z)\big)$ grows linearly, which gives $s_0 = s_1 = 2$. The last row is listed for comparison with the positively homogeneous formulation of Section 4.

4. The k-homogeneous formulation

The ReLU^k activation function is of particular interest since it is widely used in the Sobolev training [14, 11]. For gradient fitting with ReLU^k we use $k \ge 2$, so that $K(\omega) \in C^1(\overline{D})$. As we will see in this section, the induced penalty is $\psi_k(t) = t^q$ with $q = 2/(k + 1) < 1$, so it has infinite right slope at zero. It does not extend the result in Section 3.

On the other hand, k-homogeneity makes the radial part of an inner parameter redundant: every atom can be moved onto the unit sphere $\mathbb{S}^d \subset \mathbb{R}^{d+1}$ at the cost of rescaling its outer coefficient. The parameter domain of the resulting problem is compact, so the dictionary is bounded, and neither the polynomial weight w_p nor the weighted space $\mathcal{M}_p(\Omega)$ of Section 3 is required. Throughout this section we work on $\mathcal{M}(\mathbb{S}^d)$ with the total-variation norm $\|\cdot\|_{\mathcal{M}(\mathbb{S}^d)}$, which we identify with the subspace of $\mathcal{M}(\Omega)$ of measures vanishing outside $\mathbb{S}^d$.

Assumption 4.1 (k-homogeneity)**.** For some $k \ge 1$,

$$K(\lambda\omega) = \lambda^k K(\omega) \qquad (\lambda > 0,\ \omega \in \mathbb{R}^{d+1}).$$

Lemma 4.2 (The ReLU^k dictionary)**.** *Let $k \ge 2$, let $D \subset \mathbb{R}^d$ be bounded with positive Lebesgue measure, let $\rho(z) = (z_+)^k$, and let*

$$[K(\omega)](x) = (a \cdot x + b)_+^k, \qquad \omega = (a, b) \in \mathbb{R}^{d+1}.$$

Then

(1) *K satisfies Assumption 4.1 with degree k;*
(2) *$K(\omega) \in C^1(\mathbb{R}^d)$ with $\nabla K(\omega)(x) = k(a \cdot x + b)_+^{k-1} a$, and this classical gradient is also the distributional gradient of $K(\omega)$ on D;*
(3) *$K : \mathbb{S}^d \to H^1(D)$ is continuous; and*
(4) *$C_S := \sup_{\omega \in \mathbb{S}^d} \|K(\omega)\|_{H^1(D)}$ satisfies $0 < C_S < \infty$.*

Proof. (1) For $\lambda > 0$ one has $(\lambda z)_+ = \lambda z_+$, hence $(\lambda a \cdot x + \lambda b)_+^k = \lambda^k (a \cdot x + b)_+^k$ for every x.

(2) Set $h_k(z) := (z_+)^k$. On $(0, \infty)$ and on $(-\infty, 0)$ the function h_k is differentiable with $h_k'(z) = k(z_+)^{k-1}$. At $z = 0$ both one-sided difference quotients are $O(|z|^{k-1})$ and vanish in the

limit because $k \geq 2$, so $h_k'(0) = 0 = k(0_+)^{k-1}$. The formula $h_k'(z) = k(z_+)^{k-1}$ therefore holds on all of $\mathbb{R}$, and h_k' is continuous because $k-1 \geq 1$. The chain rule applied to $x \mapsto a \cdot x + b$ gives $\nabla K(\omega)(x) = k(a \cdot x + b)_+^{k-1} a$. Since D is bounded, both $K(\omega)$ and $\nabla K(\omega)$ are bounded and continuous on D; a continuously differentiable function on a neighbourhood of D lies in $H^1(D)$ and its distributional gradient is its classical gradient.

(3) Let $\omega_j = (a_j, b_j) \to \omega = (a, b)$ in $\mathbb{S}^d$ and put $R_D := \sup_{x \in D} |x| < \infty$. Then

$$\sup_{x \in D} \big|(a_j \cdot x + b_j) - (a \cdot x + b)\big| \leq |a_j - a|\, R_D + |b_j - b| \longrightarrow 0,$$

so the pre-activations converge uniformly on D. All of them lie in the compact interval $I := [-(R_D+1), R_D+1]$, because $|a_j \cdot x + b_j| \leq R_D + 1$ for every j and every $x \in D$ by $|a_j| \leq 1$, $|b_j| \leq 1$. The functions h_k and h_k' are uniformly continuous on I, whence

$$K(\omega_j) \to K(\omega) \quad \text{and} \quad h_k'(a_j \cdot x + b_j) \to h_k'(a \cdot x + b) \qquad \text{uniformly on } D.$$

Since $\nabla K(\omega_j)(x) = h_k'(a_j \cdot x + b_j)\, a_j$, $a_j \to a$, and the scalar factors are uniformly bounded on D, the gradients converge uniformly on D as well. As D has finite Lebesgue measure, uniform convergence implies convergence in $L^2(D)$, so $K(\omega_j) \to K(\omega)$ in $H^1(D)$.

(4) Finiteness follows from (3) and compactness of $\mathbb{S}^d$, since a continuous real function attains a finite maximum on a compact set. For positivity take $\widehat{\omega} = (0, 1) \in \mathbb{S}^d$, for which $[K(\widehat{\omega})](x) = 1$ for every x; hence $C_S \geq \|K(\widehat{\omega})\|_{H^1(D)} = |D|^{1/2} > 0$. □

In the remainder of the section $K : \mathbb{S}^d \to H^1(D)$ is continuous and Lemma 4.2 verifies this assumption and $C_S > 0$ for ReLU^k with $k \geq 2$, the range used for the gradient-augmented problem.

The following lemma extends the degree-one ReLU rescaling result [20, Appendix F, Proposition 3] to a k-homogeneous dictionary. We retain general positive exponents to make explicit how the homogeneity degree changes both the induced exponent and its coefficient.

Lemma 4.3 (Rescaling of a k-homogeneous network)**.** *Let Assumption 4.1 hold, and let $\xi, \zeta > 0$. For a fixed nonzero represented coefficient $\widetilde{c}$ and a unit direction $\widehat{\omega} \in \mathbb{S}^d$ with $K(\widehat{\omega}) \neq 0$, minimize*

$$\frac{|c|^\xi}{\xi} + \frac{|\omega|^\zeta}{\zeta}$$

over radial representations

$$\omega = \lambda \widehat{\omega}, \qquad \lambda > 0, \qquad c\, K(\omega) = \widetilde{c}\, K(\widehat{\omega}).$$

The minimum equals

$$C_{k,\xi,\zeta} |\widetilde{c}|^q, \qquad q := \frac{\xi\zeta}{k\xi + \zeta}, \qquad C_{k,\xi,\zeta} := \frac{1}{q}\, k^{-k\xi/(k\xi+\zeta)}. \tag{41}$$

Consequently, consider any finite representation without zero-function neurons and set

$$\widehat{\omega}_n := \frac{\omega_n}{|\omega_n|}, \qquad \widetilde{c}_n := c_n |\omega_n|^k.$$

Among all independent radial rescalings that preserve each function $c_n K(\omega_n)$, the minimum total penalty is $C_{k,\xi,\zeta} \sum_n |\widetilde{c}_n|^q$.

Proof. We reduce the rescaling constraint to a scalar minimization and solve it explicitly. Write $\omega = \lambda \widehat{\omega}$, where $\lambda = |\omega| > 0$. Since $K(\widehat{\omega}) \neq 0$, k-homogeneity makes the representation constraint equivalent to $c\lambda^k = \widetilde{c}$. Hence, for fixed $\widetilde{c}$, the radial penalty is

$$g(\lambda) = \frac{|\widetilde{c}|^\xi}{\xi \lambda^{k\xi}} + \frac{\lambda^\zeta}{\zeta}.$$

It tends to $+\infty$ at both endpoints, and

$$g'(\lambda) = \lambda^{-k\xi-1} \left(\lambda^{k\xi+\zeta} - k|\widetilde{c}|^\xi \right).$$

Thus g' changes sign exactly once, at

$$\lambda_* = \big(k|\widetilde{c}|^{\xi}\big)^{1/(k\xi+\zeta)}.$$

It is therefore the unique global minimizer. The stationarity identity gives $|\widetilde{c}|^{\xi}\lambda_*^{-k\xi} = \lambda_*^{\zeta}/k$; using the definition of q in (41), we obtain

$$\begin{aligned} g(\lambda_*) &= \left(\frac{1}{k\xi} + \frac{1}{\zeta}\right)\lambda_*^{\zeta} \\ &= \frac{1}{q}\,k^{-k\xi/(k\xi+\zeta)}|\widetilde{c}|^{q} \\ &= C_{k,\xi,\zeta}|\widetilde{c}|^{q}. \end{aligned}$$

For each normalized neuron $(\widetilde{c}_n, \widehat{\omega}_n)$, set

$$\lambda_n = \big(k|\widetilde{c}_n|^{\xi}\big)^{1/(k\xi+\zeta)}, \qquad \omega_n = \lambda_n\widehat{\omega}_n, \qquad c_n = \widetilde{c}_n\lambda_n^{-k}.$$

By k-homogeneity, $c_nK(\omega_n) = \widetilde{c}_nK(\widehat{\omega}_n)$, and the displayed rescaling preserves this function. The radial variables are independent, so applying the scalar minimum to each neuron and summing proves the final assertion. ☐

Remark 4.4 (The quadratic case and the meaning of α). For $\xi = \zeta = 2$,

$$q = \frac{2}{k+1}, \qquad C_k := C_{k,2,2} = \frac{k+1}{2}\,k^{-k/(k+1)}.$$

In particular $C_1 = 1$ and $C_2 = \frac{3}{2}\,2^{-2/3}$. Since C_k depends on k, equal nominal values of α in the penalty $\alpha\sum_n |c_n|^q$ for different powers k do *not* correspond to equal coefficients of the original separable penalty.

For $k \geq 2$ define the scalar penalty

$$\psi_k(t) := t^q, \qquad q := \frac{2}{k+1} \in (0,1), \qquad t \geq 0,$$

which is the exponent of Lemma 4.3 for $\xi = \zeta = 2$. It is continuous and strictly increasing with $\psi_k(0) = 0$ and $\psi_k'(0^+) = +\infty$, and it is subadditive:

$$\psi_k(s+t) \leq \psi_k(s) + \psi_k(t) \qquad (s, t \geq 0). \tag{42}$$

Indeed, for $s + t > 0$ the numbers $u := s/(s+t)$ and $1 - u = t/(s+t)$ lie in $[0,1]$, where $v^q \geq v$; hence $\psi_k(s) + \psi_k(t) = (s+t)^q\big(u^q + (1-u)^q\big) \geq (s+t)^q$. Iterating (42) and passing to the limit in the finite partial sums gives, for every summable sequence $(t_j)_{j\geq 1}$ of nonnegative numbers,

$$\psi_k\Big(\sum_{j\geq 1} t_j\Big) \leq \sum_{j\geq 1}\psi_k(t_j). \tag{43}$$

Definition 4.5 (Homogeneous measure penalty)**.** For $\mu \in \mathcal{M}(\mathbb{S}^d)$ with atomic and continuous parts μ_{atom} and μ_{cont} as in Section 2.1, the atom set is at most countable. Hence the following sum of nonnegative terms is well defined in $[0, +\infty]$ and independent of its enumeration. Set

$$\Phi_{\psi_k}(\mu) := \begin{cases} \displaystyle\sum_{\omega\in \text{atom}\,\mu} \psi_k\big(|\mu(\{\omega\})|\big), & \mu_{\text{cont}} = 0, \\ +\infty, & \mu_{\text{cont}} \neq 0. \end{cases}$$

The functional is nonnegative, vanishes only at the zero measure, and, whenever $\Phi_{\psi_k}(\mu) < \infty$, countable subadditivity gives

$$\Phi_{\psi_k}(\mu) \geq \psi_k\big(\|\mu\|_{\mathcal{M}(\mathbb{S}^d)}\big) = \|\mu\|_{\mathcal{M}(\mathbb{S}^d)}^{q}. \tag{44}$$

For a finite representation $\mu = \sum_{n=1}^{N} c_n\delta_{\omega_n}$, repeated locations are merged. The triangle inequality and (42) show that merging cannot increase $\sum_n \psi_k(|c_n|)$.

4.1. **The measure problem on the sphere.** For $\mu \in \mathcal{M}(\mathbb{S}^d)$ we now set

$$J_{\psi_k}(\mu) := L(\mu) + \alpha \Phi_{\psi_k}(\mu), \tag{45}$$

$$J^M_{\psi_k}(\mu) := l^M(\mu) + \alpha \Phi_{\psi_k}(\mu), \tag{46}$$

with the convention that a sum containing $+\infty$ equals $+\infty$. Since the empirical fidelity is also one half of a squared Hilbert-space norm and its dictionary is continuous on the compact sphere, all results below hold for the empirical objective as well. The empirical fidelity (6) was introduced for finite networks; for ReLU^k with $k \geq 2$ it is defined on all of $\{\mu \in \mathcal{M}(\mathbb{S}^d) : \Phi_{\psi_k}(\mu) < \infty\}$, provided the sampled data $V(x^m)$ and g^m are given. Indeed, such a μ is purely atomic with $\sum_j |c_j| < \infty$, while Lemma 4.2(2)–(3) and compactness of $\mathbb{S}^d$ bound $|[K(\omega)](x^m)|$ and $|\nabla K(\omega)(x^m)|$ uniformly in $\omega \in \mathbb{S}^d$ and m; hence $\sum_j c_j [K(\omega_j)](x^m)$ and $\sum_j c_j \nabla K(\omega_j)(x^m)$ converge absolutely, so both residuals in $l^M(\mu)$ are defined. We study the minimization problem

$$\min_{\mu \in \mathcal{M}(\mathbb{S}^d)} J_{\psi_k}(\mu). \tag{P_{ψ_k}}$$

For a finite network with distinct locations $\omega_n \in \mathbb{S}^d$, the definition of Φ_{ψ_k} gives

$$J_{\psi_k}(\mu_{\boldsymbol{\omega},c}) = L(\mu_{\boldsymbol{\omega},c}) + \alpha \sum_{n=1}^{N} |c_n|^q,$$

which is the objective minimized by the finite-dimensional solver in Algorithm 2; the empirical objective (46) restricts in the same way.

Local minimizers of (P_{ψ_k}) are understood as in Definition 3.14, with $\mathcal{M}_p(\Omega)$ replaced by $\mathcal{M}(\mathbb{S}^d)$, J by J_{ψ_k}, and the weighted measure norm by the total-variation norm.

Proposition 4.6 (Finite truncations and equality of the infima). *Let $V \in H^1(D)$, let $K : \mathbb{S}^d \to H^1(D)$ be continuous with bound C_S, let $\mu \in \mathcal{M}(\mathbb{S}^d)$ satisfy $\Phi_{\psi_k}(\mu) < \infty$, let $(\omega_j)_{j\in\mathcal{I}}$, $\mathcal{I} \subseteq \mathbb{N}$, enumerate* atom μ, *write $c_j := \mu(\{\omega_j\}) \neq 0$, and set*

$$\mu_m := \sum_{j\in\mathcal{I},\ j\leq m} c_j \delta_{\omega_j}.$$

Then

$$\mathcal{N}\mu \in H^1(D), \qquad \|\mathcal{N}\mu\|_{H^1(D)} \leq C_S \|\mu\|_{\mathcal{M}(\mathbb{S}^d)},$$
$$\|\mathcal{N}\mu_m - \mathcal{N}\mu\|_{H^1(D)} \leq C_S \|\mu_m - \mu\|_{\mathcal{M}(\mathbb{S}^d)} \longrightarrow 0.$$

Moreover, $J_{\psi_k}(\mu_m) \to J_{\psi_k}(\mu)$. Consequently the all-measures infimum and the finite-network infimum coincide:

$$\inf_{\mu\in\mathcal{M}(\mathbb{S}^d)} J_{\psi_k}(\mu) = \inf_{N\in\mathbb{N}_0} \inf_{\substack{\omega_n\in\mathbb{S}^d,\ c_n\in\mathbb{R} \\ n=1,\dots,N}} \Big[L(\mu_{\boldsymbol{\omega},c}) + \alpha \sum_{n=1}^{N} \psi_k(|c_n|) \Big]. \tag{47}$$

Proof. Since $\Phi_{\psi_k}(\mu) < \infty$, Definition 4.5 gives $\mu_{\text{cont}} = 0$, so $\mu = \sum_{j\in\mathcal{I}} c_j \delta_{\omega_j}$ with distinct ω_j and $c_j \neq 0$. Since $\sum_{j\in\mathcal{I}} |c_j| = \|\mu\|_{\mathcal{M}(\mathbb{S}^d)} < \infty$, the tail identity

$$\|\mu_m - \mu\|_{\mathcal{M}(\mathbb{S}^d)} = \sum_{j\in\mathcal{I}, j>m} |c_j| \longrightarrow 0.$$

Bounded linearity of $\mathcal{N}$ therefore gives $\mathcal{N}\mu \in H^1(D)$ and the stated norm bounds. Continuity of $u \mapsto \frac{1}{2}\|u - V\|^2_{H^1(D)}$ gives $L(\mu_m) \to L(\mu)$, while

$$\Phi_{\psi_k}(\mu_m) = \sum_{j\in\mathcal{I}, j\leq m} \psi_k(|c_j|) \longrightarrow \Phi_{\psi_k}(\mu)$$

by monotone convergence. Combining these two limits proves $J_{\psi_k}(\mu_m) \to J_{\psi_k}(\mu)$.

It remains to compare the infima; call them i_{meas} and i_{disc}, in the order in which they appear in (47). Every finite network defines an admissible measure, and its finite-dimensional objective equals the corresponding measure objective. Hence $i_{\text{meas}} \le i_{\text{disc}}$. Conversely, let μ satisfy $J_{\psi_k}(\mu) < \infty$. Its truncations μ_m are finite networks and satisfy $J_{\psi_k}(\mu_m) \to J_{\psi_k}(\mu)$. Therefore $i_{\text{disc}} \le J_{\psi_k}(\mu)$. Taking the infimum over such μ gives $i_{\text{disc}} \le i_{\text{meas}}$. □

Proposition 4.6 shows that a measure of finite penalty automatically has finite fidelity. Since conversely $J_{\psi_k}(\mu) < \infty$ contains $\alpha\Phi_{\psi_k}(\mu) < \infty$, the two conditions agree:

$$J_{\psi_k}(\mu) < \infty \quad \iff \quad \Phi_{\psi_k}(\mu) < \infty \qquad \big(\mu \in \mathcal{M}(\mathbb{S}^d)\big),$$

and either may be used as the standing finiteness hypothesis.

Proposition 4.6 is used only as a bridge between the finite-network problem and (P_{ψ_k}). It equates the two infima; attainment is established in Theorem 4.9 from the finite-dimensional argument and is not claimed here.

Theorem 4.7. *Let $k \ge 2$, $q = 2/(k+1) \in (0,1)$, $\alpha > 0$, $V \in H^1(D)$, and let $K : \mathbb{S}^d \to H^1(D)$ be continuous. Let $\bar\mu \in \mathcal{M}(\mathbb{S}^d)$ satisfy $J_{\psi_k}(\bar\mu) < \infty$ and be optimal against perturbations of the weights of its own atoms: there is $\varepsilon > 0$ such that*

$$J_{\psi_k}(\bar\mu) \le J_{\psi_k}(\bar\mu + t\delta_\omega) \qquad \text{for every } \omega \in \operatorname{atom}\bar\mu \text{ and every } |t| < \varepsilon. \tag{48}$$

Set

$$B(\bar\mu) := \|P_{\bar\mu}\|_{C(\mathbb{S}^d)}.$$

Then $\bar\mu$ is purely atomic, $B(\bar\mu) < \infty$, and every atom $\omega_n \in \operatorname{atom}\bar\mu$ with weight $c_n := \bar\mu(\{\omega_n\})$ satisfies

$$P_{\bar\mu}(\omega_n) + \alpha q |c_n|^{q-1}\operatorname{sign}(c_n) = 0. \tag{49}$$

If $B(\bar\mu) = 0$, then $\bar\mu = 0$. If $B(\bar\mu) > 0$, then

$$|c_n| \ge c_{\min}(\bar\mu) := \left(\frac{\alpha q}{B(\bar\mu)}\right)^{1/(1-q)} > 0 \qquad \text{for every } \omega_n \in \operatorname{atom}\bar\mu. \tag{50}$$

Hypothesis (48) holds in particular if $\bar\mu$ is a local minimizer of (P_{ψ_k}) in total variation in the sense specified above.

Proof. We derive (49) from a two-sided perturbation of a single atom weight, and then convert it into the lower bound (50).

Since $J_{\psi_k}(\bar\mu) < \infty$ we have $\Phi_{\psi_k}(\bar\mu) < \infty$, so $\bar\mu_{\text{cont}} = 0$ and $\bar\mu$ is purely atomic. By Proposition 4.6, applied to $\bar\mu$, $\mathcal{N}\bar\mu \in H^1(D)$, so $r_{\bar\mu} \in H^1(D)$ is well defined. The map $P_{\bar\mu}$ is continuous on $\mathbb{S}^d$, being the composition of the continuous map K with the continuous linear functional $\langle r_{\bar\mu}, \cdot\rangle_{H^1(D)}$, and $\mathbb{S}^d$ is compact, so $B(\bar\mu) < \infty$.

Fix $\omega_n \in \operatorname{atom}\bar\mu$, write $c_n = \bar\mu(\{\omega_n\}) \ne 0$, let $s \in \{-1, 1\}$, and consider $\mu_t := \bar\mu + ts\delta_{\omega_n}$ for $0 < t < \min\{\varepsilon, |c_n|\}$. The measure μ_t differs from $\bar\mu$ only in the weight of the atom ω_n, which becomes $c_n + ts \ne 0$, so $\operatorname{atom}\mu_t = \operatorname{atom}\bar\mu$ and

$$\Phi_{\psi_k}(\mu_t) - \Phi_{\psi_k}(\bar\mu) = \psi_k(|c_n + ts|) - \psi_k(|c_n|),$$

all other summands being unchanged. For the fidelity, $\mathcal{N}\mu_t = \mathcal{N}\bar\mu + tsK(\omega_n)$, so the residual is $r_{\bar\mu} + tsK(\omega_n)$ and expanding the square gives the exact identity

$$\begin{aligned} L(\mu_t) - L(\bar\mu) &= ts\langle r_{\bar\mu}, K(\omega_n)\rangle_{H^1(D)} + \frac{t^2}{2}\|K(\omega_n)\|^2_{H^1(D)} \\ &= tsP_{\bar\mu}(\omega_n) + \frac{t^2}{2}\|K(\omega_n)\|^2_{H^1(D)}. \end{aligned}$$

Because $c_n \ne 0$, the map $z \mapsto |z|^q$ is differentiable at c_n with derivative $q|c_n|^{q-1}\operatorname{sign}(c_n)$. Hypothesis (48) gives $J_{\psi_k}(\mu_t) - J_{\psi_k}(\bar\mu) \ge 0$; dividing by $t > 0$ and letting $t \downarrow 0$ yields

$$0 \le s\Big[P_{\bar\mu}(\omega_n) + \alpha q|c_n|^{q-1}\operatorname{sign}(c_n)\Big].$$

Choosing $s = 1$ and $s = -1$ gives (49).

Suppose $B(\bar{\mu}) = 0$, that is, $P_{\bar{\mu}} \equiv 0$ on $\mathbb{S}^d$. If $\bar{\mu}$ had an atom ω_n, then (49) would read $\alpha q |c_n|^{q-1} \operatorname{sign}(c_n) = 0$, which is impossible because $\alpha > 0$, $q > 0$, and $|c_n|^{q-1} \in (0, \infty)$ for $c_n \neq 0$. Hence atom $\bar{\mu} = \emptyset$, and since $\bar{\mu}$ is purely atomic, $\bar{\mu} = 0$.

Suppose $B(\bar{\mu}) > 0$. Taking absolute values in (49) gives

$$\alpha q |c_n|^{q-1} = |P_{\bar{\mu}}(\omega_n)| \leq B(\bar{\mu}).$$

Since $q - 1 = -(1-q) < 0$, this reads $|c_n|^{-(1-q)} \leq B(\bar{\mu})/(\alpha q)$, hence $|c_n|^{1-q} \geq \alpha q / B(\bar{\mu})$; raising both sides to the positive power $1/(1-q)$ preserves the inequality and gives (50). ☐

Remark 4.8. The identity $\psi_k'(0^+) = +\infty$ has the following consequences.

(1) Inserting a new atom at $\omega \notin$ atom $\bar{\mu}$ with weight ts, $s \in \{-1, 1\}$, changes the penalty by αt^q. Since $t^q/t = t^{q-1} \to \infty$ as $t \downarrow 0$, division by t produces no finite limit, and hence no pointwise condition on $P_{\bar{\mu}}(\omega)$ away from atom $\bar{\mu}$. Theorem 4.7 is restricted to existing atoms for this reason.

(2) The zero measure is a local minimizer of (P_{ψ_k}) in total variation. Indeed, $J_{\psi_k}(\mu) = \infty$ whenever $\Phi_{\psi_k}(\mu) = \infty$, while for μ with $\Phi_{\psi_k}(\mu) < \infty$ and $\|\mu\|_{\mathcal{M}(\mathbb{S}^d)} = \varepsilon$,

$$\begin{aligned} J_{\psi_k}(\mu) - J_{\psi_k}(0) &= \langle -V, \mathcal{N}\mu \rangle_{H^1(D)} + \tfrac{1}{2}\|\mathcal{N}\mu\|_{H^1(D)}^2 + \alpha \Phi_{\psi_k}(\mu) \\ &\geq -\|V\|_{H^1(D)} C_S \, \varepsilon + \alpha \varepsilon^q, \end{aligned}$$

by (44), the bound $\|\mathcal{N}\mu\|_{H^1(D)} \leq C_S \varepsilon$ of Proposition 4.6, and the Cauchy–Schwarz inequality. Since $q < 1$, the right-hand side is positive for every sufficiently small $\varepsilon > 0$.

An insertion criterion based on an infinitesimal weight perturbation is therefore unavailable for this penalty, and Algorithm 2 uses a finite-step decrease test instead.

4.2. **Global existence and a uniform width bound.**

Theorem 4.9 (Existence and width bound). *Let $k \geq 2$, $q = 2/(k+1) \in (0, 1)$, $\alpha > 0$, let $V \in H^1(D)$ with $V \neq 0$, and let $K : \mathbb{S}^d \to H^1(D)$ be continuous with $C_S = \sup_{\omega \in \mathbb{S}^d} \|K(\omega)\|_{H^1(D)} > 0$; the latter holds for* ReLU^k *by Lemma 4.2(4). Then (P_{ψ_k}) admits a global minimizer, and its minimum value equals the finite-network infimum on the right-hand side of (47). Every global minimizer $\bar{\mu}$ is a finite atomic measure satisfying*

$$\#\,\mathrm{atom}\,\bar{\mu} \leq N_{\max} := \frac{\|V\|_{H^1(D)}^2}{2\alpha\, c_{\min,\mathrm{glob}}^q}, \qquad c_{\min,\mathrm{glob}} := \left(\frac{\alpha q}{\|V\|_{H^1(D)}\, C_S} \right)^{1/(1-q)}.$$

Remark 4.10. The excluded cases are immediate. If $V = 0$, then $\mu = 0$ is the unique global minimizer by the nonnegativity of the fidelity and the strict positivity of Φ_{ψ_k} away from zero. If $C_S = 0$, then $K \equiv 0$, so the fidelity is constant and the same penalty argument again makes $\mu = 0$ the unique global minimizer. Thus the atom bound is zero in both cases. The assumptions $V \neq 0$ and $C_S > 0$ are needed only to ensure that $c_{\min,\mathrm{glob}} \in (0, \infty)$ and $N_{\max} < \infty$ are well defined.

Proof. We first obtain minimizers at each fixed width by compactness, then derive one width estimate for both the existence argument and arbitrary global minimizers.

For $N \in \mathbb{N}_0$, set

$$F_N(\boldsymbol{\omega}, c) := L(\mu_{\boldsymbol{\omega}, c}) + \alpha \sum_{n=1}^{N} |c_n|^q, \qquad m_N := \inf F_N,$$

with $F_0 = m_0 = \frac{1}{2}\|V\|_{H^1(D)}^2$. For $N \geq 1$, continuity of K, scalar multiplication, and finite addition show that $(\boldsymbol{\omega}, c) \mapsto \sum_{n=1}^{N} c_n K(\omega_n)$ is continuous into $H^1(D)$; hence F_N is continuous.

Any minimizing sequence may be chosen in the sublevel $F_N \le m_0 + 1$. Since the fidelity is nonnegative, this sublevel satisfies

$$|c_n| \le R := \left(\frac{m_0+1}{\alpha}\right)^{1/q} \qquad (n = 1, \dots, N).$$

It therefore lies in the compact set $(\mathbb{S}^d)^N \times [-R, R]^N$. Compactness gives a convergent subsequence, and continuity shows that its limit attains m_N. The case $N = 0$ is immediate, and we have $m_{N+1} \le m_N$.

We use the following estimate twice. Suppose that a measure μ satisfies $J_{\psi_k}(\mu) \le m_0$ and the atomwise optimality hypothesis (48). Theorem 4.7 shows that μ is purely atomic. The nonnegativity of the penalty gives $\|r_\mu\|_{H^1(D)} \le \|V\|_{H^1(D)}$, and hence

$$B(\mu) \le \|V\|_{H^1(D)} C_S.$$

If $\mu \neq 0$, Theorem 4.7 gives $B(\mu) > 0$ and

$$|\mu(\{\omega\})| \ge \left(\frac{\alpha q}{B(\mu)}\right)^{1/(1-q)} \ge c_{\min,\text{glob}} \qquad (\omega \in \operatorname{atom} \mu).$$

Consequently, for every finite $A \subseteq \operatorname{atom} \mu$,

$$\# A\, \alpha c_{\min,\text{glob}}^q \le \alpha \sum_{\omega \in A} |\mu(\{\omega\})|^q \le \alpha \Phi_{\psi_k}(\mu) \le J_{\psi_k}(\mu) \le \frac{1}{2}\|V\|^2_{H^1(D)}.$$

An infinite atom set would contain finite subsets of arbitrarily large cardinality. Thus $\operatorname{atom} \mu$ is finite and its cardinality is at most $N_{\max}$; the same conclusion is immediate when $\mu = 0$.

Set $N_{\text{cut}} := \lfloor N_{\max} \rfloor$. The estimate above applies to every minimizer defining m_N, since each coefficient can be varied independently. Thus, for $N \ge N_{\text{cut}}$, that minimizer has width at most N_{cut}, so $m_{N_{\text{cut}}} \le m_N$. The reverse inequality follows from the monotonicity of (m_N). Hence

$$\inf_{N \in \mathbb{N}_0} m_N = m_{N_{\text{cut}}}.$$

Since $m_{N_{\text{cut}}}$ is attained, Proposition 4.6 gives a global minimizer of (P_{ψ_k}) with the same value. Every global minimizer satisfies the hypotheses of the estimate above and therefore has at most $N_{\max}$ atoms. □

Corollary 4.11 (Finite support of total-variation local minimizers)**.** *Let the hypotheses of Theorem 4.7 on k, q, α, V, K hold and let $\bar\mu \neq 0$ be a local minimizer of (P_{ψ_k}) in total variation. Then $B(\bar\mu) > 0$, the measure $\bar\mu$ is finite atomic, and*

$$\# \operatorname{atom} \bar\mu \le \frac{J_{\psi_k}(\bar\mu)}{\alpha\, c_{\min}(\bar\mu)^{\,q}}.$$

Proof. By Theorem 4.7, $B(\bar\mu) = 0$ would force $\bar\mu = 0$; hence $B(\bar\mu) > 0$ and every atom weight satisfies $|c_n| \ge c_{\min}(\bar\mu) > 0$. Since ψ_k is nondecreasing and the fidelity is nonnegative,

$$N\alpha\, c_{\min}(\bar\mu)^{\,q} \le \alpha \Phi_{\psi_k}(\bar\mu) \le J_{\psi_k}(\bar\mu) < \infty$$

with $N = \# \operatorname{atom} \bar\mu \in \mathbb{N}_0 \cup \{\infty\}$, which forces $N < \infty$ and gives the stated bound. □

The estimate of Corollary 4.11 depends on the minimizer through both $J_{\psi_k}(\bar\mu)$ and $B(\bar\mu)$ and is therefore an a posteriori bound. It is not computable in advance, unlike the a priori bound $N_{\max}$ of Theorem 4.9, which is uniform over all global minimizers and computable from $\|V\|_{H^1(D)}$, C_S, α and q alone. On a global minimizer the two are comparable and the a posteriori bound is the sharper one, since $J_{\psi_k}(\bar\mu) \le \frac{1}{2}\|V\|^2_{H^1(D)}$ and $c_{\min}(\bar\mu) \ge c_{\min,\text{glob}}$ there.

5. Algorithms

This section describes the computational pipeline: the training data are generated via the open-loop problem, candidate neurons are inserted greedily, and insertion alternates with finite-dimensional optimization of the current network.

Data generation via the open-loop problem. Set $U := L^2(t_0, T; \mathbb{R}^{m_u})$ and assume that for every $u \in U$ the state equation in (P) has a unique solution $y(u)$ on $[t_0, T]$ and that the control-to-state map $u \mapsto y(u)$ is continuously Fréchet differentiable from U into $C([t_0, T]; \mathbb{R}^d)$. The objective of ($P$) is then the reduced functional $u \mapsto \mathcal{C}(u; t_0, x)$ on U, and the standard adjoint (Lagrangian) calculus — see, e.g., [12] — gives its gradient

$$\nabla_u \mathcal{C}(u; t_0, x) = 2\eta u + g(y(u))^\top p \quad \text{in } U,$$

where p solves the costate equation of (TPBVP) backward in time along $(y(u), u)$ in place of (y^*, u^*). Each gradient evaluation therefore costs one forward solve of the state equation and one backward solve of the costate equation, and the implicit derivative of $u \mapsto y(u)$ is never formed. For the data-driven approach it is essential that the initial dataset $\{x^j, V(0, x^j), g^j\}_{j=1}^M$ is *causality-free*: the samples are generated without approximate values at neighboring points. [13] surveys generation methods. As in [1], we minimize $\mathcal{C}(\cdot; t_0, x)$ with Barzilai–Borwein step sizes. Since (P) is not convex, the iteration returns a stationary control rather than a certified global minimizer; the converged initial costate supplies g^j, subject to that reservation and the differentiability qualification of Section 1.

The two insertion algorithms. We use two greedy insertion algorithms of primal–dual active point (PDAP) type, in the sense of the accelerated conditional-gradient method for measure spaces of [21]. Algorithm 1 follows the nonconvex variant of [20], whose scalar penalty satisfies $\phi'(0) < \infty$ as in Assumption 3.2; Algorithm 2 is an adaptation of it to the penalty ψ_k, whose infinite slope at the origin is explicitly excluded by that reference.

In both algorithms the correction step freezes the locations $\boldsymbol{\omega}$ and applies a semismooth Newton method to the resulting finite-dimensional coefficient problem. The two corrections use different coefficient coordinates and different scalar maps, so they are specified in their respective subsections below. In either case the method is local and its output is retained only when it does not increase the post-insertion objective.

For the sampled fidelity, write $P_\mu^M(\omega) := Dl^M(\mu)[\delta_\omega]$, and let $\|K(\omega)\|_M$ denote the sampled value–gradient norm associated with l^M.

For a joint insertion $\sum_{i=1}^r c_i \delta_{\omega_i}$ at distinct new locations the penalty is additive, but the fidelity change contains pairwise cross terms between the inserted features, which no individual test controls. For $N_{\text{ins}} = 1$ the individual decrease statement applies to every accepted candidate, while Corollary 5.2 additionally requires an exact global maximizer. For $N_{\text{ins}} > 1$ the individual estimates do not compose, and neither a decrease per outer iteration nor monotonicity of the objective along the iterates is claimed. Restoring a batch decrease guarantee requires recomputing the residual after each insertion, or testing the joint increment of the whole batch.

5.1. Algorithm 1: the log penalty on the normalized measure.

Theorem 5.1. *Suppose that Assumption 3.1 holds, $p > s_1$, and $\alpha > 0$. Let ϕ satisfy Assumptions 3.2 and 3.3. For $\mu \in \mathcal{M}_p(\Omega)$ define*

$$\Delta(\mu, \omega) := \max\{|P_{p,\mu}(\omega)| - \alpha L_\phi, 0\},$$
$$\Delta(\mu) := \sup_{\omega \in \Omega} \Delta(\mu, \omega) = \max\{\|P_{p,\mu}\|_\infty - \alpha L_\phi, 0\}.$$

For a finite atomic μ, define

$$R(\mu) := \max\left\{1, \left(\frac{2^{s_1+1}C_K\|r_\mu\|_{H^1(D)}}{\alpha L_\phi}\right)^{1/(p-s_1)}\right\}. \tag{51}$$

Every ω with $\Delta(\mu,\omega) > 0$ belongs to $\overline{B}_{R(\mu)}$. If $\Delta(\mu) > 0$, then $B_p > 0$ and $|P_{p,\mu}|$ attains its maximum at some $\omega^* \in \overline{B}_{R(\mu)}$, with $\Delta(\mu,\omega^*) = \Delta(\mu)$. For every ω with $\Delta(\mu,\omega) > 0$, the insertion

$$\kappa(\omega) := -\frac{\Delta(\mu,\omega)}{B_p^2}\operatorname{sign}\big(P_{p,\mu}(\omega)\big), \qquad c(\omega) := \frac{\kappa(\omega)}{w_p(\omega)}, \tag{52}$$

satisfies

$$J\big(\mu + c(\omega)\delta_\omega\big) \le J(\mu) - \frac{\Delta(\mu,\omega)^2}{2B_p^2}. \tag{53}$$

Proof. We first establish attainment of the certificate maximum. $p > s_1$ and (4) give $K_p \in C_0(\Omega; H^1(D))$, hence $B_p < \infty$ and $P_{p,\mu} \in C_0(\Omega)$. If $\Delta(\mu) > 0$, then $\|P_{p,\mu}\|_\infty > \alpha L_\phi > 0$. Choose a closed ball outside which $|P_{p,\mu}| < \|P_{p,\mu}\|_\infty/2$; this is possible because $P_{p,\mu}$ vanishes at infinity. The continuous function $|P_{p,\mu}|$ attains its maximum on that compact ball. Moreover, $B_p = 0$ would imply $P_{p,\mu} = 0$, so $B_p > 0$.

We now bound the change caused by an arbitrary normalized insertion $\kappa\delta_\omega$ with $\kappa \in \mathbb{R}$. The explicit atomic–continuous representation of Φ_ϕ, the monotonicity and subadditivity of ϕ, and Lemma 3.6 give

$$\begin{aligned}
\Phi_\phi(\mu_p + \kappa\delta_\omega) - \Phi_\phi(\mu_p) &\le \phi(|\kappa|) && \text{by subadditivity at the point } \omega\\
&\le L_\phi|\kappa| && \text{by the tangent bound at zero.}
\end{aligned} \tag{54}$$

Since $(\mu + \kappa w_p(\omega)^{-1}\delta_\omega)_p = \mu_p + \kappa\delta_\omega$, the quadratic fidelity has the exact expansion

$$\begin{aligned}
&J\left(\mu + \frac{\kappa}{w_p(\omega)}\delta_\omega\right) - J(\mu)\\
&\quad\le \kappa P_{p,\mu}(\omega) + \frac{\kappa^2}{2}\|K_p(\omega)\|_{H^1(D)}^2 + \alpha L_\phi|\kappa| && \text{by (54)}\\
&\quad\le \kappa P_{p,\mu}(\omega) + \frac{B_p^2\kappa^2}{2} + \alpha L_\phi|\kappa| && \text{by the definition of } B_p.
\end{aligned} \tag{55}$$

Fix $\omega \in \Omega$ with $\Delta(\mu,\omega) > 0$. Then $|P_{p,\mu}(\omega)| = \alpha L_\phi + \Delta(\mu,\omega)$, and substitution of $\kappa = \kappa(\omega)$ from (52) into (55) yields

$$\begin{aligned}
&J\big(\mu + c(\omega)\delta_\omega\big) - J(\mu)\\
&\quad\le -|\kappa(\omega)|\big(\alpha L_\phi + \Delta(\mu,\omega)\big) + \frac{B_p^2|\kappa(\omega)|^2}{2} + \alpha L_\phi|\kappa(\omega)| && \text{since } \operatorname{sign}\kappa(\omega) = -\operatorname{sign} P_{p,\mu}(\omega)\\
&\quad= -|\kappa(\omega)|\Delta(\mu,\omega) + \frac{B_p^2|\kappa(\omega)|^2}{2} && \text{the threshold terms cancel}\\
&\quad= -\frac{\Delta(\mu,\omega)^2}{2B_p^2} && \text{by the choice of } \kappa(\omega).
\end{aligned}$$

This proves (53); at $\omega = \omega^*$ it gives the decrease $\Delta(\mu)^2/(2B_p^2)$.

Finally, we derive the radius from (4). If $|\omega| \ge 1$, then Cauchy–Schwarz gives

$$\begin{aligned}
|P_{p,\mu}(\omega)| &\le \|r_\mu\|_{H^1(D)}\|K_p(\omega)\|_{H^1(D)} && \text{by Cauchy–Schwarz}\\
&\le C_K\|r_\mu\|_{H^1(D)}\frac{(1+|\omega|)^{s_1}}{1+|\omega|^p} && \text{by (4)}\\
&\le 2^{s_1}C_K\|r_\mu\|_{H^1(D)}|\omega|^{-(p-s_1)} && \text{since } |\omega| \ge 1.
\end{aligned}$$

If $|\omega| > R(\mu)$, the last expression is strictly smaller than $\alpha L_\phi/2$ by (51). This proves the claimed localization of every accepted candidate and every maximizer. ☐

Corollary 5.2. *Let $(\mu_k)_{k\geq 0}$ be finite atomic iterates with $J(\mu_0) < \infty$, and write $\Delta_k := \Delta(\mu_k)$. If $\Delta_k > 0$, insert at an exact maximizer given by Theorem 5.1. Then correct the coefficients, accepting the corrected iterate only if it does not increase the objective. Then*

$$\sum_{k=0}^{\infty} \Delta_k^2 \leq 2B_p^2 J(\mu_0),$$
$$\Delta_k \longrightarrow 0,$$
$$\min_{0\leq k<n} \Delta_k \leq B_p \sqrt{\frac{2J(\mu_0)}{n}} \qquad (n \geq 1).$$

These conclusions do not assert convergence of (μ_k) to a global minimizer or an objective-gap rate.

Proof. If $B_p = 0$, then $\Delta_k = 0$ for every k and the claims are immediate. Otherwise Theorem 5.1 and the correction hypothesis give, for every $n \geq 1$,

$$\sum_{k=0}^{n-1} \frac{\Delta_k^2}{2B_p^2} \leq J(\mu_0) - J(\mu_n) \leq J(\mu_0).$$

Letting $n \to \infty$ gives the first claim, and summability gives $\Delta_k \to 0$. Finally,

$$\min_{0\leq k<n} \Delta_k^2 \leq \frac{1}{n}\sum_{k=0}^{n-1} \Delta_k^2 \leq \frac{2B_p^2 J(\mu_0)}{n},$$

which proves the stated rate. ☐

For the sampled data, Algorithm 1 minimizes

$$J^M(\boldsymbol{\omega}, c) := l^M(\mu_{\boldsymbol{\omega},c}) + \alpha \sum_{n=1}^{N} \phi_\gamma\big(w_p(\omega_n)|c_n|\big). \tag{56}$$

Its normalized insertion profile is

$$P^M_{p,\mu_t}(\omega) := \frac{P^M_{\mu_t}(\omega)}{w_p(\omega)}.$$

For fixed nodes, introduce the normalized coefficients $u_n := w_p(\omega_n)c_n$. The outer-weight correction then minimizes

$$l^M\left(\sum_{n=1}^{N} \frac{u_n}{w_p(\omega_n)}\delta_{\omega_n}\right) + \alpha \sum_{n=1}^{N} \phi_\gamma(|u_n|).$$

Theorem 5.1 and Corollary 5.2 apply verbatim with the sampled value–gradient inner product in place of the $H^1(D)$ inner product. The empirical search radius is therefore given by (51), using the sampled residual norm and the corresponding feature-growth constants.

The insertion step. The theorem requires finding a maximizer of $|P^M_{p,\mu_t}(\omega)|$. Finding a global maximizer is generally intractable; for ReLU dictionaries it is NP-hard [2, Section 3.3]. The practical multistart search is proposed in the PDAP algorithm in the paper [20]. At the current iterate $\mu_t = \mu_{\boldsymbol{\omega}^{(t)},c^{(t)}}$ a candidate ω is therefore accepted when

$$\Delta(\mu_t, \omega) > 0, \qquad \text{that is,} \qquad |P^M_{p,\mu_t}(\omega)| > \alpha L_\phi. \tag{57}$$

Theorem 5.1 provides two additional pieces of information. First, it gives the bound $R(\mu_t)$ for sampling the initial points at each iteration. Second, because the semismooth Newton method is local, it gives an initial outer weight that already guarantees a decrease.

For a nonhomogeneous activation set $R_{\text{search}}(\mu_t) := \min\{R(\mu_t), e^5\}$. Each multistart initial point has norm at most $R_{\text{search}}(\mu_t)$. Starting from each ω_0, one unconstrained L-BFGS solve jointly optimizes all components of ω for $|P^M_{p,\mu_t}(\omega)|$. After optimization, candidates with $|\omega| > R_{\text{search}}(\mu_t)$ are discarded. This avoids constrained optimization.

Before running the SSN solver, the algorithm calculates

$$A^M_p(\omega) := \left\| \frac{K(\omega)}{w_p(\omega)} \right\|^2_M, \qquad \kappa_A(\omega) := -\frac{\Delta(\mu_t, \omega)}{A^M_p(\omega)} \operatorname{sign}\big(P^M_{p,\mu_t}(\omega)\big), \qquad c_A(\omega) := \frac{\kappa_A(\omega)}{w_p(\omega)}. \tag{58}$$

The coefficient $c_A(\omega)$ is used to initialize the subsequent semismooth Newton correction, which optimizes all outer weights. The decrease statement holds for every accepted candidate, not only for a global maximizer of $|P^M_{p,\mu_t}|$.

Writing $h_\gamma(z) := \phi_\gamma(|z|) - |z|$ and $\tau = (1 + \alpha\gamma)^{-1}$, the semismooth Newton solve uses the coordinatewise residual

$$u = \operatorname{prox}_{\tau\alpha|\cdot|}\left(u - \tau \left[\nabla_u l^M \left(\sum_{n=1}^N \frac{u_n}{w_p(\omega_n)} \delta_{\omega_n} \right) + \alpha \nabla h_\gamma(u) \right] \right). \tag{59}$$

After the solve, the physical outer weights are recovered as $c_n = u_n / w_p(\omega_n)$; the proximal map in (59) is ordinary soft thresholding.

Algorithm 1 uses the scalar penalty ϕ_γ from (32) and the objective (56), with the insertion condition (57).

Algorithm 1 Iterative node insertion for J^M

1: Choose T_{out}, N_{trial}, N_{ins}, $\varepsilon_{\text{prune}}$, and an initial network $\boldsymbol{\omega}^{(0)}, c^{(0)}$ of width $N(0)$; set $t = 0$.
2: **while** $t < T_{\text{out}}$ **do**
3: Set $R_{\text{search}} = \min\{R(\mu_t), e^5\}$. Sample N_{trial} points ω_0 with norm bounded by R_{search}.
4: From each ω_0, run one unconstrained joint L-BFGS maximization of $|P^M_{p,\mu_t}(\omega)|$.
5: Discard candidates with $|\omega| > R_{\text{search}}$, then remove near-duplicates.
6: Keep the candidates satisfying (57); terminate if there is none.
7: Insert at most N_{ins} nodes with the largest $\Delta(\mu_t, \omega)$, warm-start outer weight by (58).
8: Optimize the outer weights of the enlarged network by the semismooth Newton method.
9: Remove nodes with $|c_n| \le \varepsilon_{\text{prune}}$ (resulting in width $N(t+1)$).
10: $t = t + 1$.
11: **end while**

5.2. **Algorithm 2: the fractional-exponent penalty on the sphere.** Algorithm 2 uses $\psi_k(z) = z^q$, $q = 2/(k+1)$, on the unit sphere. For $q < 1$, the zero network is a local minimizer and the infinitesimal insertion condition of Algorithm 1 is unavailable (Remark 4.8). We therefore test the actual change produced by inserting one atom with a finite coefficient.

We use the scalar proximal map

$$\operatorname{prox}_{\lambda|\cdot|^q}(v) \in \operatorname*{arg\,min}_{z \in \mathbb{R}} \left\{ \frac{1}{2}(z - v)^2 + \lambda |z|^q \right\}, \tag{60}$$

with zero selected when the minimum is attained both at zero and at a nonzero point. For $0 < q < 1$, put

$$\bar{z} = [2\lambda(1-q)]^{1/(2-q)}, \qquad \bar{v} = \frac{2-q}{2(1-q)} \bar{z}.$$

Then the selected proximal value is zero when $|v| \le \bar{v}$; when $|v| > \bar{v}$, it is $\operatorname{sign}(v) z$, where z is the largest positive solution of

$$z + \lambda q z^{q-1} = |v|.$$

The implementation evaluates this value in closed form for $q \in \{1/2, 2/3\}$, using respectively a cubic in $\sqrt{z}$ and a quartic in $z^{1/3}$. At the endpoint $q = 1$, (60) is ordinary soft thresholding.

For a candidate $\omega \notin \operatorname{atom} \mu_t$, write

$$P := P^M_{\mu_t}(\omega), \qquad A := \|K(\omega)\|^2_M.$$

If A or $|P|$ is numerically zero, the candidate is discarded. Otherwise the exact one-atom objective increment is

$$\Delta J^M_{\psi_k}(c;\omega) = cP + \frac{1}{2}Ac^2 + \alpha|c|^q. \tag{61}$$

Completing the square in (61) gives its selected global minimizer directly:

$$c^*(\omega) := \operatorname{prox}_{(\alpha/A)|\cdot|^q}\left(-\frac{P}{A}\right). \tag{62}$$

We accept the candidate precisely when $c^*(\omega) \neq 0$ and $\Delta J^M_{\psi_k}(c^*(\omega);\omega) < 0$. Thus the same coefficient both certifies the finite-step decrease and initializes the subsequent correction.

After insertion, let m_t be the smallest magnitude among the nonzero outer weights. For $q < 1$, the correction fixes the proximal scale

$$\lambda_t := \min\left\{\alpha, \frac{\rho_{\text{prox}} m_t^{2-q}}{2(1-q)}\right\}, \qquad \rho_{\text{prox}} = 0.5, \tag{63}$$

for the whole semismooth Newton solve. Every nonzero warm-start coefficient is strictly above the switching magnitude, since $\bar{z} \le \rho_{\text{prox}}^{1/(2-q)} m_t < m_t$. Define, coordinatewise,

$$c(z) := \operatorname{prox}_{\lambda_t|\cdot|^q}(z), \qquad G_t(z) := \frac{\alpha}{\lambda_t}\big(z - c(z)\big) + \nabla_c l^M(\mu_{\boldsymbol{\omega},c(z)}). \tag{64}$$

The coefficient correction applies semismooth Newton to $G_t(z) = 0$. For $q = 1$ we use soft thresholding with $\lambda_t = \alpha$. For $q < 1$, an exact undamped Newton iteration on a fixed smooth branch has the usual local rate under nonsingularity; no convergence rate is claimed for the implemented globalizations. The corrected coefficients are retained only when they do not increase the post-insertion objective.

The implementation uses $k \in \{2,3\}$, corresponding to $q \in \{2/3, 1/2\}$; $k = 1$, $q = 1$ is retained as the ReLU–ℓ^1 baseline.

Algorithm 2 Iterative node insertion for $J^M_{\psi_k}$

Choose T_{out}, N_{trial}, N_{ins}, $\varepsilon_{\text{prune}}$, and an initial network $\boldsymbol{\omega}^{(0)}, c^{(0)}$ of width $N(0)$; set $t = 0$.
while $t < T_{\text{out}}$ **do**
  Sample N_{trial} random nodes in $\mathbb{S}^d$ and add the normalized representatives of at most $N_{\text{trial}}/2$ existing support points as additional starts.
  For each start run ordinary L-BFGS on an unconstrained parameter whose profile is evaluated after normalization onto $\mathbb{S}^d$. Normalize the final points, remove cosine near-duplicates, and repeat refinement after a merge, for at most five rounds.
  Discard final candidates that coincide numerically with the current support.
  For each candidate compute $c^*(\omega)$ from (62), and keep it when $c^*(\omega) \neq 0$ and $\Delta J^M_{\psi_k}(c^*(\omega);\omega) < 0$; terminate if there is none.
  Retain the at most N_{ins} kept candidates with the smallest $\Delta J^M_{\psi_k}(c^*;\omega)$ and insert each with outer weight c^*.
  Apply the semismooth Newton correction (64), using (63); accept the result only if its objective is no larger than the post-insertion objective, and otherwise retain the post-insertion coefficients.
  Remove nodes with $|c_n| \le \varepsilon_{\text{prune}}$ (resulting in width $N(t+1)$).
  $t = t + 1$.
end while

6. Numerical examples

The examples compare the activation functions and regularizers on a smooth value function and on a value function with a discontinuous gradient. All runs use seed 42. Each reported parameter choice is run once, and the validation sets are random subsets of the PMP samples. Input normalization is computed from the full sample before the split.

For a fitted value function $\widehat{V}$, the relative errors are

$$\mathrm{Err}_{L^2} := \frac{\|\widehat{V} - V\|_{L^2}}{\|V\|_{L^2}},$$
$$\mathrm{Err}_{H^1} := \frac{\left(\|\widehat{V} - V\|_{L^2}^2 + \|\nabla\widehat{V} - \nabla V\|_{L^2}^2\right)^{1/2}}{\left(\|V\|_{L^2}^2 + \|\nabla V\|_{L^2}^2\right)^{1/2}}.$$

Both norms are computed empirically on the validation or evaluation set specified for each experiment.

Van der Pol Oscillator. We consider the optimal control of the Van der Pol oscillator:

$$\min_{u \in L^2(0,T;\mathbb{R})} \mathcal{C}(u;0,x) = \int_0^T \left(y_1^2(t) + y_2^2(t) + \eta u^2(t)\right) dt$$

subject to

$$\begin{cases} \frac{dy_1}{dt} = y_2, \\ \frac{dy_2}{dt} = -y_1 + y_2(1 - y_1^2) + u, \\ (y_1(0), y_2(0)) = (x_1, x_2), \end{cases}$$

In this setting, $d = 2$, $m_u = 1$, $\eta = 0.1$, $T = 3$, and

$$f(y) = \begin{pmatrix} y_2 \\ -y_1 + y_2(1 - y_1^2) \end{pmatrix}, \quad g(y) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad h(y) = y_1^2 + y_2^2.$$

By the optimality condition, $u^* = -\frac{1}{2\eta} g(y^*)^\top p^* = -\frac{1}{2\eta} p_2^*$. The two-point boundary value problem (TPBVP) becomes

$$\begin{cases} \dot{y}_1^* = y_2^*, \\ \dot{y}_2^* = -y_1^* + y_2^*(1 - y_1^{*2}) + u^*, \\ -\dot{p}_1^* = 2y_1^* - p_2^*(1 + 2y_1^* y_2^*), \\ -\dot{p}_2^* = 2y_2^* + p_1^* + p_2^*(1 - y_1^{*2}), \\ y^*(0) = x, \quad p^*(T) = 0. \end{cases} \tag{65}$$

For each initial condition $x = (x_1, x_2)$ sampled on the grid $[-3,3]^2$, we solve (65) using a Barzilai–Borwein gradient method with backtracking line search. The control is represented in a Legendre polynomial basis $u(t) = \sum_{i=0}^{N_b-1} c_i P_i(t)$ with $N_b = 30$ coefficients. The optimization yields the dataset $\{x^m, V(x^m), g^m\}_{m=1}^M$, where $V(x^m) = \mathcal{C}(u^*;0,x^m)$ and $g^m := p^{*,m}(0)$; at differentiability points, $g^m = \nabla V(x^m)$.

We solve the open-loop problem on a 30×30 grid over $[-3,3]^2$ ($M = 900$ data points). All experiments use $T_{\mathrm{out}} = 150$ outer iterations and $N_{\mathrm{trial}} = 50$ sampled candidate nodes per iteration. Each iteration inserts the single selected candidate ($N_{\mathrm{ins}} = 1$), so the one-candidate decrease statement applies. Corollary 5.2 remains conditional on the selected candidate being an exact global maximizer and is not claimed for the multistart search. The loop terminates when no candidate clears the insertion threshold, as returned by the finite multistart search. This is a numerical stopping rule, not a certificate that the global profile condition holds; some cells terminate early and others reach T_{out}.

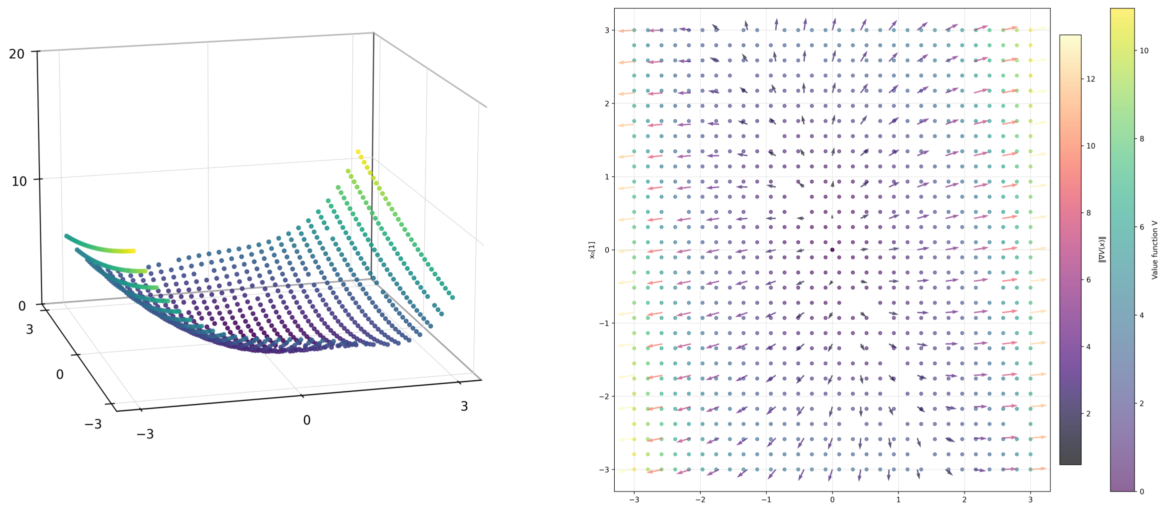


FIGURE 2. Left: value function $V(x)$ over $[-3,3]^2$. Right: $V(x)$ (dot color) and costate labels g^m (arrows); arrow color indicates $|g^m|$.

6.0.1. *Effect of the moment order p.* The moment order enters the objective through the weight $w_p(\omega) = 1 + |\omega|^p$ inside the scalar penalty: an atom at ω is charged $\phi_\gamma\big(w_p(\omega)|c|\big)$ rather than $\phi_\gamma(|c|)$, so a fixed coefficient becomes more expensive the further the atom sits from the origin. The exponent p is therefore an active parameter of the objective, not a device that acts only through a separate coefficient.

Observed support radius. For a finite atomic μ write

$$R_{\max}(\mu) := \max\{|\omega| : \omega \in \operatorname{atom}\mu\}$$

for the radius of its support. Lemma 3.19 places that support in a compact superlevel set of $|\bar{P}_p|$, so $R_{\max}$ is finite; the lemma does not itself supply a numerical radius, and the values below are measured. We use Algorithm 1 with $\gamma = 1$, equal value and gradient weights, and seed 42, for softplus, tanh, the Gaussian and GELU^2, over

$$\alpha \in \{10^{-4}, 10^{-5}\}, \qquad p \in \{2.01, 2.5, 3, 4\},$$

on both benchmarks. Figure 3 shows the resulting sixteen (problem, activation, α) families. In every family $R_{\max}$ decreases monotonically over the tested orders. The largest contraction is for GELU^2 on the pendulum at $\alpha = 10^{-5}$, from 148.06 at $p = 2.01$ to 6.77 at $p = 4$; across all families the largest radius at $p = 4$ is 11.99.

We also record the radius $R_{0.95}$, defined for a discrete measure $\mu = \sum_j c_j \delta_{\omega_j}$ by

$$R_{0.95}(\mu) := \inf\left\{R \geq 0 : |\mu|(B_R) \geq 0.95\,|\mu|(\Omega)\right\},$$

and the support size N. Unlike $R_{\max}$, both depend on the fitted coefficient distribution. Each decreases monotonically in twelve of the sixteen families, rather than all sixteen, so their behavior is reported separately rather than inferred from the support theorem.

The computable search radius. Theorem 5.1 supplies the radius (51) outside which no candidate can satisfy the insertion condition. Its constants are available before the fit: the growth exponents s_0, s_1 and the constant C_ρ of Assumption 3.1 are properties of the activation, and the remaining factor follows from the sample extent, so $C = C_\rho\big(A_M^{s_0} + A_M^{s_1-1}\big)$ with $A_M = \max\{1, \max_m \sqrt{1+|x^m|^2}\}$. Nothing is calibrated from the fit.

On the Van der Pol data ($A_M = \sqrt{3}$, hence $C = 2.73$ for softplus) the radius is 56.5 at $p = 4$, $\alpha = 10^{-4}$ and 121.7 at $p = 4$, $\alpha = 10^{-5}$, while for $p \leq 3$ it exceeds the implementation's own

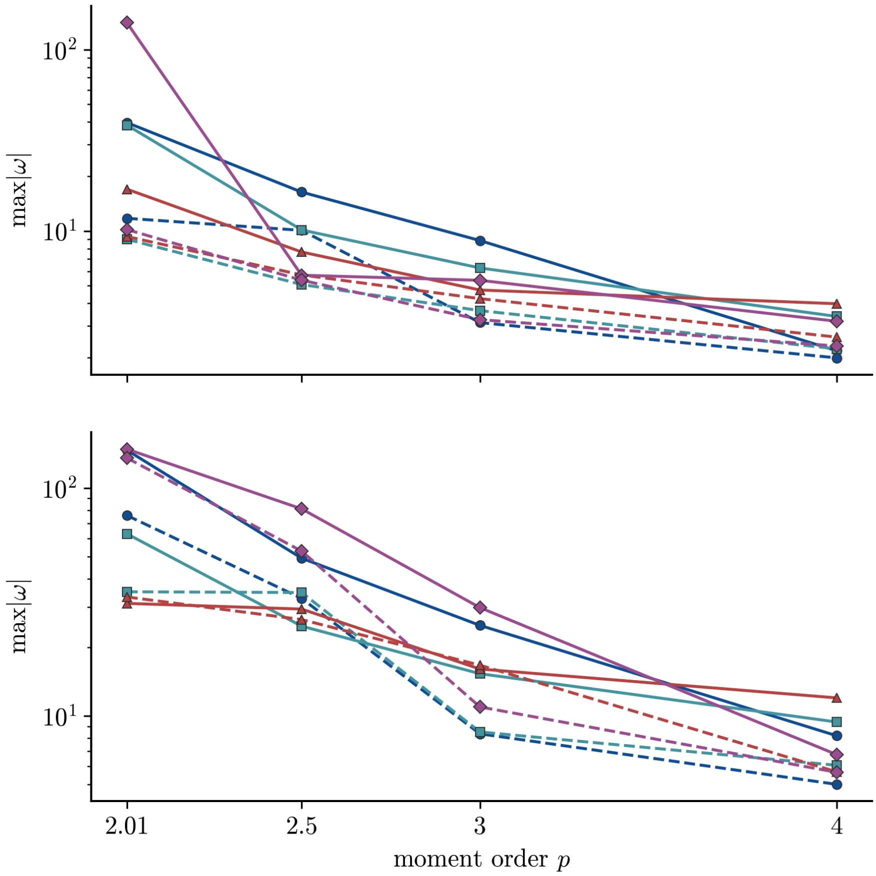


FIGURE 3. Support radius $R_{\max}$ against the moment order p, for softplus, tanh, the Gaussian and GELU^2 at $\alpha = 10^{-4}$ and $\alpha = 10^{-5}$; upper panel Van der Pol, lower panel pendulum swing-up. Colour identifies the activation and line style the value of α. All sixteen support-radius curves decrease monotonically; all use seed 42.

bound $\lambda \leq e^5$. The radius is therefore informative only at the largest tested p, and only for the activations with $s_1 = 1$; for GELU^2, where $s_1 = 2$, the exponent $1/(p - s_1)$ never becomes small enough on this grid.

The fixed-versus-theorem ablation shows a small but nonzero effect. Among the thirty-two paired sequential-insertion Van der Pol runs, eight change when $\min\{R(\mu_t), e^5\}$ replaces the fixed e^5 search filter: two at $p = 3$ and six at $p = 4$. The relative H^1 differences are at most 2.8×10^{-3}, and their sign is mixed. Because the radius determines both the distribution of random starts and the admissible final candidates, the change does not require a retained atom to sit on the boundary. The computable radius becomes operational only at the largest moment orders on this grid; it does not systematically improve or degrade the fit.

6.0.2. *Algorithm 1: the log penalty on the normalized measure.* We investigate the effect of including gradient data for softplus, tanh, and the Gaussian. Table 1 compares fitting the values alone with fitting both values and gradients.

Choice of activation function in gradient training. We compare softplus, tanh, and the Gaussian. Because the gradient-augmented loss contains both the value and gradient errors, $K(\omega)$ contributes to the approximation of V and $\nabla K(\omega)$ contributes to the approximation of ∇V. Figure 4 contrasts the three activations through their value, first derivative, and second

TABLE 1. Effect of gradient-augmented fitting with $\alpha = 10^{-4}$, $\gamma = 10$, $p = 2.01$, seed 42. The same parameters are used for all three activations, so the comparison is not confounded with per-activation tuning.

Activation	L^2 trained	H^1 trained
softplus	1.04×10^{-1}	4.24×10^{-1}
tanh	9.58×10^{-2}	4.17×10^{-1}
Gaussian	7.16×10^{-2}	4.17×10^{-1}

(A) Effect of gradient training on Err_{L^2}

Activation	L^2 trained	H^1 trained
softplus	5.82×10^{-1}	1.03×10^{-1}
tanh	5.97×10^{-1}	1.01×10^{-1}
Gaussian	5.37×10^{-1}	9.83×10^{-2}

(B) Effect of gradient training on Err_{H^1}

derivative. The softplus derivative increases from 0 to 1, the derivative of tanh tends to zero at both ends of the real line, and the Gaussian derivative is localized and changes sign. These derivatives generate different columns $a\,\rho'(a \cdot x + b)$ in the gradient fitting problem.

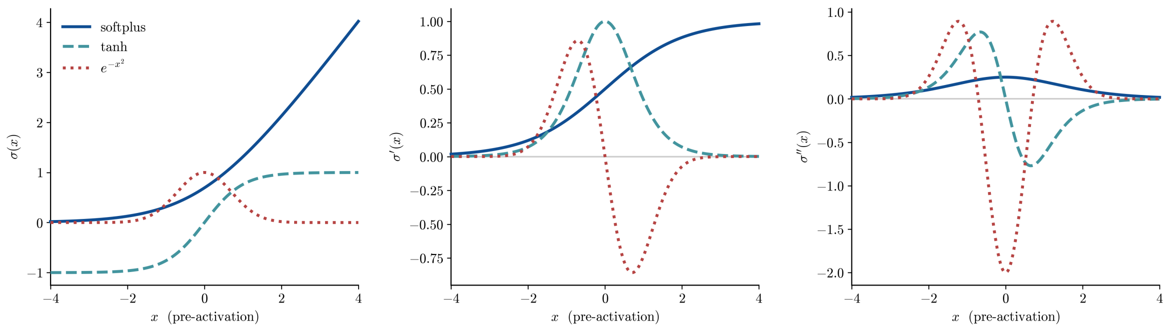
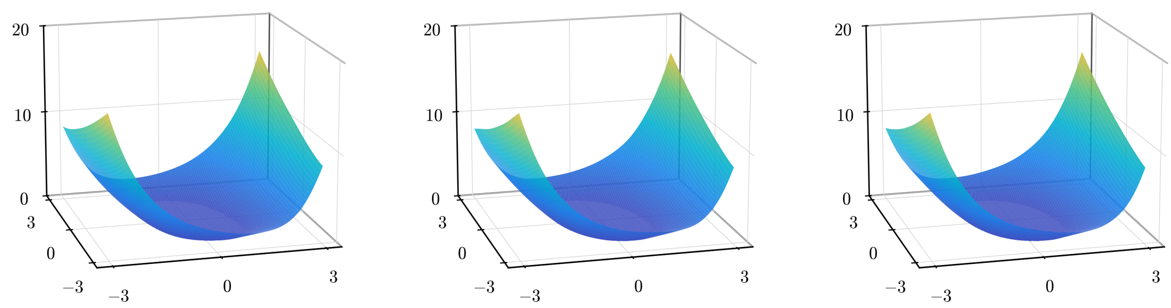


FIGURE 4. The three smooth profiles used by Algorithm 1 — softplus, tanh, and the Gaussian e^{-z^2} — shown as ρ, ρ', and ρ''. Under the gradient-augmented loss, the first derivative is the basis that reconstructs ∇V.

For the tested parameters, the Gaussian has the smallest relative H^1 error. All three approximate the smooth value function, with different support sizes.



(A) softplus ($N = 16$, $\mathrm{Err}_{H^1} = 0.103$).

(B) tanh ($N = 38$, $\mathrm{Err}_{H^1} = 0.101$).

(C) Gaussian ($N = 34$, $\mathrm{Err}_{H^1} = 0.098$).

FIGURE 5. Learned value functions under the H^1 loss using the parameter values in Table 1. All three approximate the smooth value surface; they differ in support size and accuracy.

TABLE 2. Support size and validation error for the parameter values in Table 1.

Activation	N	Err_{L^2}
softplus	6	1.04×10^{-1}
tanh	10	9.58×10^{-2}
Gaussian	10	7.16×10^{-2}

(A) L^2 trained

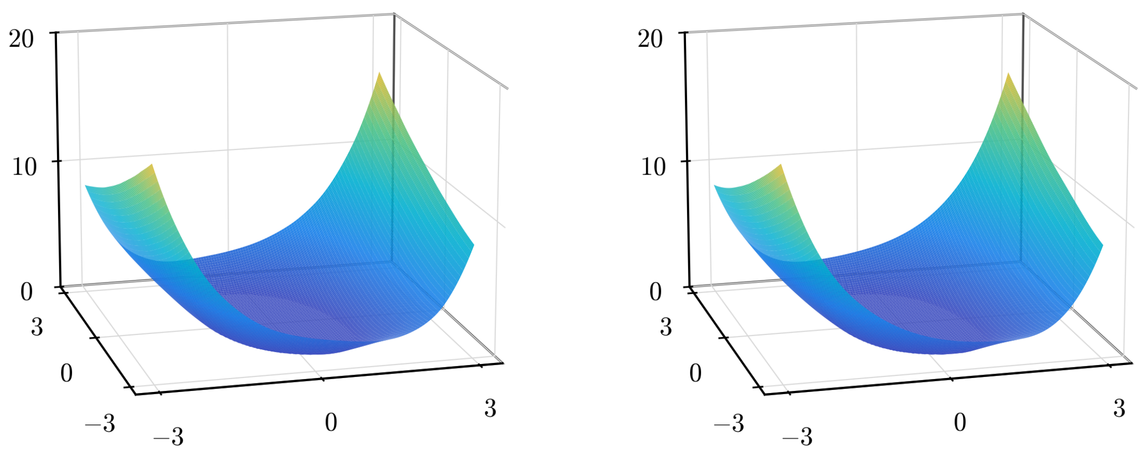
Activation	N	Err_{H^1}
softplus	16	1.03×10^{-1}
tanh	38	1.01×10^{-1}
Gaussian	34	9.83×10^{-2}

(B) H^1 trained

With the H^1 loss, all three gradient errors fall substantially: from 5.96×10^{-1} to 4.36×10^{-2} for softplus, from 6.12×10^{-1} to 4.22×10^{-2} for tanh, and from 5.51×10^{-1} to 3.57×10^{-2} for the Gaussian. Fitting values alone leaves the gradient essentially unapproximated, and the resulting feedback law would be unusable.

The Gaussian is most accurate (9.83×10^{-2}) and uses 34 atoms; tanh reaches 1.01×10^{-1} with 38, and softplus 1.03×10^{-1} with only 16. The three are separated by less in accuracy than in support size, so the ordering by Err_{H^1} alone understates the difference between them.

6.0.3. *Algorithm 2: $|c|^q$-penalty with* ReLU^k *activation.* Algorithm 2 uses the profile $\rho(z) = \mathrm{ReLU}(z)^k$ with the penalty ψ_k, namely $\alpha \sum_i |c_i|^q$ with $q = 2/(k+1)$. The swept parameter is the power k; all fits below use $\alpha = 10^{-5}$.



(A) $k = 2$
($N = 45$, $\mathrm{Err}_{H^1} = 0.100$).

(B) $k = 3$
($N = 27$, $\mathrm{Err}_{H^1} = 0.099$).

FIGURE 6. H^1-trained networks with $\alpha = 10^{-5}$ for $k \in \{2, 3\}$.

Under the L^2 loss the gradient error stays large for both powers ($\mathrm{Err}_{H^1} \approx 4\text{–}5 \times 10^{-1}$): the value-only fit ignores ∇V, so the atoms are placed without regard to the gradient field. The H^1 loss drives the gradient error down to $\approx 1.0 \times 10^{-1}$ for both powers (Table 3).

Changing k changes both the activation and the induced exponent $q = 2/(k+1)$. At the same H^1 error scale, the $k = 3$ fit uses 27 atoms instead of 45 for $k = 2$. Under the L^2 loss the fits use 12–13 atoms and have $\mathrm{Err}_{H^1} \approx 4.1\text{–}4.5 \times 10^{-1}$ (Table 3 (a)).

TABLE 3. Algorithm 2: network size and validation error vs. the atom power k ($\alpha = 10^{-5}$).

Power	N	Err_{L^2}	Err_{H^1}
$k = 2$	13	2.78×10^{-2}	4.09×10^{-1}
$k = 3$	12	3.67×10^{-2}	4.53×10^{-1}

(A) L^2 trained

Power	N	Err_{L^2}	Err_{H^1}
$k = 2$	45	4.17×10^{-1}	9.96×10^{-2}
$k = 3$	27	4.15×10^{-1}	9.87×10^{-2}

(B) H^1 trained

6.0.4. *Summary.* Figure 7 plots the running best relative H^1 error along the selected fits. At the shared Algorithm 1 operating point, softplus uses 16 atoms at error 1.03×10^{-1} and the Gaussian uses 34 at 9.83×10^{-2}. The lowest-error Algorithm 2 checkpoints use 75 atoms at 9.84×10^{-2} for ReLU^2 and 40 atoms at 9.69×10^{-2} for ReLU^3. Thus all four reach the same accuracy scale, with different support sizes.

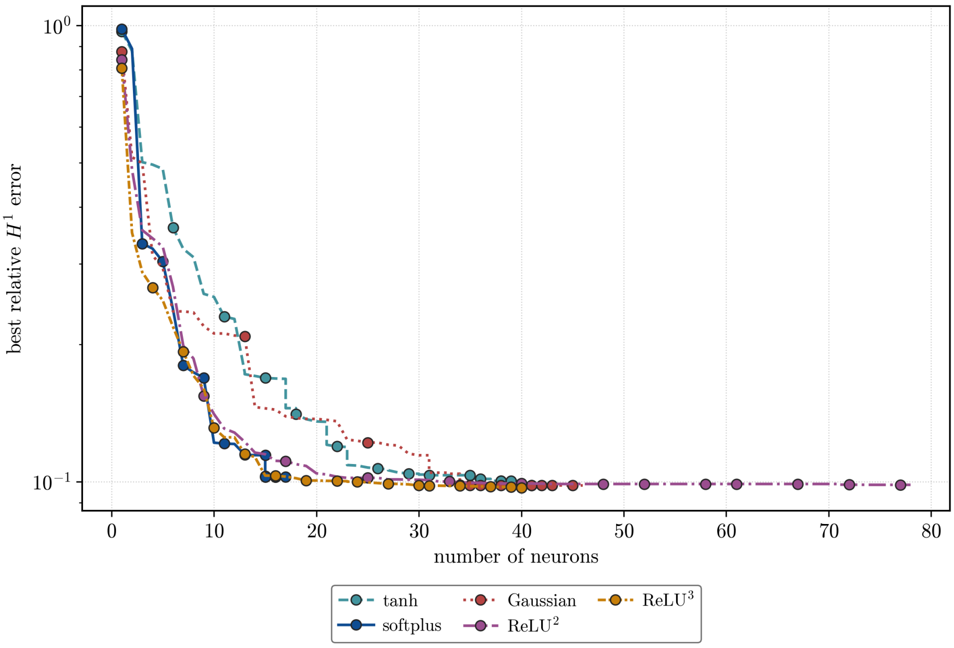
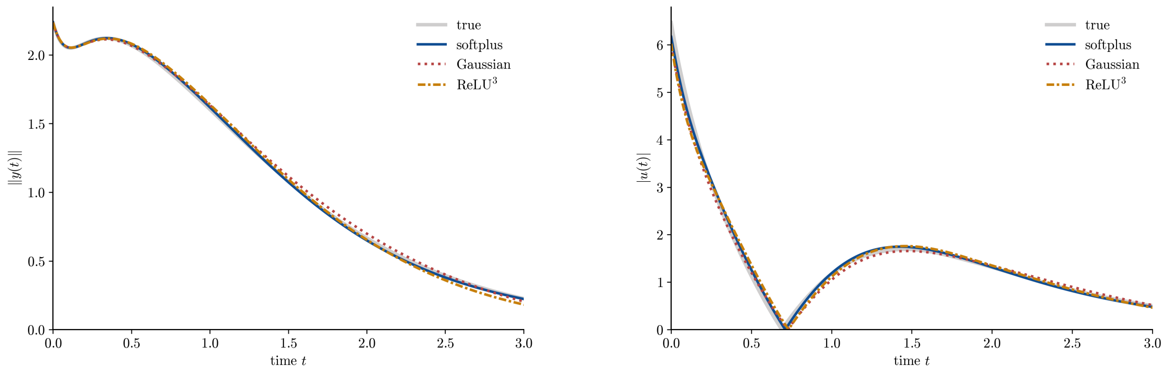


FIGURE 7. Running best relative H^1 validation error against support size. Non-homogeneous profiles use the parameter values stated in Table 1; homogeneous profiles use $\alpha\psi_k$ at $\alpha = 10^{-6}$ for the displayed $k \in \{2, 3\}$ checkpoints.

We synthesize feedback laws from $y_0 = (2, 1)$ and roll them out over the horizon $T = 12$ with fixed-step RK4, zero-order-hold controls clipped to $|u| \leq 50$ as a numerical guard. The reference law interpolates the time-zero costate component of the dataset with a C^1 Clough–Tocher interpolant and applies it as a stationary feedback. The finite-horizon optimal feedback is time-dependent and the rollout horizon exceeds the data horizon $T = 3$. Therefore its rollout cost 6.48 is not $V(0, y_0)$, and the comparison is not a test of finite-horizon optimality. Figure 8 shows $\|y(t)\|$ and $|u(t)|$ for softplus, the Gaussian, and ReLU^k ($k = 3$), beside this reference. All three drive the state to the origin and track the reference control, at essentially its closed-loop cost (6.48, 6.50, 6.50 vs. 6.48).

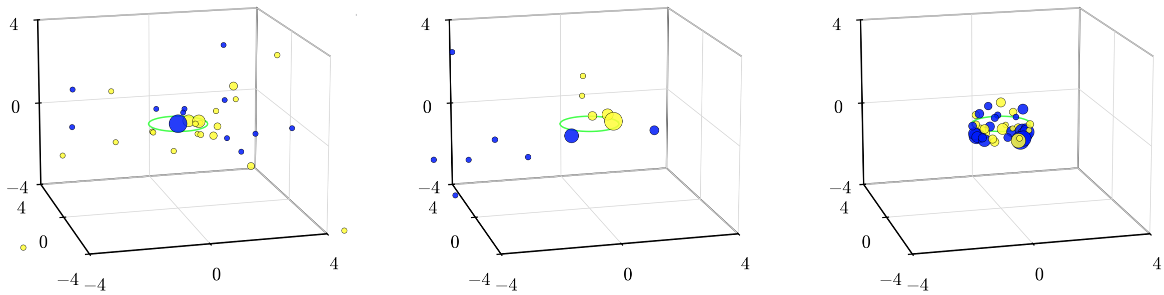
Figure 9 shows the inner parameters for the Gaussian, softplus, and ReLU^k ($k = 3$). The two algorithms produce structurally different representations: Algorithm 2 constrains its atoms to the



(A) State norm $\|y(t)\|$. (B) Control $|u(t)|$.

FIGURE 8. Evolution of the state and control from $y_0 = (2, 1)$ under the feedback synthesized from each fitted value function and under the stationary reference law obtained from the time-zero costates.

unit sphere $\mathbb{S}^2$, while Algorithm 1 leaves them free — the Gaussian atoms in particular span a wide range of norms.



(A) Gaussian. (B) softplus. (C) $\mathrm{ReLU}^3 + \alpha\,\psi_3$.

FIGURE 9. Inner parameters (a_1, a_2, b), with the unit circle in the plane $b = 0$ shown in light green. The two nonhomogeneous fits use $\alpha = 10^{-4}$, $\gamma = 10$, $p = 2.01$; dot color gives the sign and dot size the magnitude of the outer weight.

Pendulum Swing-Up. The question in this example is how a sparse gradient-trained network approximates a continuous value function whose gradient jumps across a switching set. A finite network with a continuously differentiable activation has a continuous gradient and therefore cannot reproduce the jump exactly. Rectified homogeneous activations can place changes of slope along hyperplanes. We compare these two classes at the support sizes returned by Algorithms 1 and 2.

Figure 10 contrasts three activation functions. The rectified linear unit ReLU has a step derivative, so a single neuron can represent a gradient jump on its hyperplane. The derivative of ReLU^2 is continuous and piecewise linear, whereas the softplus derivative is smooth. The positive homogeneity of the rectified functions allows the fractional-exponent formulation of Section 4.

The example we consider follows from [10]. We write the state as

$$x = (x_1, x_2)^\top = (\theta, \dot\theta)^\top,$$

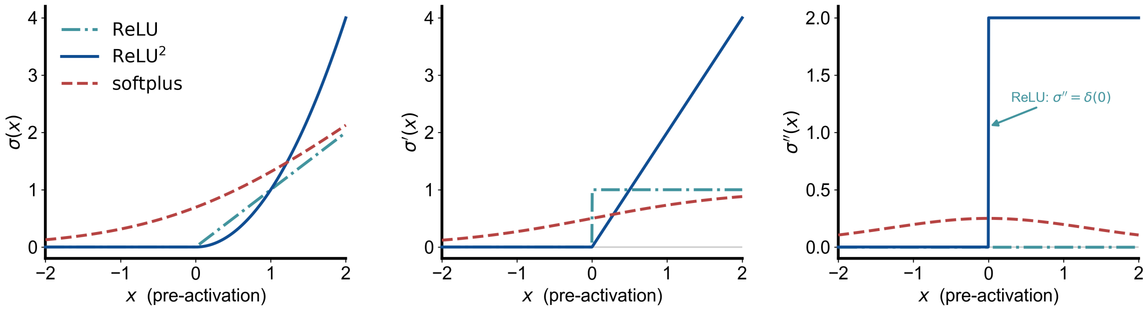


FIGURE 10. The profiles ρ, ρ', and ρ'' for ReLU, ReLU^2, and softplus. The derivative of ReLU jumps, the derivative of ReLU^2 has a kink, and the softplus derivative is smooth. The second distributional derivative of ReLU is a Dirac measure at the origin.

the pendulum angle and angular velocity. (The letter ω remains reserved for the network parameters.) We consider the optimal control problem

$$\min_{u \in L^2(0,\infty;\mathbb{R})} \mathcal{C}(u;0,x) = \int_0^\infty \left(\sin^2\theta(t) + (\cos\theta(t) - 1)^2 + \dot\theta^2(t) + \eta u^2(t) \right) dt$$

subject to

$$\begin{cases} m\ell^2\,\ddot\theta = -b\dot\theta + mg\ell\sin\theta + u, \\ (\theta(0), \dot\theta(0)) = (x_1, x_2). \end{cases}$$

The target equilibria are $x_{U,j} := (2j\pi, 0)$, $j \in \mathbb{Z}$; for a selected target we write $x_U = x_{U,j}$.

In this setting the state dimension is $d = 2$ and the control is scalar. The physical and cost parameters are $m = \ell = 1$, $b = 0.1$, $g = 9.8$, and $\eta = 1$. To initialize the backward integration, we use the local linear-quadratic regulator (LQR) approximation near x_U. Let $\widetilde{x} = x - x_U$ be the local coordinate around the upright equilibrium. Linearizing the dynamics at $(x_U, 0)$ gives

$$\dot{\widetilde{x}} = A\widetilde{x} + Bu, \quad A = \begin{pmatrix} 0 & 1 \\ \frac{g}{\ell} & -\frac{b}{m\ell^2} \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ \frac{1}{m\ell^2} \end{pmatrix}.$$

The running cost is locally approximated by

$$|\widetilde{x}|^2 + \eta u^2.$$

The local infinite-horizon LQR value function is

$$V^\infty(\widetilde{x}) = \widetilde{x}^\top P \widetilde{x},$$

where P is the positive definite solution of the algebraic Riccati equation

$$A^\top P + PA - \frac{1}{\eta} PBB^\top P + I_2 = 0.$$

The superscript ∞ refers to the infinite-horizon LQR problem. For $\varepsilon > 0$, we define the local LQR sublevel set and boundary by

$$\mathcal{E}_\varepsilon = \{\widetilde{x} : V^\infty(\widetilde{x}) \le \varepsilon\}, \qquad \partial\mathcal{E}_\varepsilon = \{\widetilde{x} : V^\infty(\widetilde{x}) = \varepsilon\}.$$

The corresponding state-costate system becomes

$$\begin{cases} m\ell^2\,\ddot{\theta}^* = -b\dot{\theta}^* + mg\ell\sin\theta^* + u^*, \\ -\dot{p}_1^* = 2\sin\theta^* + \dfrac{g}{\ell}\cos\theta^*\,p_2^*, \\ -\dot{p}_2^* = 2\dot{\theta}^* + p_1^* - \dfrac{b}{m\ell^2}p_2^*, \\ x^*(0) = x, \quad x^*(T_x) - x_U \in \partial\mathcal{E}_\varepsilon, \quad p^*(T_x) = \nabla V^\infty(x^*(T_x) - x_U). \end{cases} \tag{66}$$

In the experiment we use $\varepsilon = 2\times 10^{-4}$, and on $\partial\mathcal{E}_\varepsilon$ the terminal costate is

$$p^*(T_x) = \nabla V^\infty(\widetilde{x}^*(T_x)) = 2P\widetilde{x}^*(T_x).$$

For the data generation, we integrate 2000 PMP paths backward from the local LQR boundary $\partial\mathcal{E}_\varepsilon$. Along each path, the accumulated running cost plus the boundary value ε gives $V(x)$, and the costate gives the label $g = p$. At differentiability points, $g = \nabla V(x)$. The backward integration stops when the accumulated value reaches 100; the training samples are restricted to values below 50. This yields the dataset

$$\{x^m, V(x^m), g^m\}_{m=1}^M.$$

The construction provides samples on both sides of the gradient discontinuity. For $j \in \mathbb{Z}$, let $V_{2j\pi}(x)$ be the fixed-target value obtained by minimizing over paths that reach the local LQR neighbourhood of $(2j\pi, 0)$. The stationary value is

$$V(x) = \min_{j\in\mathbb{Z}} V_{2j\pi}(x).$$

In the region where the targets $(0,0)$ and $(2\pi,0)$ are minimizing, this reduces to $V(x) = \min\{V_0(x), V_{2\pi}(x)\}$. Define the representative switching set between these adjacent targets by

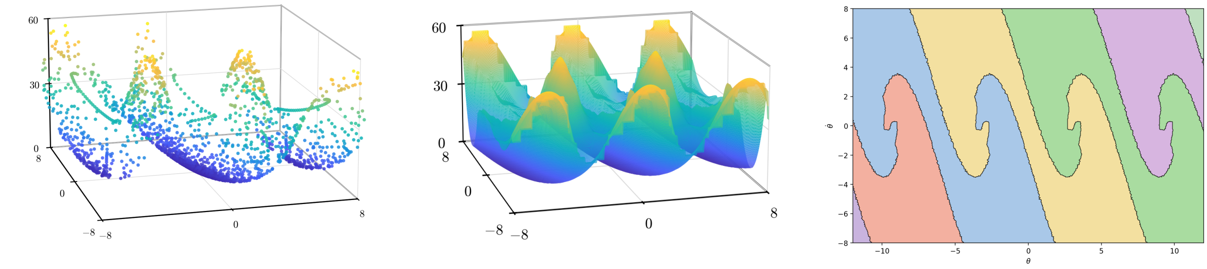
$$S_{0,2\pi} := \{x \in \mathbb{R}^2 : V_0(x) = V_{2\pi}(x) = V(x)\}.$$

On the numerical domain $D = [-8,8]^2$, the periodic switching set and its closed δ-neighbourhood are

$$S := D \cap \bigcup_{j\in\mathbb{Z}} \big(S_{0,2\pi} + (2j\pi, 0)\big), \qquad S_\delta := \{x \in D : \operatorname{dist}(x,S) \le \delta\}, \qquad \delta > 0.$$

The regional errors below use $\delta = 0.3$. At points of S, two adjacent fixed-target values agree while their gradients may differ; the sampled PMP costates provide the corresponding one-sided gradients, without asserting a classical gradient on S.

The generator locates $S_{0,2\pi}$ by intersecting equal-value contours with their 2π-shifted copies and uses periodic translates in D. To obtain samples on both sides of S, the raw paths are shifted by $\pm 2\pi$ in θ. A shifted point within distance 0.5 of S is retained when its value agrees with the corresponding fixed-target value V_0 or $V_{2\pi}$ and is at least 0.3 below the first-order extrapolation of the other value. The latter is computed from the 32 nearest raw samples within radius 0.1. This adds 900 samples, 300 on one side of S and 600 on the other. The final dataset contains 3,900 samples: 3,000 from the backward PMP paths and 900 near S. It therefore contains costate labels from both sides of the discontinuity. Figure 11 shows the dataset generated.



(A) value samples $\{x^m, V(x^m)\}$. (B) value surface $V(x)$. (C) regions associated with the periodic targets.

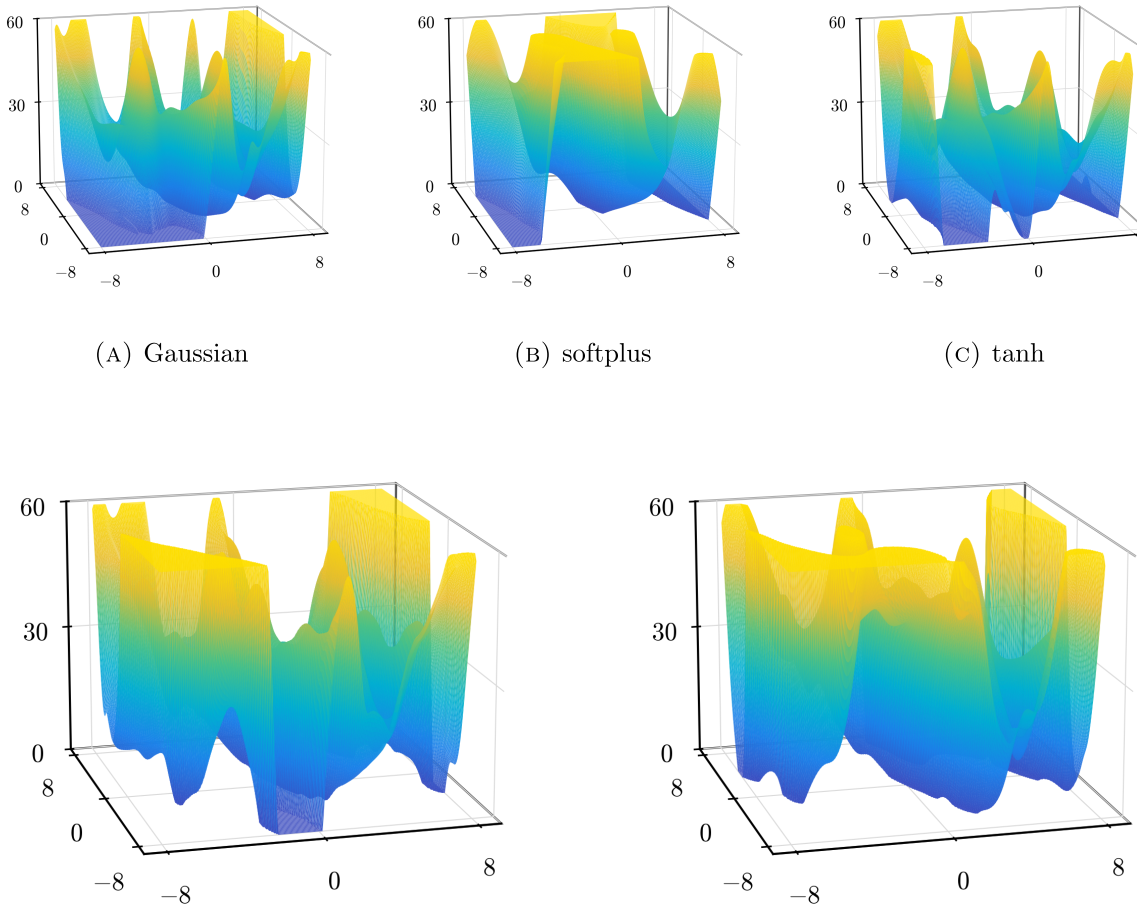
FIGURE 11. Optimal value function $V(x)$ over the $(\theta, \dot{\theta})$ plane, 2π-periodic in θ. (a) The raw (x, V) samples produced by the PMP generator. (b) The reconstructed value surface. (c) Regions where V_0 or another periodic fixed-target value is minimal; their boundaries form S, across which ∇V may be discontinuous.

6.0.5. *Approximation near the switching set.* Figure 12 shows the fitted surfaces on the same physical window and value range as the value function in Figure 11b. The cross-section below compares the approximations near S.

Let x_S be the selected regular point of S in the region with the highest sample density, and let n be a unit normal to S at x_S. Figure 13 shows the cross-section $x(s) = x_S + sn$. Along this section, $V = \min\{V_0, V_{2\pi}\}$. It has a kink at $s = 0$, and the normal gradient $n \cdot \nabla V$ jumps there. The training set contains samples on both sides of this point. No fitted model reproduces the magnitude of the jump: all undershoot the steep gradient on the side $s < 0$. ReLU^2 has the smallest error on $S_{0.3}$ and develops the sharpest change of slope near $s = 0$; the smooth Gaussian, softplus, and tanh fits interpolate through the discontinuity. The higher rectified powers remain smooth enough in their first derivative to miss the jump at the fitted budgets.

We next ask whether adding samples near S reduces the error on $S_{0.3}$. We compare 6,000-sample sets containing 23%, 40%, or 60% of their samples near S with a set obtained by adding 2,000 such samples to the first set. The Gaussian and ReLU^2 networks are evaluated on the same disjoint set of approximately 9.4×10^5 points.

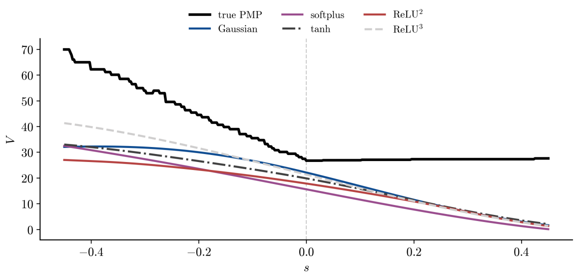
For the tested sets, increasing the proportion of samples near S produces no systematic or material reduction in the error on $S_{0.3}$ (Table 4). Thus the observed difficulty is not resolved by additional local samples.



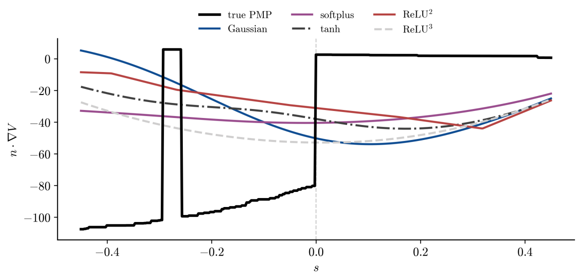
(A) Gaussian (B) softplus (C) tanh

(D) ReLU^2 (E) ReLU^3

FIGURE 12. Learned value surfaces $\widehat{V}(x)$ over the $(\theta, \dot{\theta})$ window $[-8, 8]^2$, displayed on the common range $\widehat{V} \in [0, 60]$. The ReLU^k models are trained with the fractional-exponent penalty ψ_k, and the nonhomogeneous models with (56) using the parameter values reported in Table 5.



(A) value V along the normal.



(B) normal gradient $n \cdot \nabla V$ along the normal.

FIGURE 13. Normal cross-section of the switching curve, parametrized by the normal distance s. Left: the true value (dashed) is $\min\{V_0, V_{2\pi}\}$ on this cross-section and has a kink at $s = 0$. Right: the true normal gradient jumps at $s = 0$; ReLU^2 develops the sharpest fitted change of slope, while the smooth models interpolate through the discontinuity. The rectangular excursion of the reference curve near $s \approx -0.3$ is a nearest-neighbor artifact in the evaluation of $\min\{V_0, V_{2\pi}\}$, not a feature of V.

Model	Training set	$S_{0.3}$	$D \setminus S_{0.3}$	N
Gaussian	6k, 23% near S	2.78×10^{-1}	1.36×10^{-1}	73
	6k, 40% near S	2.76×10^{-1}	1.58×10^{-1}	143
	6k, 60% near S	2.99×10^{-1}	1.79×10^{-1}	145
	6k + 2k near S	2.96×10^{-1}	1.65×10^{-1}	79
ReLU^2	6k, 23% near S	3.30×10^{-1}	2.94×10^{-1}	140
	6k, 40% near S	2.93×10^{-1}	2.46×10^{-1}	139
	6k, 60% near S	3.56×10^{-1}	2.94×10^{-1}	145
	6k + 2k near S	3.54×10^{-1}	2.65×10^{-1}	105

TABLE 4. Relative H^1 error on $S_{0.3}$ and $D \setminus S_{0.3}$. For each model and training set, the table reports the result with the smallest error on $S_{0.3}$ among the three tested values of α, with every entry in that row taken from the same run. The Gaussian uses $(\gamma, p) = (0, 2.01)$ and $\alpha \in \{10^{-3}, 10^{-4}, 10^{-5}\}$; ReLU^2 uses ψ_2 and $\alpha \in \{10^{-4}, 10^{-5}, 10^{-6}\}$. Errors are computed after un-normalizing, so the two blocks share one definition. The symbol N denotes the support size.

6.0.6. *Validation and regional errors.* Figure 14 tracks the best relative H^1 validation error reached as neurons are inserted, for each activation paired with its penalty. The three nonhomogeneous activations use the parameter values in Table 5; the rectified powers use ψ_k with $\alpha = 10^{-6}$. The Gaussian reaches the smallest validation error, approximately 3.61×10^{-1}, followed by ReLU^2 at 3.75×10^{-1}, tanh at 4.06×10^{-1}, ReLU^3 at 4.56×10^{-1}, and softplus at 5.31×10^{-1}.

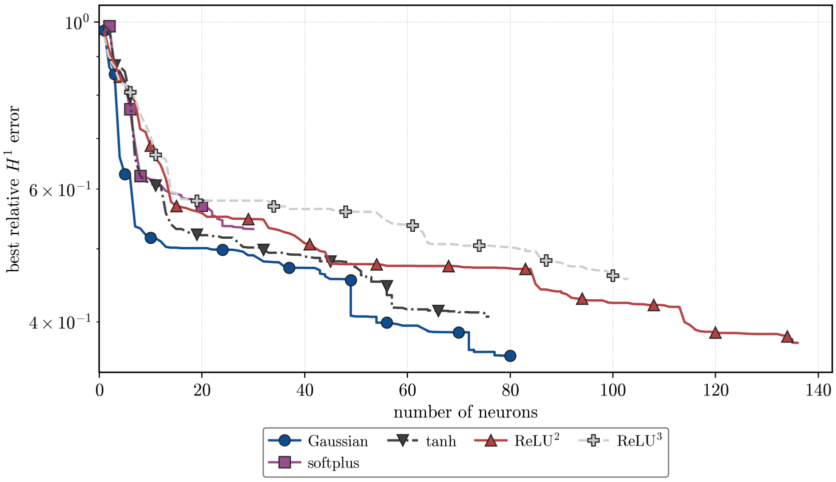


FIGURE 14. Best relative H^1 validation error against neuron count. The non-homogeneous activations use (56); the homogeneous activations use $J^M_{\psi_k}$ with $\alpha = 10^{-6}$ for $k \in \{2, 3\}$.

Figure 15 reports the relative H^1 error on a separate evaluation set of approximately 9.6×10^5 points. Training rows are excluded, and for each run we use the recorded iterate with the smallest training objective. Errors are reported on $S_{0.3}$ and $D \setminus S_{0.3}$. The Gaussian has the smallest errors in both regions (0.304 and 0.191). The corresponding values are 0.340 and 0.250 for ReLU^2, 0.358 and 0.197 for tanh, 0.409 and 0.294 for ReLU^3, and 0.501 and 0.417 for softplus.

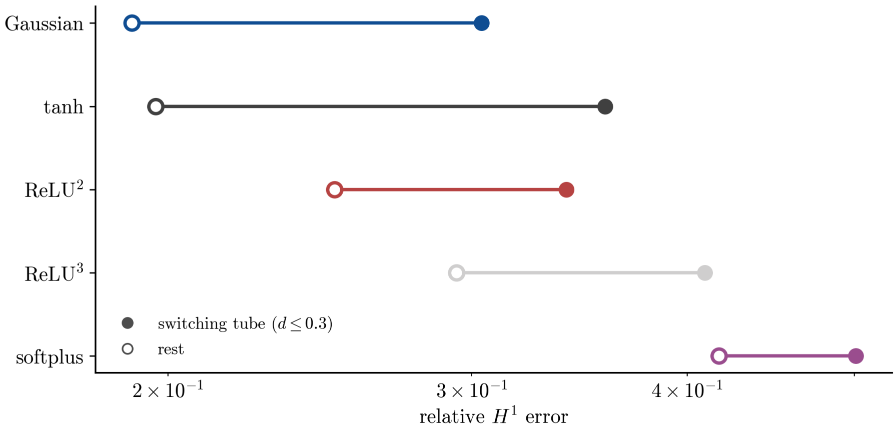


FIGURE 15. Relative H^1 error on $S_{0.3}$ (filled markers) and $D \setminus S_{0.3}$ (open markers), with rows ordered by the latter error.

Figure 16 draws the line $\{a \cdot x + b = 0\}$ associated with each atom in the physical $(\theta, \dot\theta)$ plane, with opacity proportional to the outer weight $|c_n|$, for ReLU^2 (left) and Gaussian (right); the switching curve is shown in black. The lines with the largest coefficients for ReLU^2 are approximately parallel to the diagonal parts of S. The Gaussian lines have a wider range of orientations. Each Gaussian atom $e^{-(a\cdot x+b)^2}$ is constant along its line.

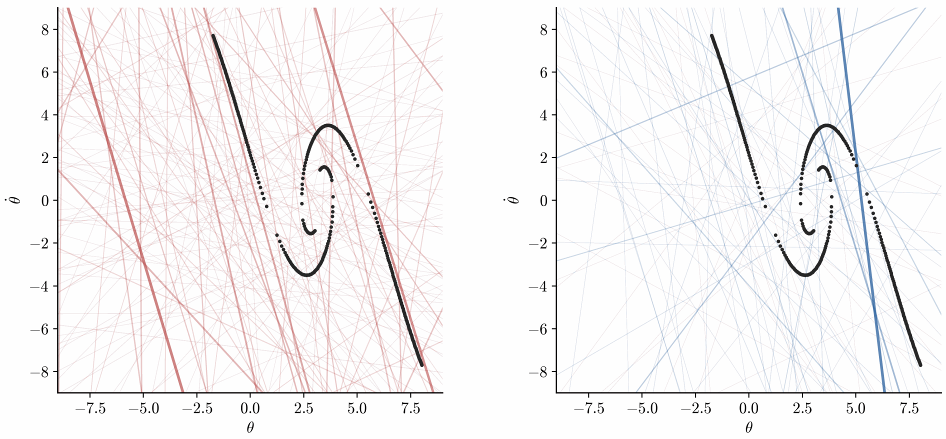


FIGURE 16. Lines $\{a \cdot x + b = 0\}$ in the $(\theta, \dot\theta)$ plane (opacity $\propto |c_n|$) for ReLU^2 (left) and Gaussian (right); the switching curve is black. ReLU^2's strongest ridges run approximately parallel to the switching set; the Gaussian atoms have a wider range of orientations.

6.0.7. *Feedback synthesis across the switching set.* Minimizing the Hamiltonian in u gives the optimal control in feedback form,

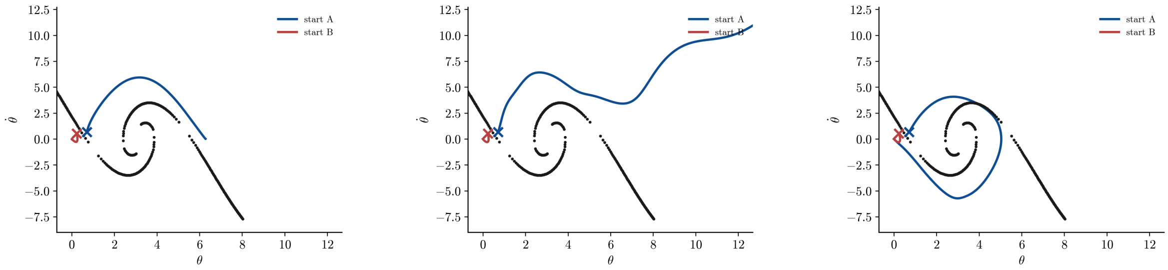
$$u(x) = -\frac{1}{2\,\eta\, m\ell^2}\,\partial_{\dot\theta} V(x);$$

replacing V by a fitted $\widehat{V}$ yields a candidate feedback law. We test each law in closed loop from two initial states placed on either side of the switching curve, $A = (0.71, 0.68)$ and $B = (0.23, 0.53)$, integrating the dynamics over the horizon $T = 10$ with fixed-step RK4 ($\Delta t = 5 \times 10^{-3}$), zero-order-hold controls, and the numerical bound $|u| \le 30$ (the OCP itself is unconstrained). The optimal paths from the two starts are different. From A, the optimal law swings the pendulum over the top to the upright at $\theta = 2\pi$, while from B it brakes directly to the upright at $\theta = 0$. Thus it must select the path associated with $V_{2\pi}$ at A and the path associated with V_0 at B. The reference law uses the 40 nearest raw PMP samples to compare V_0 and $V_{2\pi}$ and applies the feedback formula using the costate associated with the smaller value.

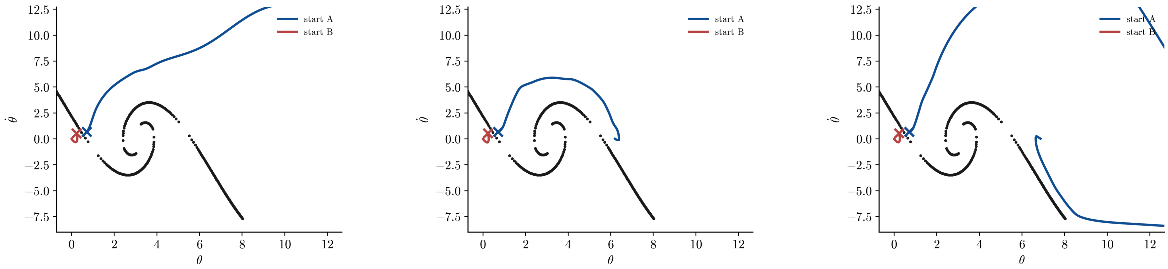
Two fitted laws reach the upright neighbourhood from both starts, ReLU^2 and softplus, and they do so at different costs (Table 5). From B, ReLU^2 brakes to the upright at essentially the reference cost (10.3 vs. 10.2); from A it reaches the upright with cost 76.3 rather than the reference cost 26.2. Softplus reaches the upright at costs 69.8 and 11.2. The Gaussian, tanh, and ReLU^3 satisfy the terminal criterion only from B. Figure 18 shows the control signal from B: the ReLU^2 law is closest to the reference, while all five fitted laws that reach the upright from B produce distinct transient controls.

In summary, the Gaussian gives the smallest validation and regional errors, while ReLU^2 gives the rollout closest to the reference. The ReLU^2 law reaches the upright from both starts, at cost 10.3 from B and 76.3 from A against the reference 10.2 and 26.2. Softplus also reaches the upright from both starts, but with a larger cost from A; the Gaussian, tanh, and ReLU^3 succeed only from B.

The regional approximation error and the binary terminal outcome contain different information. The Gaussian and tanh fits are more accurate than softplus on both reported regions, yet softplus makes the correct branch decision from A while they do not. Conversely, ReLU^2 has larger



(A) reference (PMP) (B) Gaussian (C) softplus

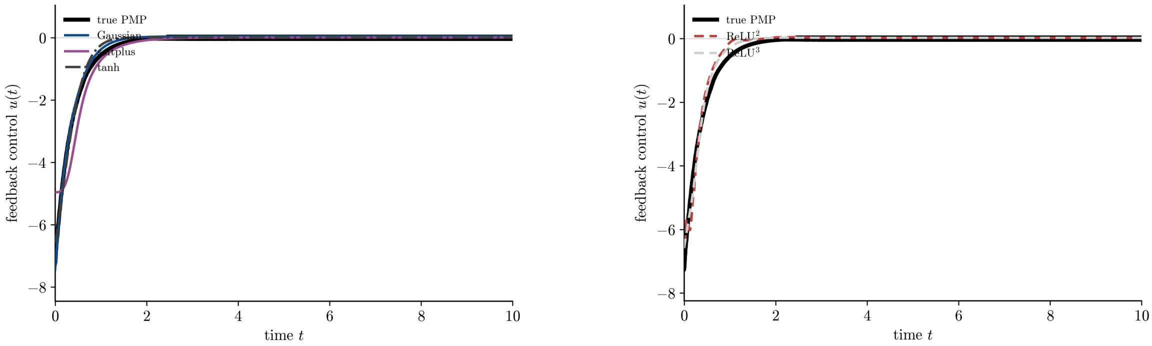


(D) tanh (E) $\mathrm{ReLU}^2 + \psi_2$ (F) $\mathrm{ReLU}^3 + \psi_3$

FIGURE 17. Closed-loop paths in the $(\theta, \dot\theta)$ plane from the two starts A (blue) and B (red) on either side of the switching curve (black), one panel per feedback law; all panels share the same axes. The reference law swings over the top from A and brakes to $\theta = 0$ from B. Softplus and ReLU^2 reach an upright neighbourhood from both starts; the other fitted laws reach one only from B.

	start $A = (0.71, 0.68)$		start $B = (0.23, 0.53)$	
Feedback law	cost	reaches upright	cost	reaches upright
reference (PMP)	26.2	yes	10.2	yes
Gaussian	8451.8	no	10.1	yes
softplus	69.8	yes	11.2	yes
tanh	7822.2	no	10.2	yes
$\mathrm{ReLU}^2 + \psi_2$	76.3	yes	10.3	yes
$\mathrm{ReLU}^3 + \psi_3$	1609.3	no	10.2	yes

TABLE 5. Closed-loop rollout cost over the horizon $T = 10$ and terminal outcome for each feedback law from the two starts on either side of the switching curve; "reaches upright" means the final state satisfies $\mathrm{dist}(\theta, 2\pi\mathbb{Z}) < 0.4$ and $|\dot\theta| < 0.4$ at $T = 10$. This condition concerns the terminal state of one finite rollout and does not assert asymptotic stability. The reference law compares V_0 and $V_{2\pi}$ using the nearest PMP samples. The middle block uses Algorithm 1 with $\alpha = 10^{-4}$, $\gamma = 10$, $p = 2.01$; the lower block uses Algorithm 2 at the most accurate α for each k. For ReLU^3 from A the rollout leaves the useful operating region and accumulates a very large cost.



(A) Algorithm 1.

(B) ReLU^k models.

FIGURE 18. Feedback control $u(t)$ from start B, per law, beside the reference (black). The reference brakes to $\theta = 0$ with u rising from ≈ -7 to 0; ReLU^2 follows it most closely.

regional errors than the Gaussian and tanh away from the switching set, yet gives the rollout closest to the reference. Thus the switching-set error diagnoses the difficulty of approximating the discontinuous gradient, but a reliable feedback assessment still requires a closed-loop test.

7. CONCLUSION AND OUTLOOK

We developed sparse gradient-augmented neural-network approximations of value functions from PMP data on an unbounded parameter space. For nonhomogeneous profiles, the adopted objective is

$$J(\mu) = L(\mu) + \alpha \Phi_\phi(\mu_p), \qquad \mu_p = w_p \mu, \qquad K_p = \frac{K}{w_p}.$$

The finite right slope $L_\phi = \phi'(0+)$ determines the cost of the continuous part in the measure penalty. Theorem 3.9 gives its disjoint-test representation, and Corollary 3.10 gives weak-* lower semicontinuity and bounded total-variation sublevels. The normalization isometry in Theorem 3.13 places the parameter weight inside the scalar penalty and makes the normalized dictionary vanish at infinity under a polynomial growth gap.

For value growth $s_0 < p$, normalized weak-* compactness gives a global minimizer (Theorem 3.16). Under full H^1 growth $s_1 < p$, Theorem 3.17 gives the global inequality

$$|\bar{P}_p(\omega)| \le \alpha L_\phi.$$

With strict bounded-interval curvature, every finite-objective local minimizer has no continuous part, distinct normalized atoms satisfy the quantitative separation in Theorem 3.20, and its number of neurons is bounded by a covering number. Corollary 3.21 shows that every global minimizer is finite atomic and that the measure-theoretic minimum equals an attained finite-network minimum. For exact global maximization of $|P_{p,\mu}|$ followed by nonincreasing corrections, Theorem 5.1 gives squared one-step descent, while Corollary 5.2 gives convergence of the excess above the insertion threshold to zero and a best-iterate rate of order $n^{-1/2}$. The sufficient local condition is given in Theorem 3.23.

For positively k-homogeneous profiles, radial scaling is absorbed into the outer coefficient. The equivalent problem on the unit sphere has the fractional-power penalty $|c|^{2/(k+1)}$, together with the existing first-order conditions, global existence result, and explicit width bound. This infinite-slope case is structurally linked to, but is not obtained by substitution into, the finite-slope theory of Section 3.

For the Van der Pol oscillator, the moment order changes the observed support statistics and the theorem search radius; the numerical trends are summarized in Section 6.0.1. The Gaussian gives the smallest H^1 error among the tested nonhomogeneous profiles, while softplus uses fewer atoms, and all three synthesize a feedback law that stabilizes the system at close to the reference cost. The homogeneous ReLU^k networks obtain comparable H^1 errors with smaller supports for larger k.

For the pendulum swing-up, the ReLU^2 errors in Table 4 are larger on $S_{0.3}$ than on its complement for every training set; the same ordering holds for the Gaussian, whose switching-region error is about 1.7–2.0 times its complementary error. Adding training samples near S produces no systematic or material reduction in the error on $S_{0.3}$ at the tested network widths. The Gaussian gives the smallest regional errors, while ReLU^2 gives the closed-loop path closest to the reference. Softplus also reaches an upright neighbourhood from both initial states, although at a larger cost from A; the Gaussian, tanh, and ReLU^3 do so only from B. The regional error and the closed-loop branch decision are therefore complementary diagnostics rather than interchangeable rankings.

The experiments are two-dimensional, and the behavior in higher dimensions remains to be studied. For targets with a gradient discontinuity, different weights on samples near and away from S may also be considered.

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